Gasoline Profiles and Impact of Ethanol Blending in Asia

October 2025

U.S. GRAINS & BIOPRODUCTS COUNCIL

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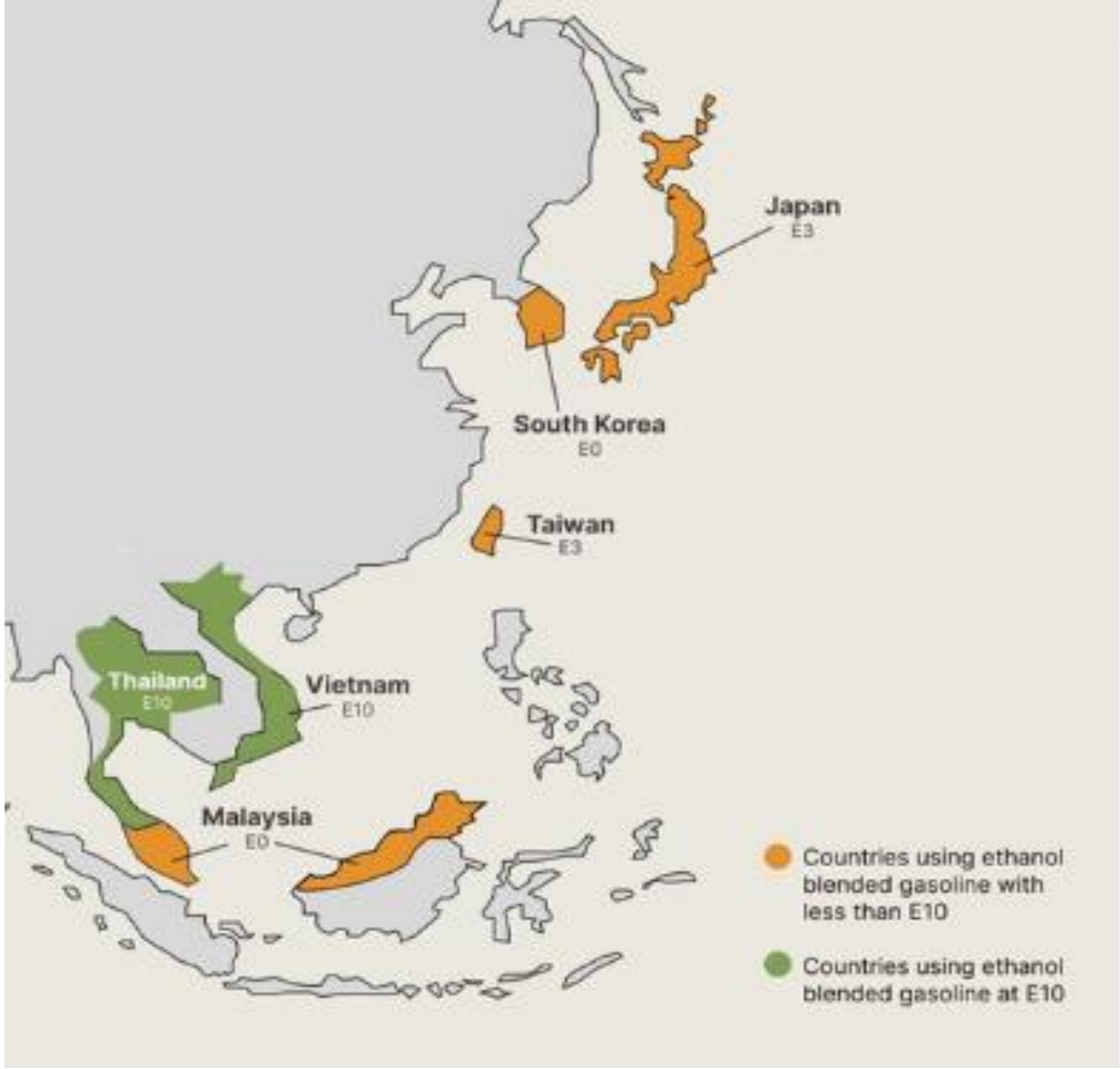
Ethanol Blending in Asia

There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

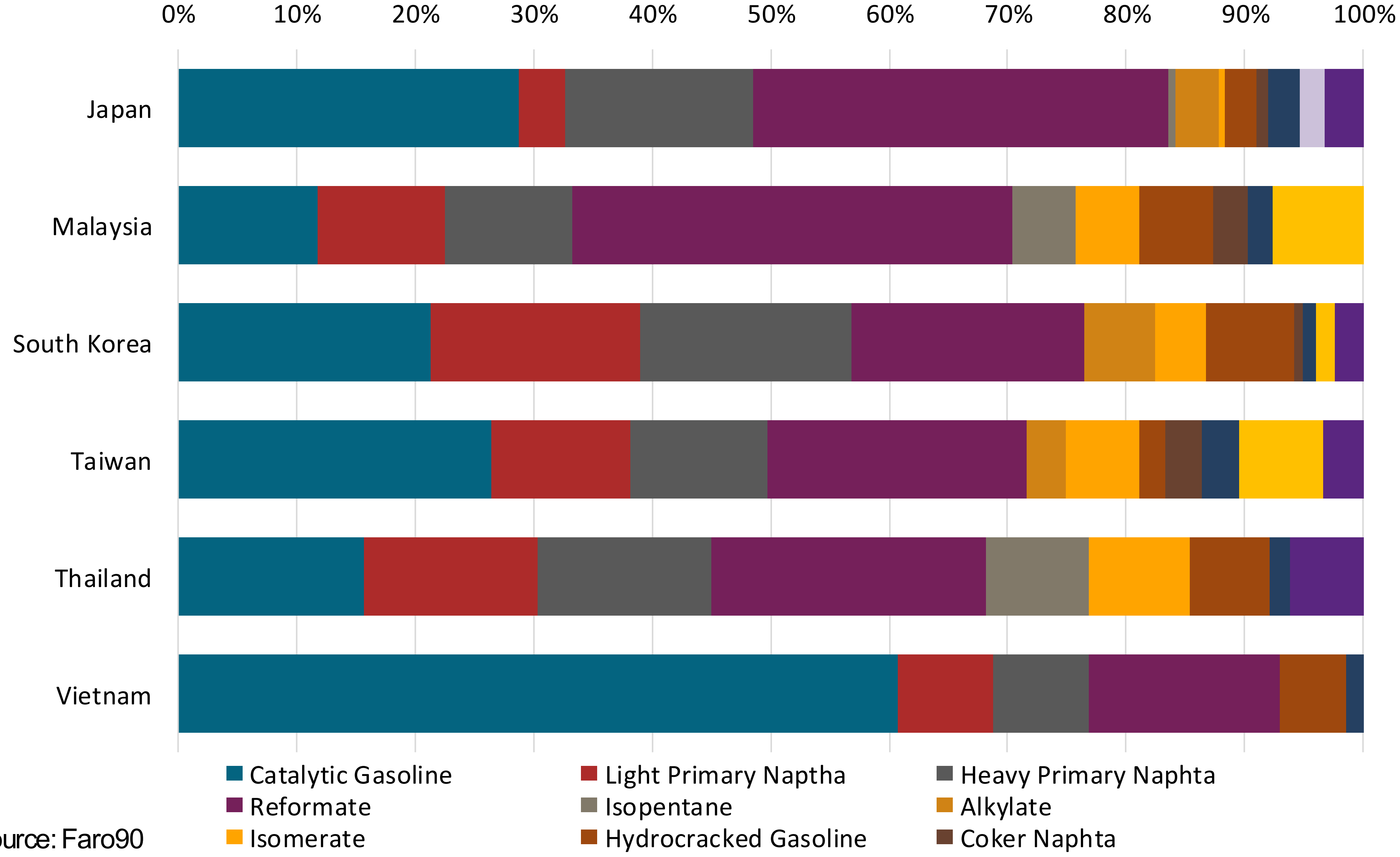


Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Gasoline Component Blending in Asia

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naptha
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

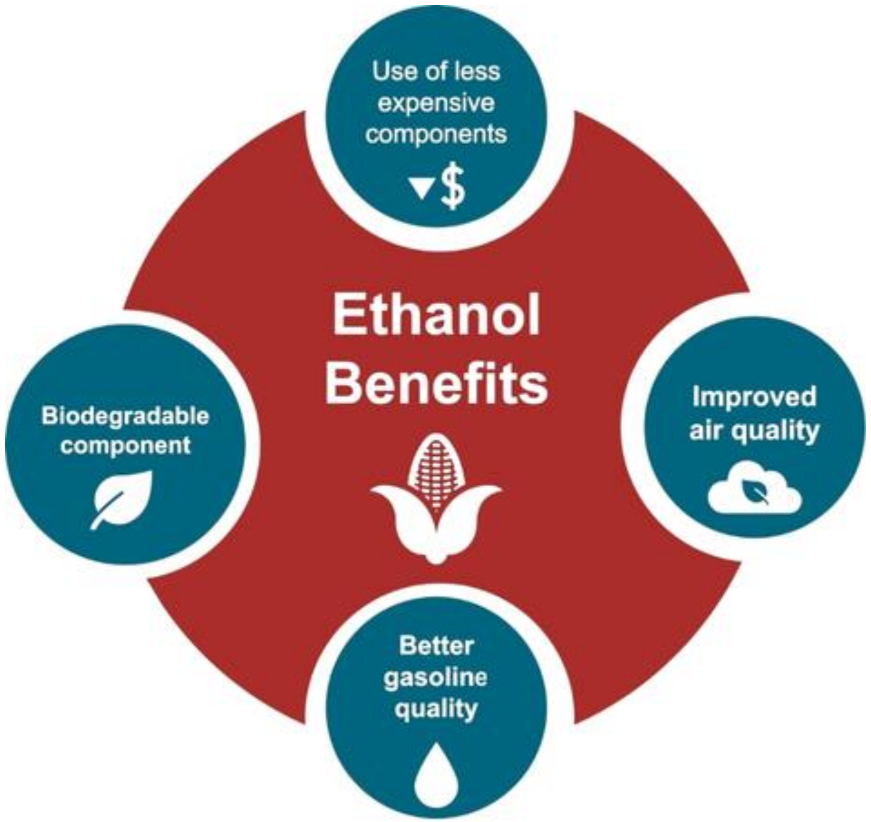
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Vehicle Emission Impact for Ethanol Gasoline Blending

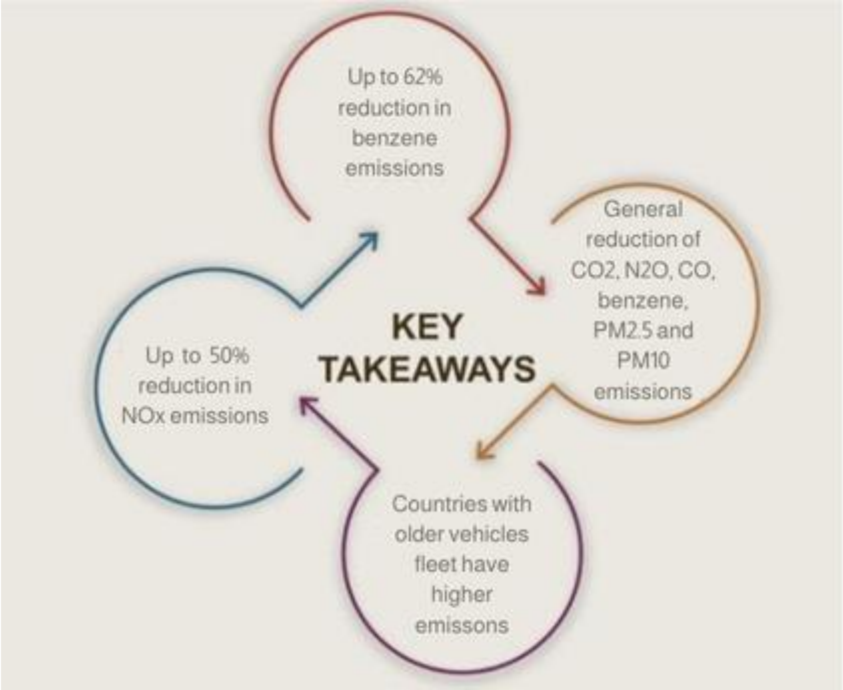
The model used in this analysis takes as a reference the [International Vehicle Emissions Model (IVE).](#)
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



Source: Bureau of transportation statistics.

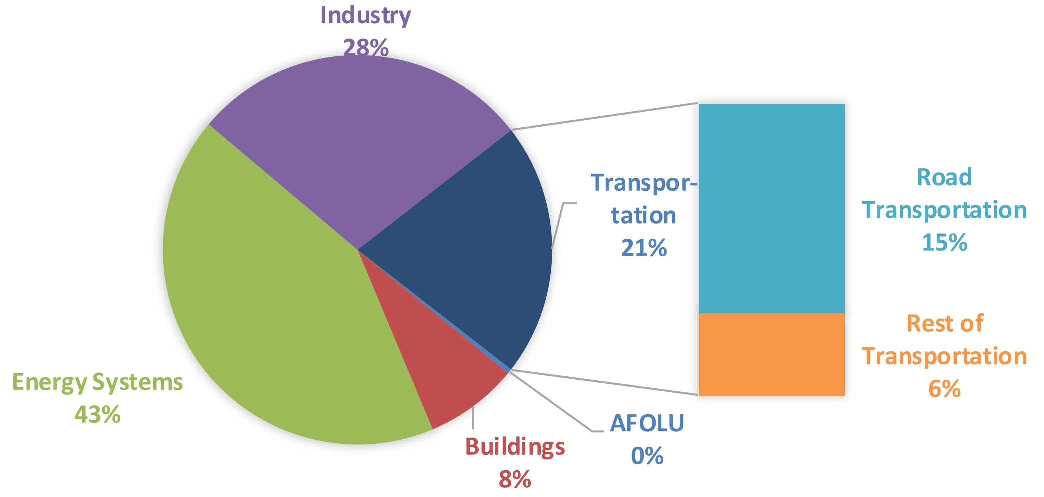
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

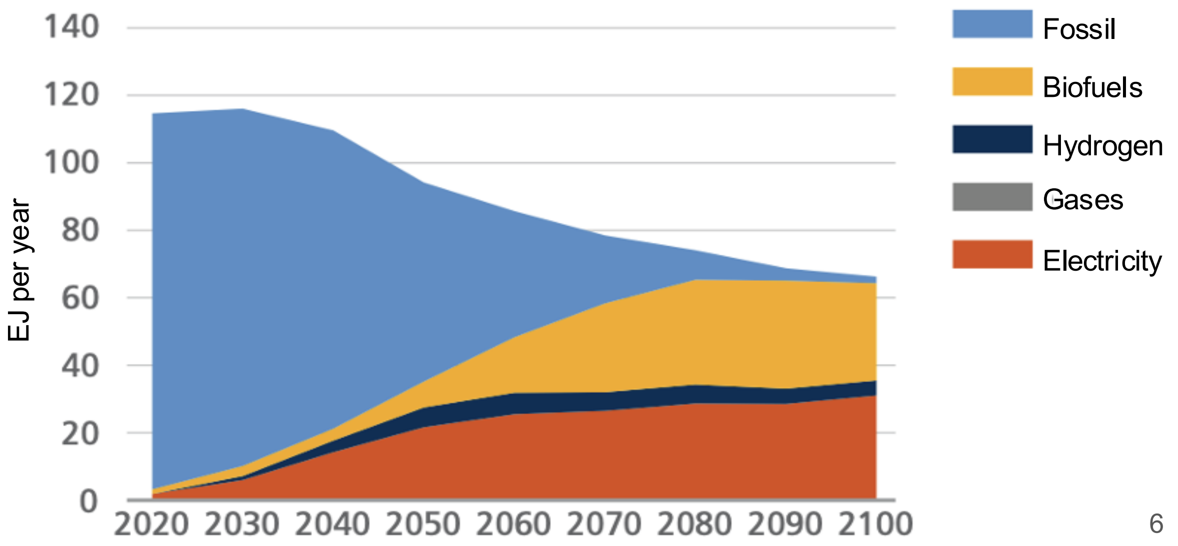
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Case Studies

1. Japan
2. Malaysia
3. South Korea
4. Taiwan
5. Thailand
6. Vietnam

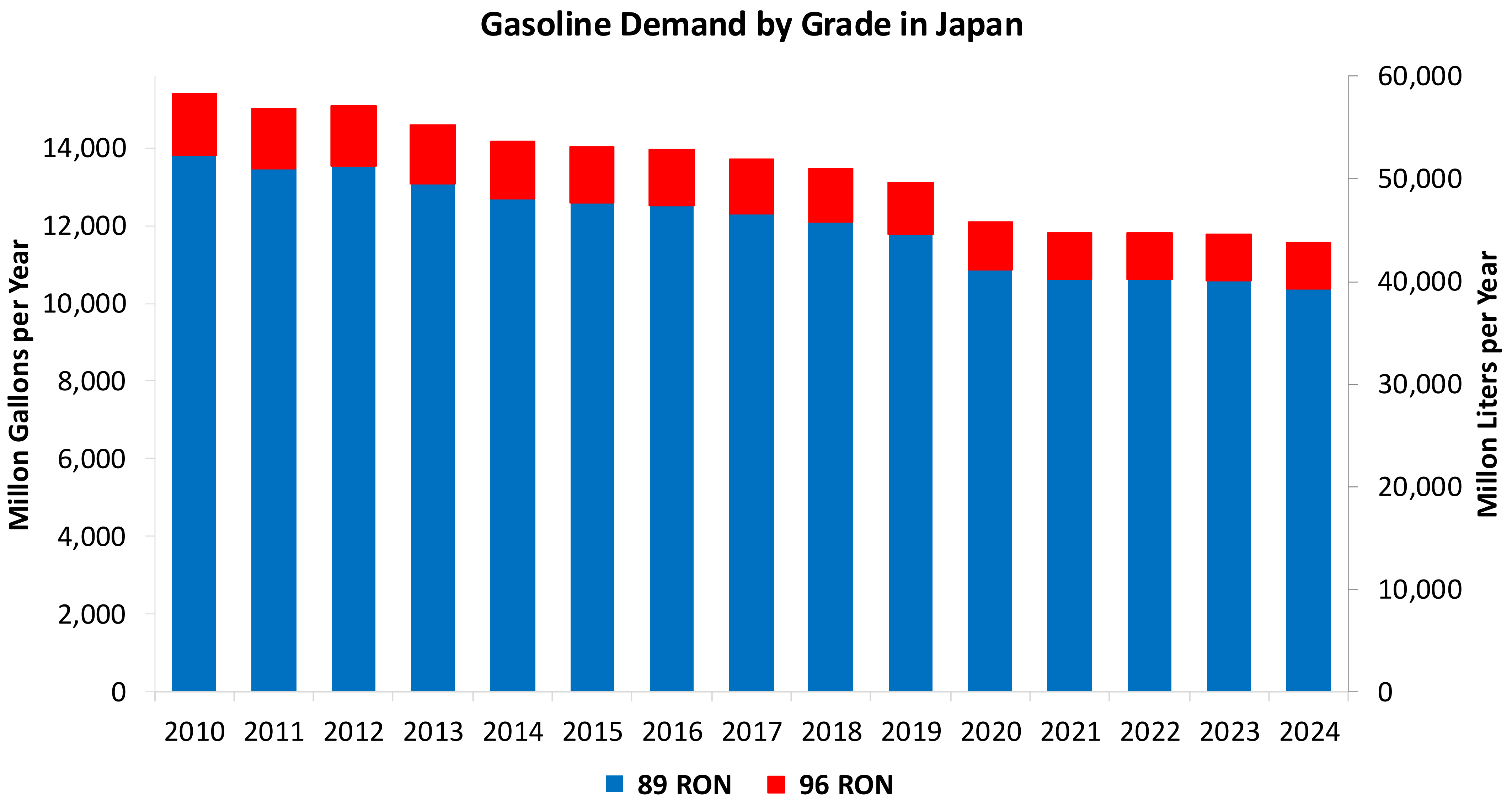
Ethanol Gasoline Blending in Japan

In 2024, gasoline consumption in Japan was 43,826 million liters (11,578 million gallons) and growth rate was -2.5%. Gasoline production and net imports were 42,543 and 1,283 million liters (11,239 and 339 million gallons) correspondingly. Japan complies with the JIS standards and offers two main grades: RON 89 and RON 96) with an average octane of 89.7 RON.

Japan currently allows E3. Last year's consumption was 830 million liters (219 million gallons) equivalent to 1.9%, primarily as the component ethyl tert-butyl ether (ETBE). There is no local production. Ethanol is imported as a blendstock for ETBE production and as ETBE. The Ministry of Economy, Trade and Industry (METI) aiming for E10 (10% ethanol) by 2030 and E20 (20% ethanol) by 2040. This initiative requires infrastructure improvements, as most existing gas stations are only equipped for the E3 blend.

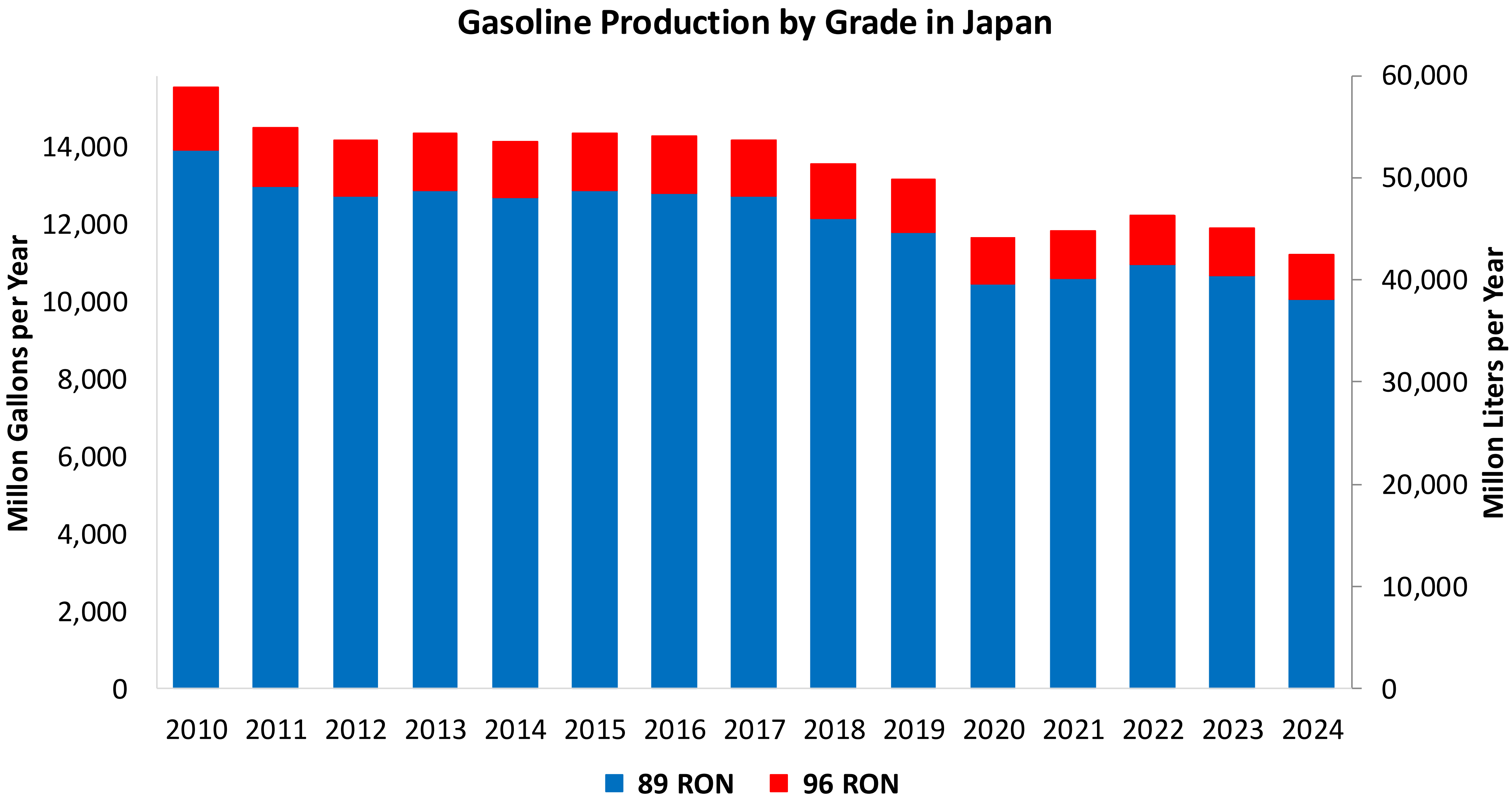
Source: METI, USDA

Gasoline Demand in Japan



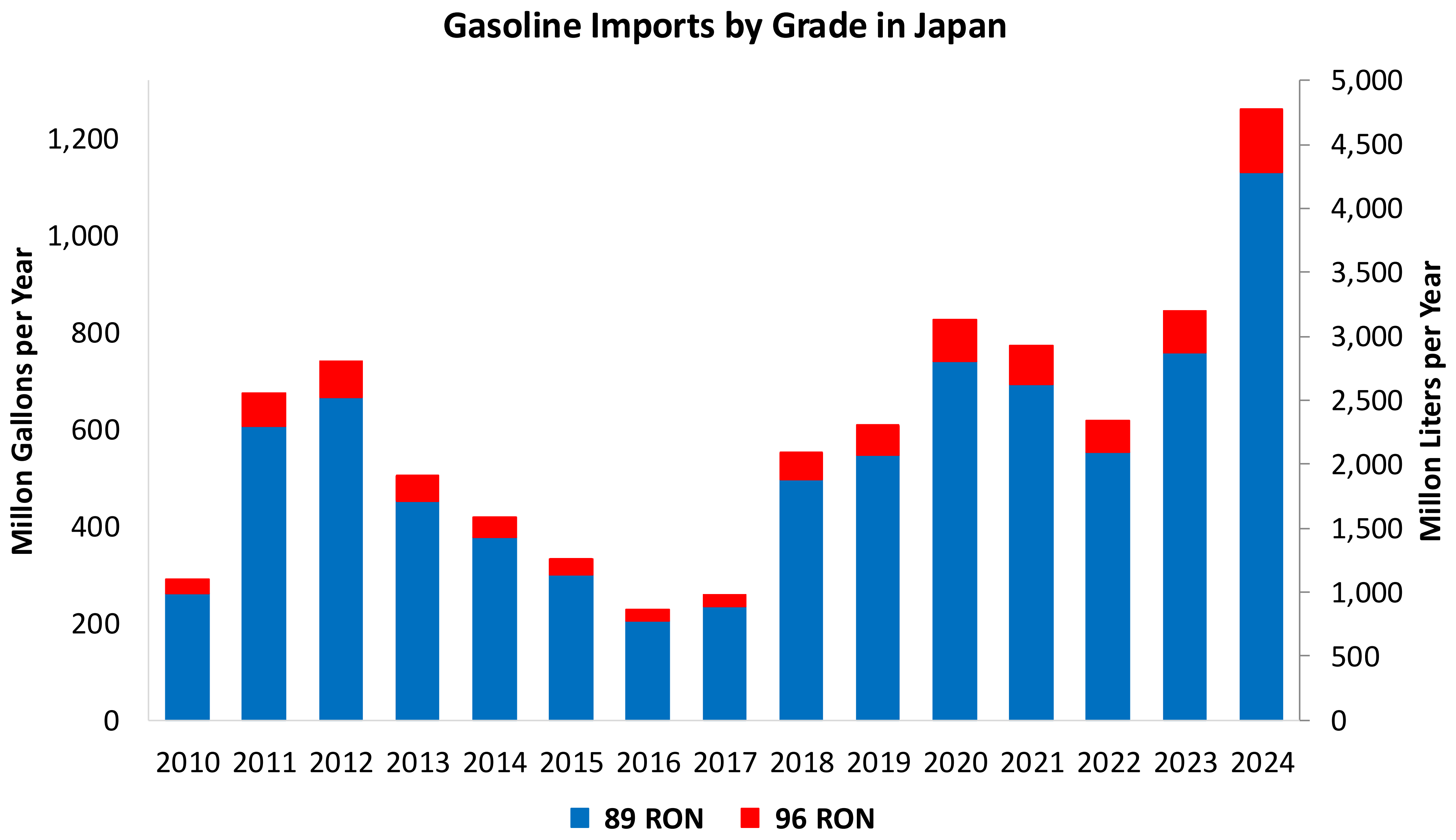
Source: METI

Gasoline Production in Japan



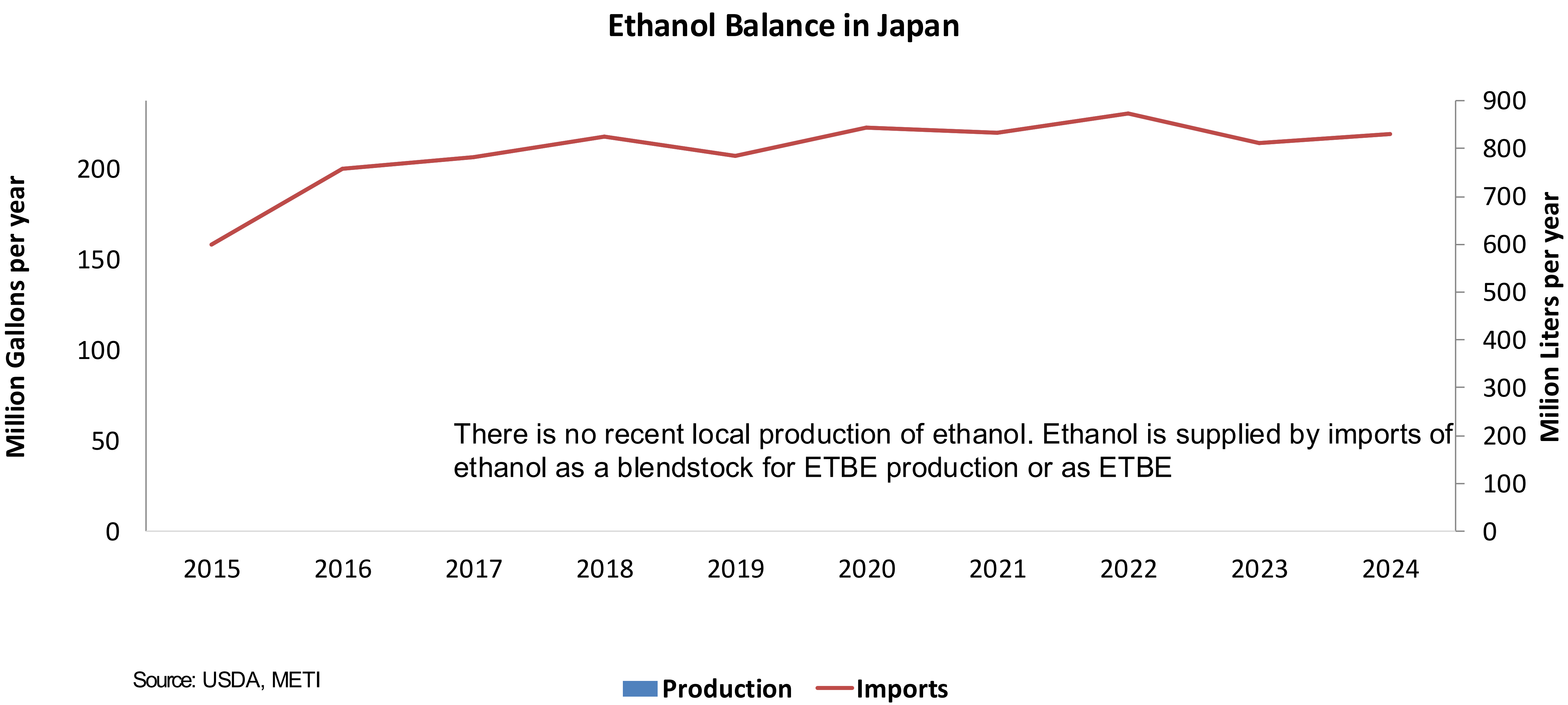
Source: METI

Gasoline Imports in Japan



Source: METI

Ethanol Balance in Japan



Source: USDA, METI

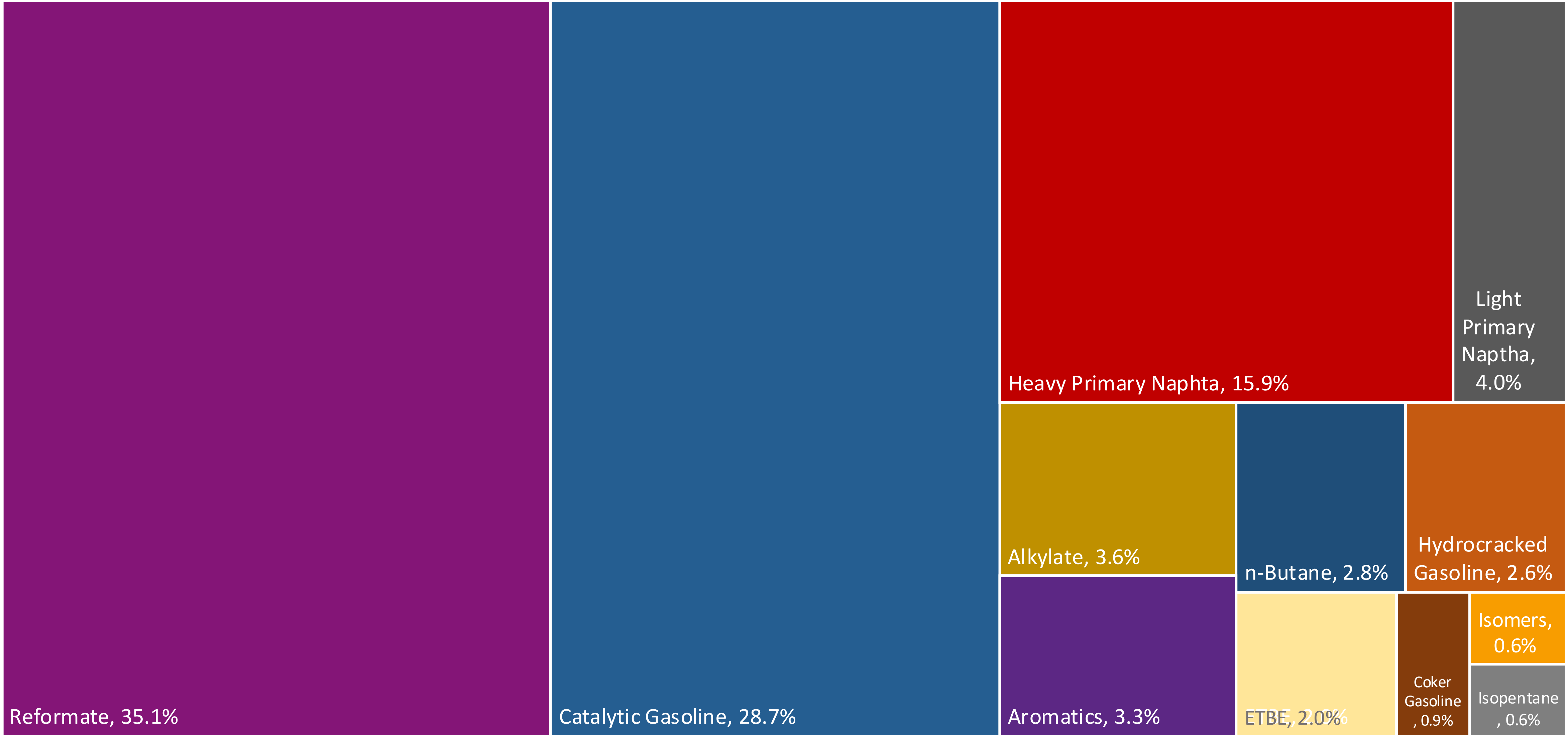
Gasoline Quality in Japan

Name	JIS K 2202: 2023 - Law on the Quality Control of Gasoline and Other Fuels							
Implementation Date	2023							
Applicability	Japan							
Grade	Regular		Premium		E10 Regular *		E10 Premium *	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	89.0		95.0		90.0		96.0	
MON								
AKI								
Benzene (%vol)		1.0		1.0		1.0		1.0
Aromatics (% vol)		-		-		-		-
Olefins (%vol)		-		-		-		-
Lead (g/L)		-		-		-		-
Manganese (mg/L)		-		-		-		-
Sulfur (ppm)		10		10		10		10
Oxygen (%w)		1.3		1.3		3.5		3.5
Ethanol (%vol)		3.0		3.0	3	10.0	3.0	10.0
MTBE (%vol)		7.0		7.0				
PVR 37.8°C (psi)	6.4	11.3	6.4	11.3	6.4	11.3	6.4	11.3
Carbon Intensity (gCOe/MJ)	0	88.74	-	88.74	0	88.74	-	88.74
Kerosene (%vol)	-	4.0	-	4.0	-	4.0	-	4.0

*Not Mandatory. A 55% Carbon Intensity Reduction target has been established and will be revised once LCA has been completed

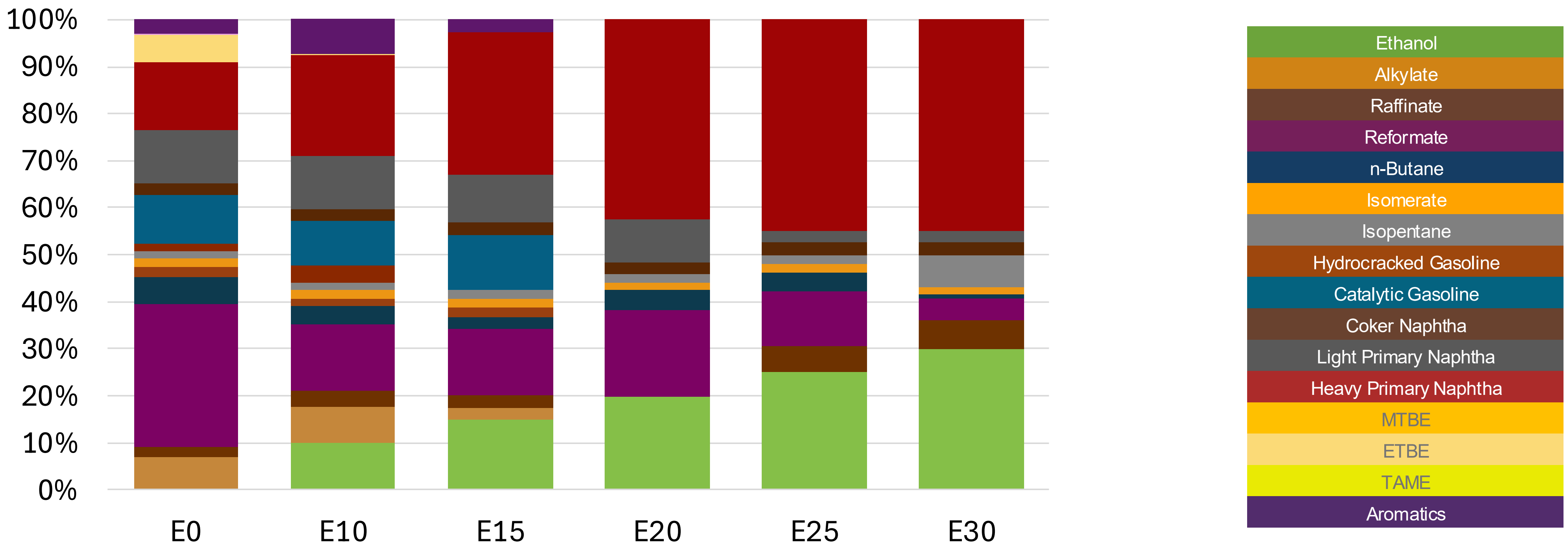
Source: METI

Japan Available Blending Components



Source: Faro90

Japan – Gasoline 1 – Constant Octane

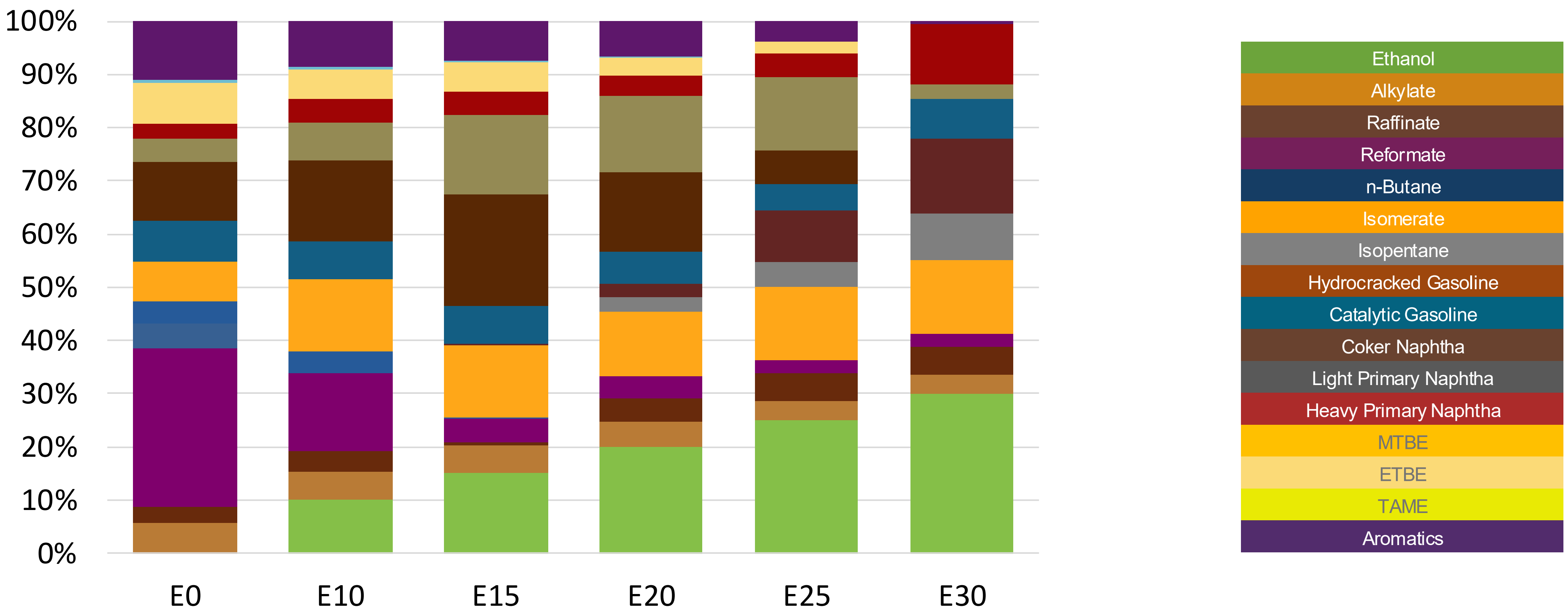


Octane (RON)	89.0	89.0	89.0	89.0	90.0	90.7
Price (US$/gal)	$ 2.223	$ 1.969	$ 1.826	$ 1.678	$ 1.611	$ 1.530

Average CIF 2024 Prices

Source: Faro90

Japan - Gasoline 2 - Constant Octane

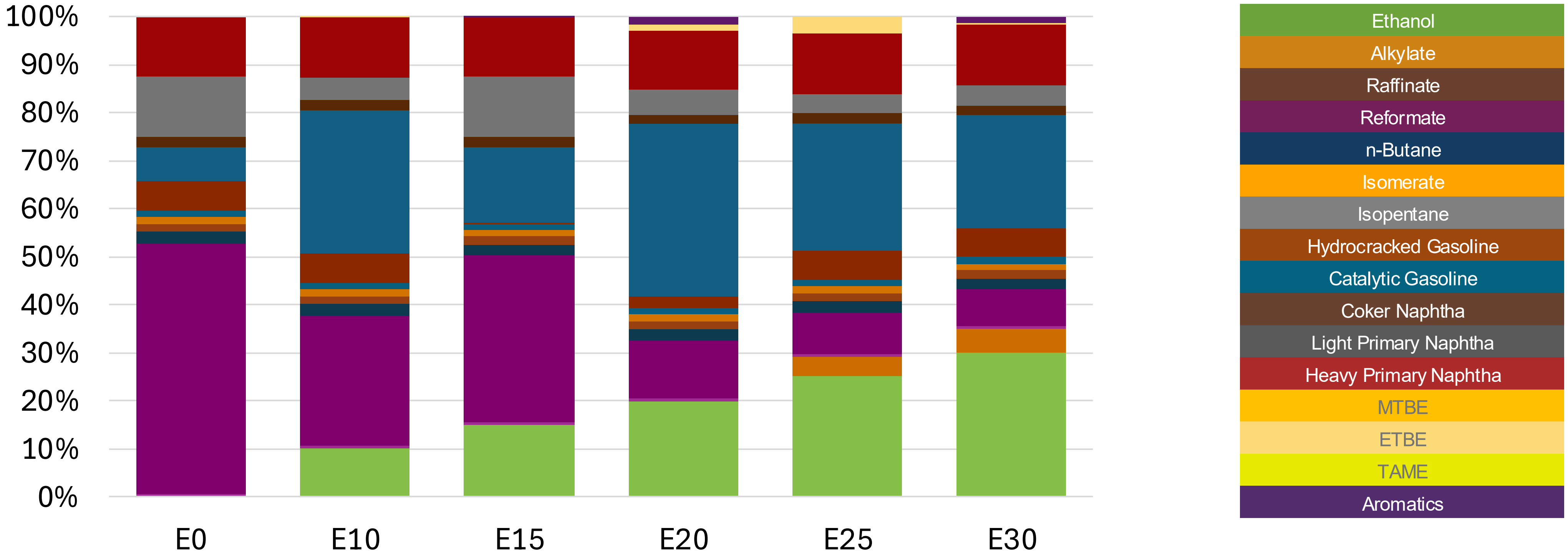


Octane (RON)	96.0	96.0	96.0	96.0	96.0	96.0
Price (US$/gal)	$ 2.421	$ 2.344	$ 2.265	$ 2.180	$ 2.090	$ 1.984

Average CIF 2024 Prices

Source: Faro90

Japan – Gasoline 1 – Increased Octane

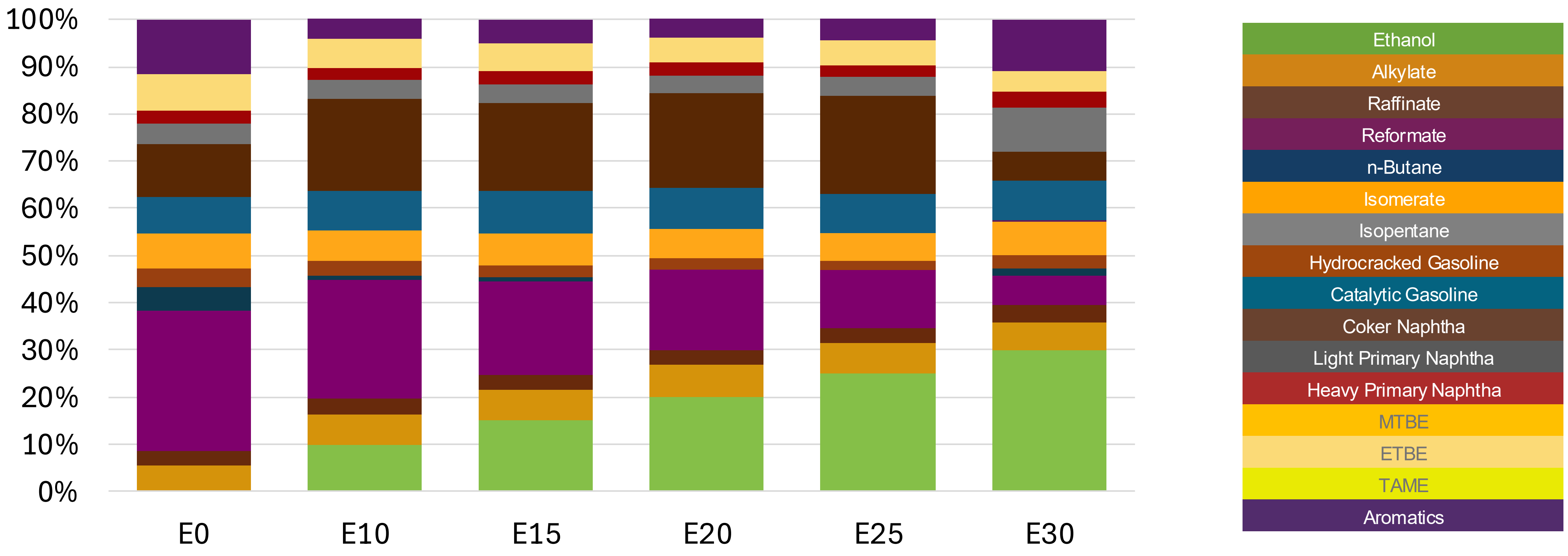


Octane (RON)	89.2	93.3	95.2	97.1	98.8	100.0
Price (US$/gal)	$ 2.223	$ 2.223	$ 2.223	$ 2.223	$ 2.223	$ 2.223

Average CIF 2024 Prices

Source: Faro90

Japan – Gasoline 2 – Increased Octane



Octane (RON)	96.0	97.6	99.1	100.5	102.0	103.2
Price (US$/gal)	$ 2.421	$ 2.421	$ 2.421	$ 2.421	$ 2.421	$ 2.421

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Japan

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	191,360
		Single FI	3-Ways		>8 years	819,341
		Multi FI	3-Ways		>8 years	4,375,203
			Euro III		>8 years	5,105,687
			Euro IV		>8 years	8,486,532
			Euro V		>8 years	2,778,754
			Euro 6		>8 years	6,882,393
					4-8 years	8,213,163
					<4 years	6,372,261
			Hybrid		All	13,710,087
		Electric			All	84,483
	Medium	Carburator	3-Ways	PCV Tank	>8 years	179,914
		Single FI	3-Ways		>8 years	1,038,575
		Multi FI	3-Ways		>8 years	1,069,752
			Euro III		>8 years	1,025,436
			Euro IV		>8 years	2,063,688
			Euro V		>8 years	844,450
			Euro 6		>8 years	1,954,564
					4-8 years	2,176,458
					<4 years	2,037,277
			Hybrid		All	3,929,919
		Electric			All	294,731
Sml Engines	Light	4-Cycle	None	PCV	>8 years	237,867
			Euro III		>8 years	346,124
			Euro IV		>8 years	882,856
			Euro V		>8 years	278,271
			Euro 6		>8 years	702,464
					4-8 years	694,349
					<4 years	921,228
			Hybrid		All	178,979
		Electric			All	189,122
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	54,410
		Fuel-Injection	3-Ways		>8 years	637,153
			Euro III		>8 years	164,112
			Euro IV		>8 years	411,358
			Euro V		>8 years	198,040
			Euro 6		>8 years	863,691
					4-8 years	1,320,080
					<4 years	1,174,293
			Hybrid		All	284,413
		Electric			All	321,973

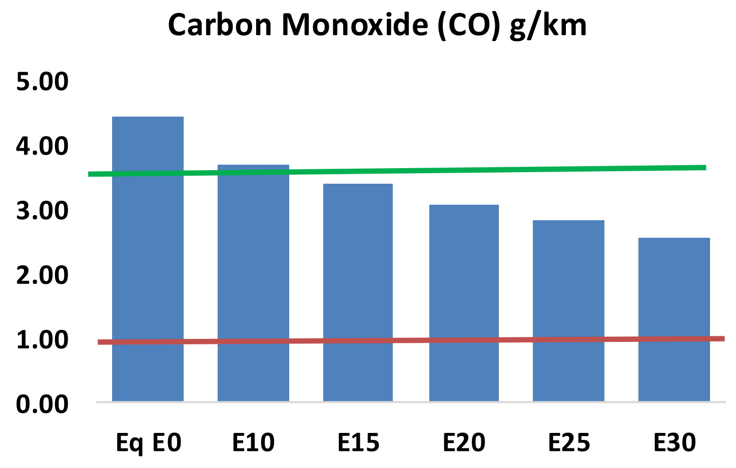
0 2,000,000 4,000,000 6,000,000 8,000,000 10,000,000 12,000,000 14,000,000 16,000,000

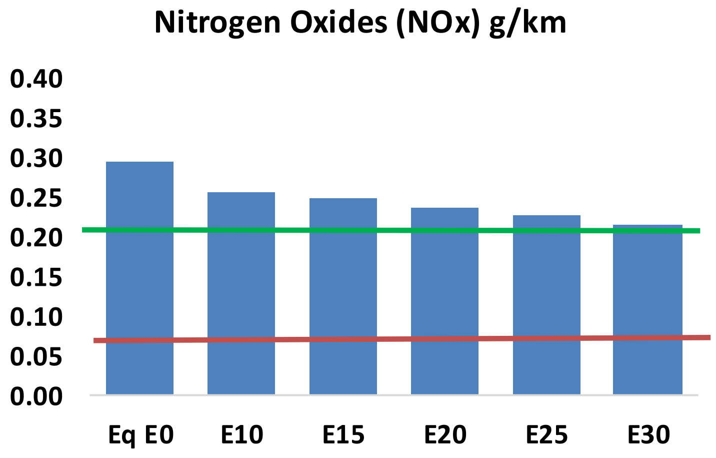
Vehicle Fleet: 83,494,811
Gasoline Vehicle Fleet: 76,775,958

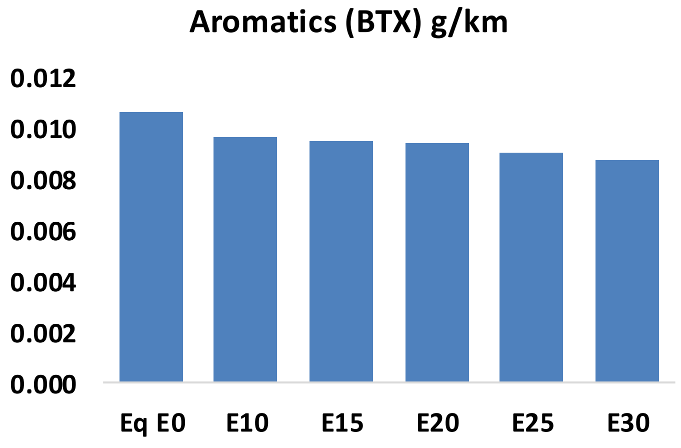
Source: AIRIA

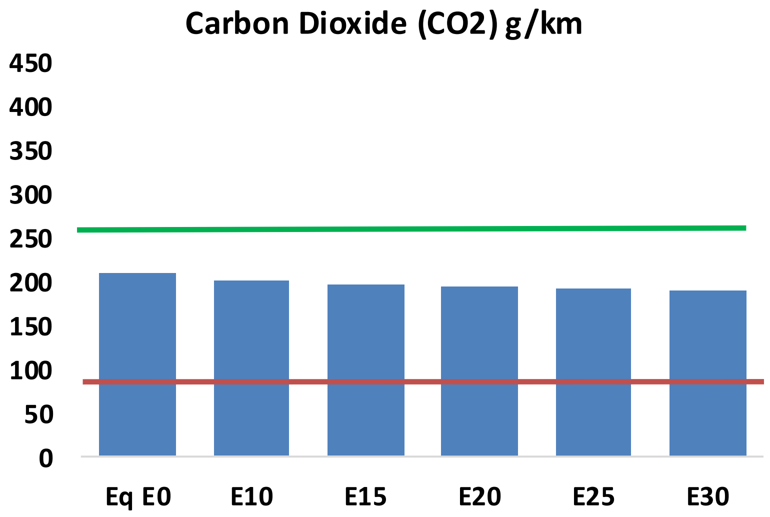
Japan – Vehicular Emissions

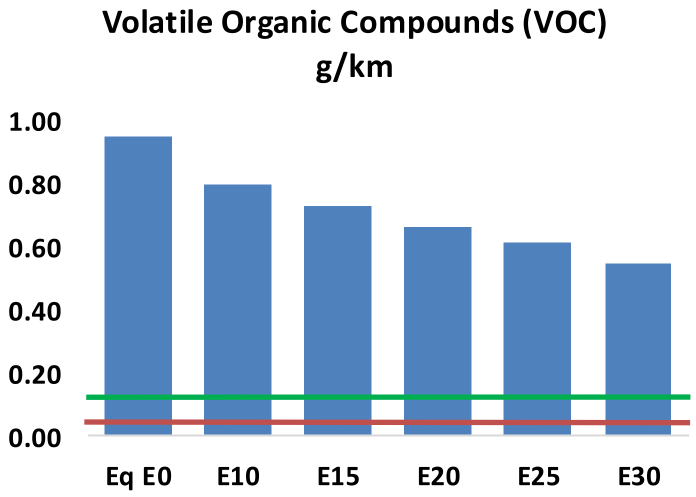
TIER III BIN 125 United States
Euro 6 Standard

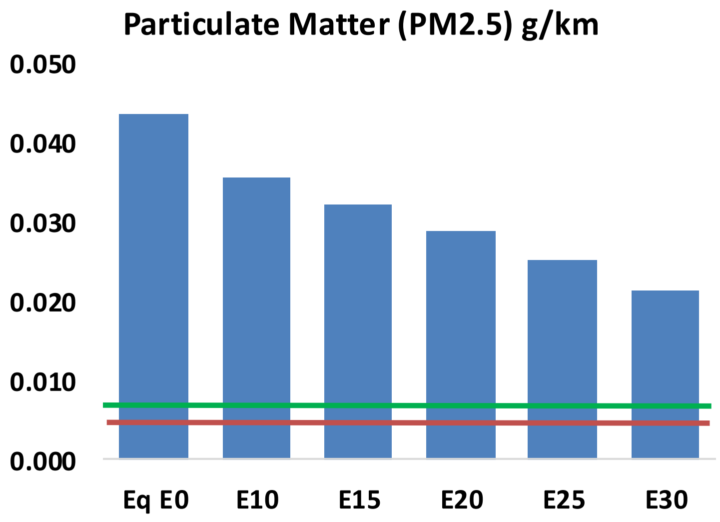












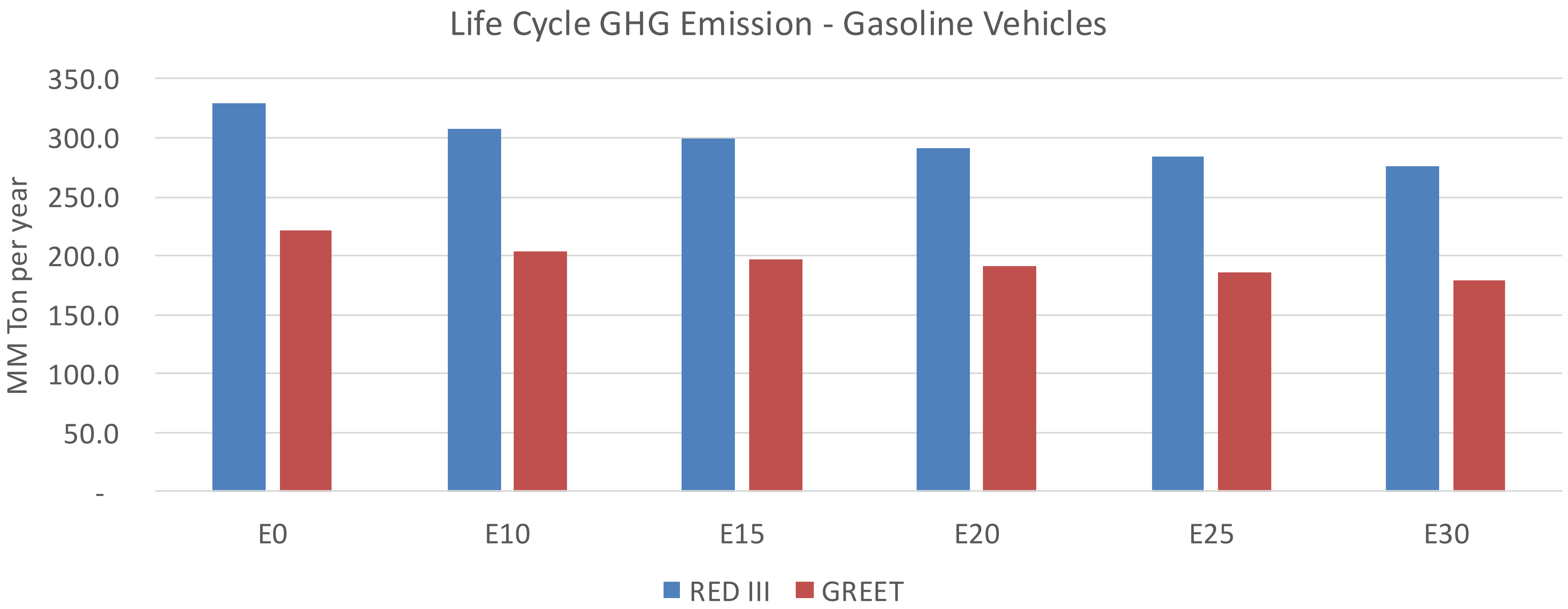
Source: Faro90

Japan – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	7,996,709	7,320,077	6,643,444	6,076,898	5,501,211	5,056,863	4,584,871	-17%	-31%
CO2	375,859,078	366,194,131	357,066,124	349,887,216	346,334,811	344,557,634	338,206,446	-5%	-8%
VOC	1,702,687	1,563,867	1,425,046	1,309,743	1,192,997	1,103,012	983,689	-16%	-30%
NOx	529,045	488,485	460,752	446,675	423,941	406,147	385,180	-13%	-20%
SOx	4,260	3,938	3,616	3,343	3,082	2,799	2,501	-15%	-28%
NH3	146,209	144,761	143,314	143,994	145,614	146,753	147,700	-2%	0%
N2O	7,310	7,270	7,241	7,276	7,426	7,509	7,594	-1%	2%
CH4	345,533	345,533	345,533	350,716	358,317	364,882	371,102	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	6,553	6,230	5,906	5,675	5,460	5,299	5,067	-10%	-17%
Acetaldehyde	23,643	27,135	39,815	60,634	82,607	94,394	111,537	68%	249%
Formaldehyde	91,090	88,560	103,236	121,221	126,997	140,493	152,942	13%	39%
Benzene	19,076	18,332	17,370	17,095	16,953	16,214	15,690	-9%	-11%
PM2.5	22,257	20,234	18,211	16,430	14,688	12,820	10,866	-18%	-34%
PM10	55,998	50,908	45,817	41,337	36,954	32,255	27,337	-18%	-34%

Source: Faro90

Japan – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/year)	1,102.2
Target @ 2035 (MMT/year)	570.4
Delta	531.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.3%	11.2%	13.9%	16.1%	19.2%
RED II/III	0.0%	6.8%	9.3%	11.6%	13.6%	16.2%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.5%	4.7%	5.8%	6.7%	8.0%
RED II/III	0.0%	4.2%	5.7%	7.2%	8.4%	10.1%

Source: Greet, RED II/III, climateactiontracker.org, F90

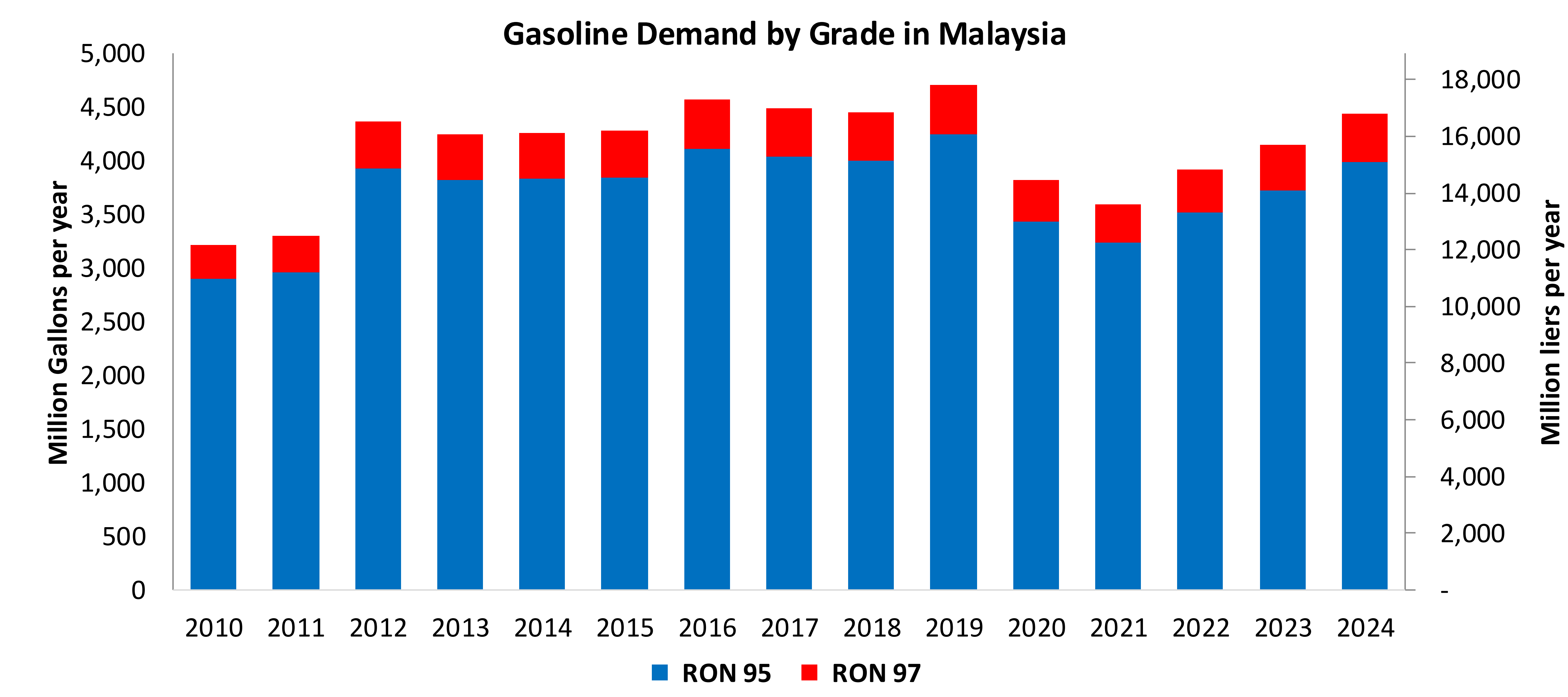
Ethanol Gasoline Blending in Malaysia

In 2024, gasoline consumption in Malaysia was 16,784 million liters (4,434 million gallons) and growth rate was -1.2%. Gasoline production and net imports were 5,436 and 11,348 million liters (1,436 and 2,998 million gallons) correspondingly. Malaysia offers two main grades: RON 95 and RON 97) with an average octane of 95.2 RON.

Government support for biofuels has so far targeted biodiesel. The Biofuel Industry Act of 2007 excluded ethanol as the source of alternative fuels under the National Biofuels Policy of 2006. As there is no policy support for ethanol, there is no production of fuel ethanol and minimal consumption. In 2024, ethanol consumption in Malaysia was 16 million liters (4 million gallons) representing 0.1% of gasoline demand.

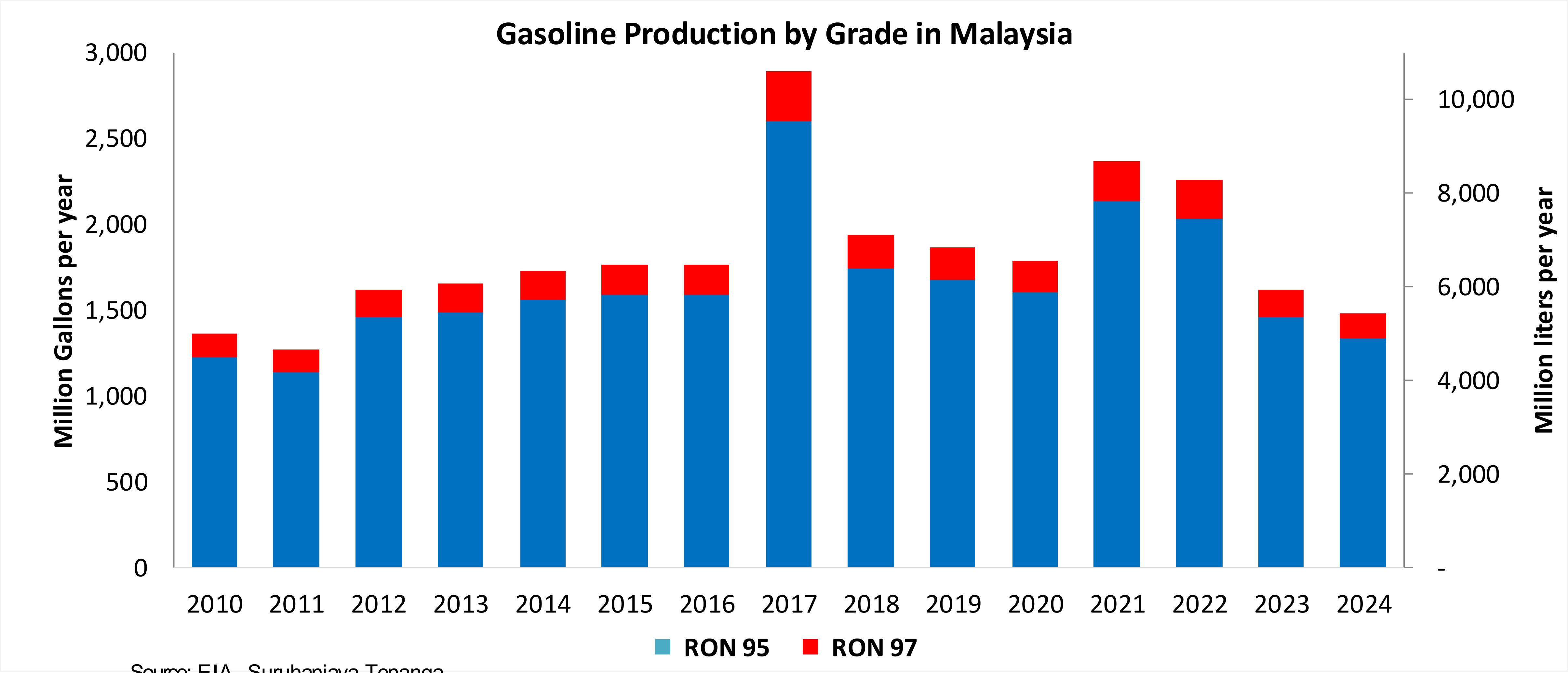
Source: EIA, Suruhanjaya Tenanga, USDA, USGrains

Gasoline Demand in Malaysia



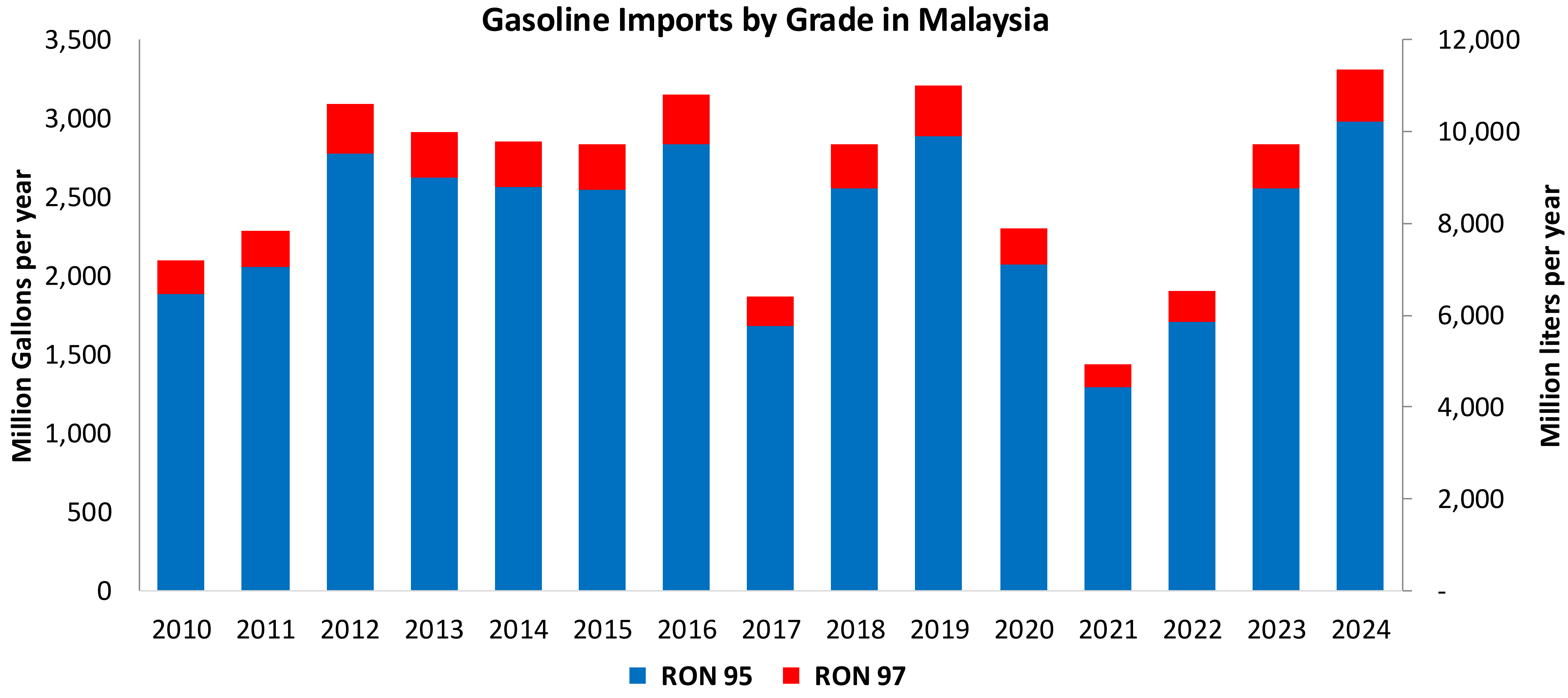
Source: EIA, Suruhanjaya Tenanga

Gasoline Production in Malaysia



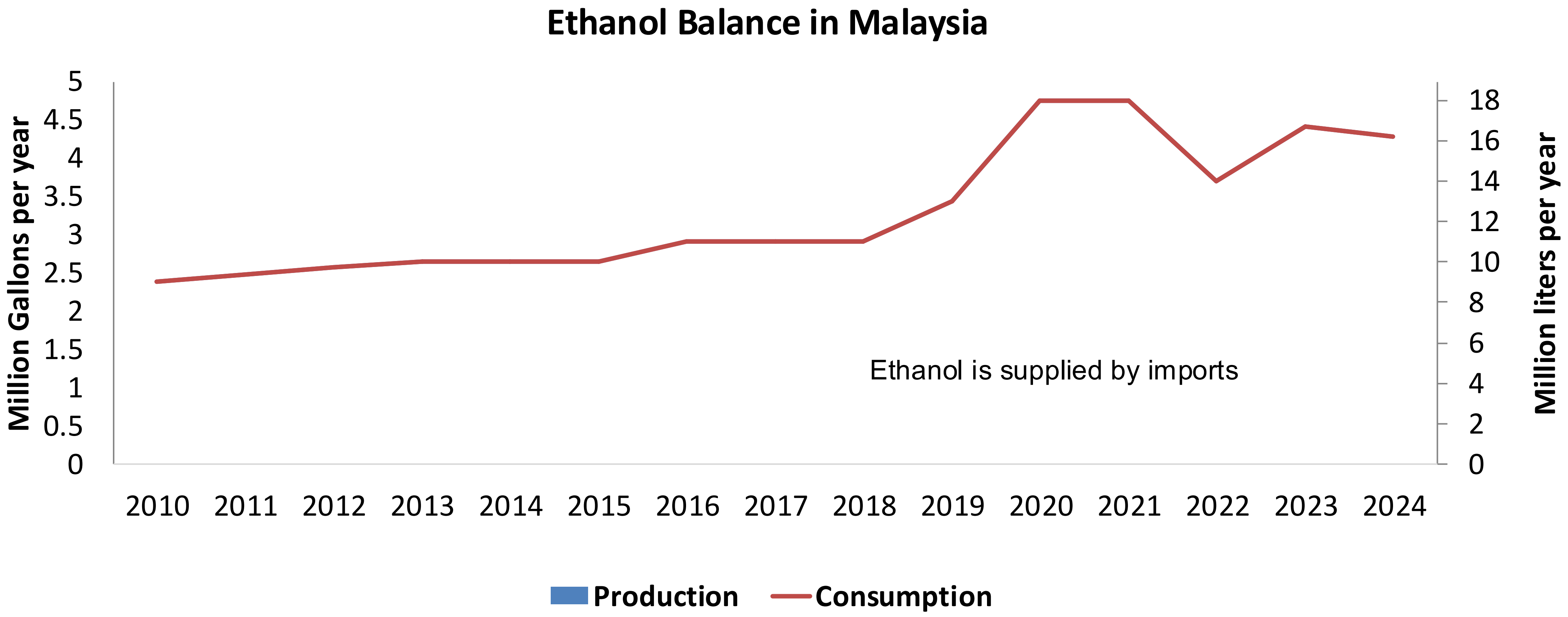
Source: EIA, Suruhanjaya Tenanga

Gasoline Net Imports in Malaysia



Source: EIA, Suruhanjaya Tenanga

Ethanol Balance in Malaysia



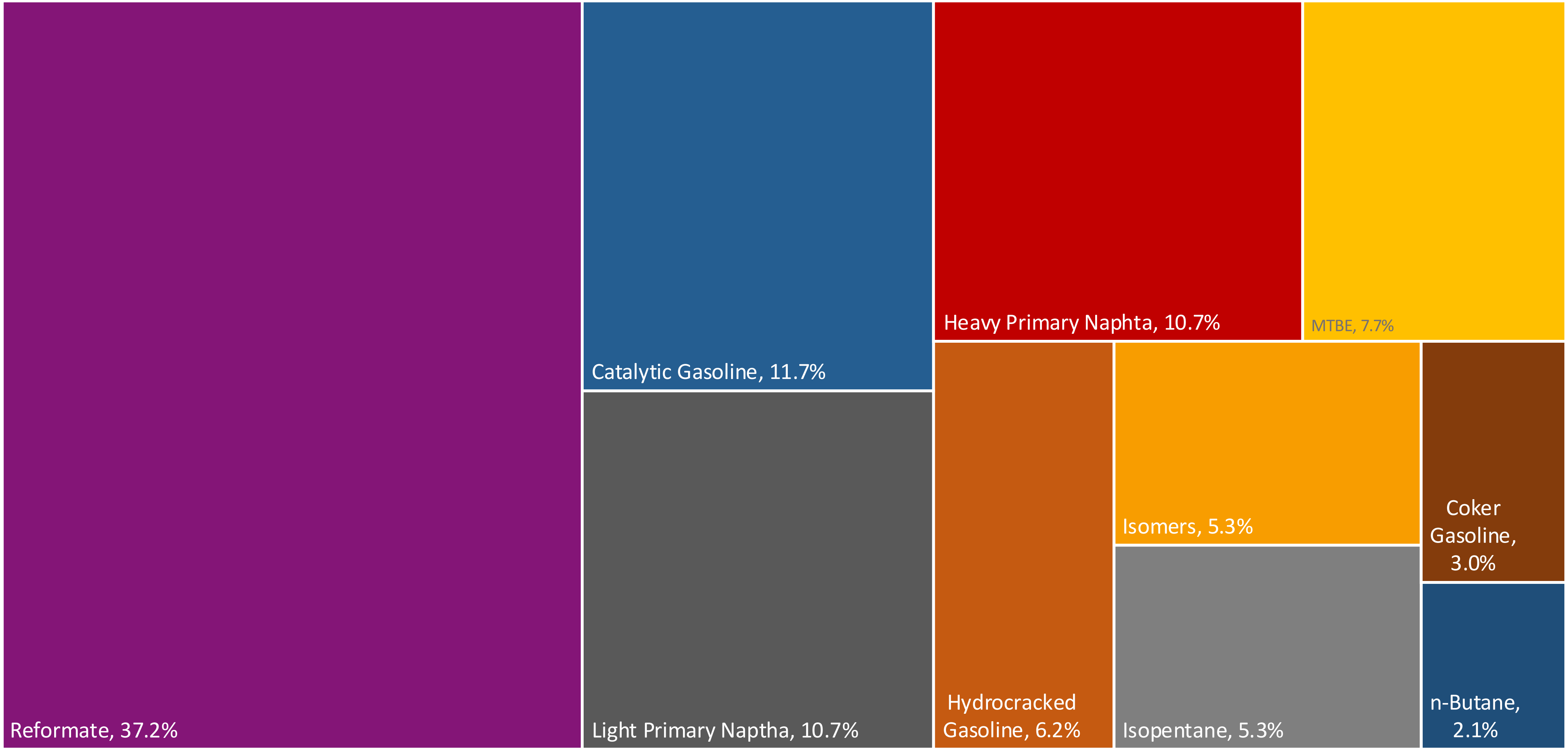
Source: USDA, USGrains

Gasoline Quality in Malaysia

Name	MS 118-3:2011			
Implementation Year	2011			
Applicability	Nationwide			
Grade	RON 95		RON 97	
	Min	Max	Min	Max
RON	95.0		97.0	
MON				
AKI				
Benzene (%vol)		1.0		1.0
Aromatics (% vol)		35.0		35.0
Olefins (%vol)		18.0		18.0
Lead (g/L)		0.013		0.013
Manganese (mg/L)				
Sulfur (ppm)		50		50
Oxygen (%v)				
Ethanol (%vol)				
MTBE (%vol)				
RVP (psi)	60.0	70.0	60.0	70.0

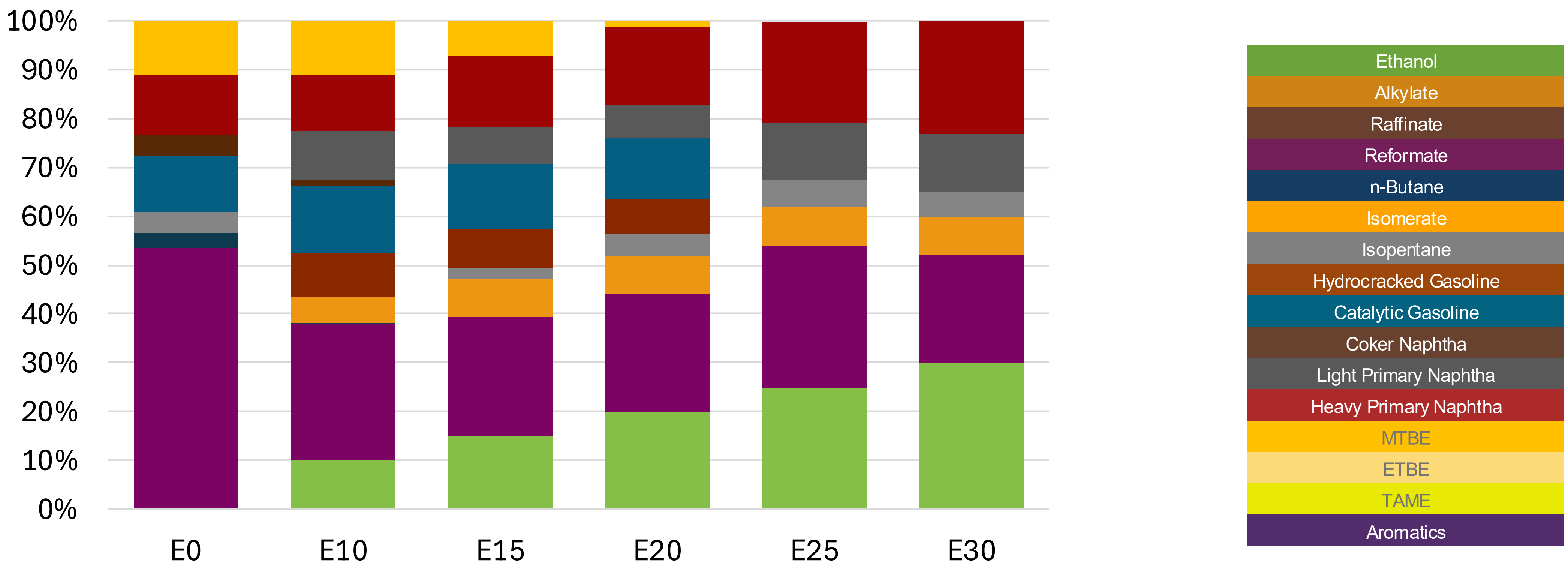
Source: Department of Standards Malaysia, Euro 4 Spec

Malaysia Available Blending Components



Source: Faro90

Malaysia – Gasoline 95 – Constant Octane

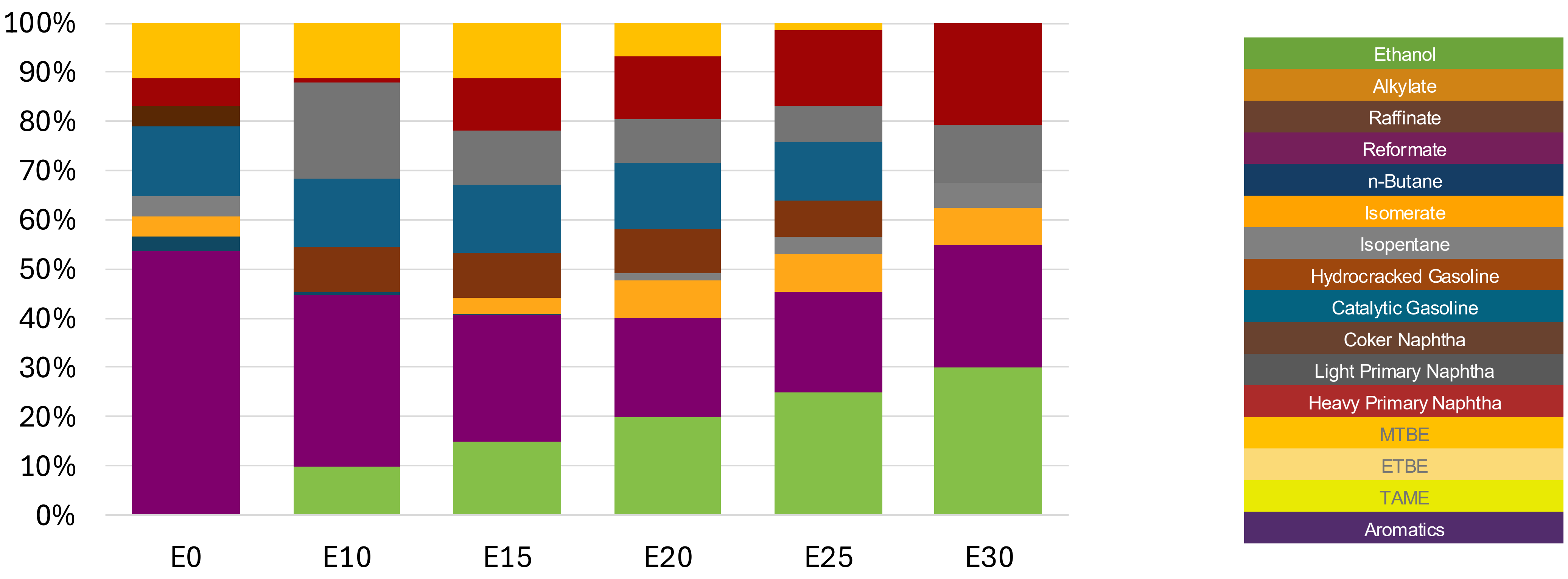


Octane (RON)	95.0	95.0	95.0	95.0	95.6	96.6
Price (US$/gal)	$ 2.360	$ 2.232	$ 2.164	$ 2.102	$ 2.025	$ 1.975

Average CIF 2024 Prices

Source: Faro90

Malaysia - Gasoline 97 - Constant Octane

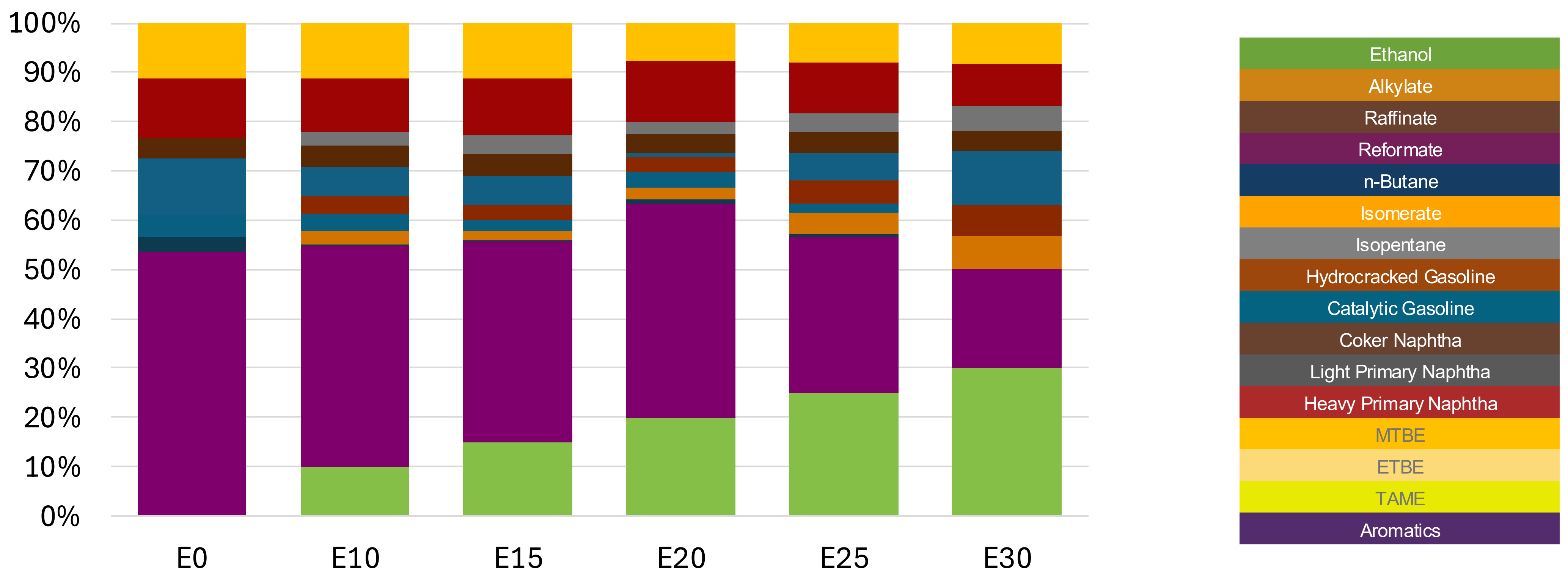


Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.5
Price (US$/gal)	$ 2.466	$ 2.345	$ 2.249	$ 2.187	$ 2.116	$ 2.033

Average CIF 2024 Prices

Source: Faro90

Malaysia – Gasoline 95 – Increased Octane

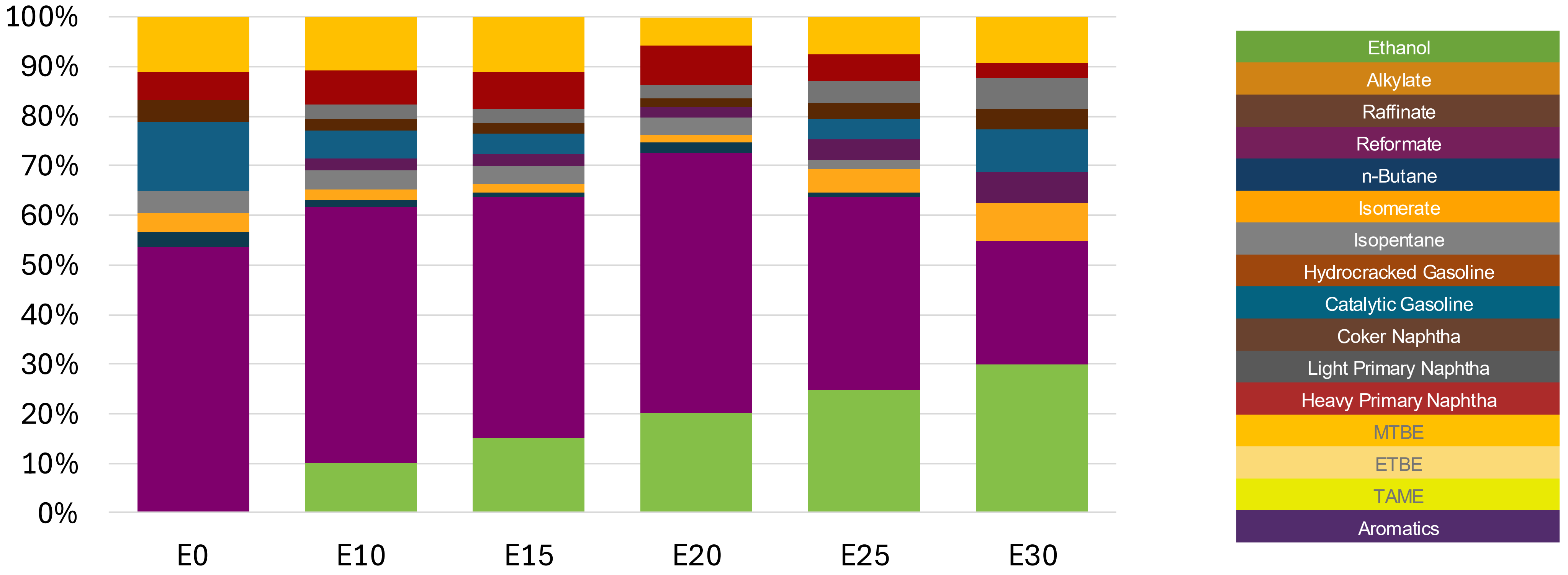


Octane (RON)	95.0	97.8	99.4	100.5	101.2	103.2
Price (US$/gal)	$ 2.360	$ 2.360	$ 2.360	$ 2.360	$ 2.360	$ 2.360

Average CIF 2024 Prices

Source: Faro90

Malaysia – Gasoline 97 – Increased Octane



Octane (RON)	97.0	99.8	101.3	102.1	102.9	104.9
Price (US$/gal)	$ 2.466	$ 2.466	$ 2.466	$ 2.466	$ 2.466	$ 2.466

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Malaysia

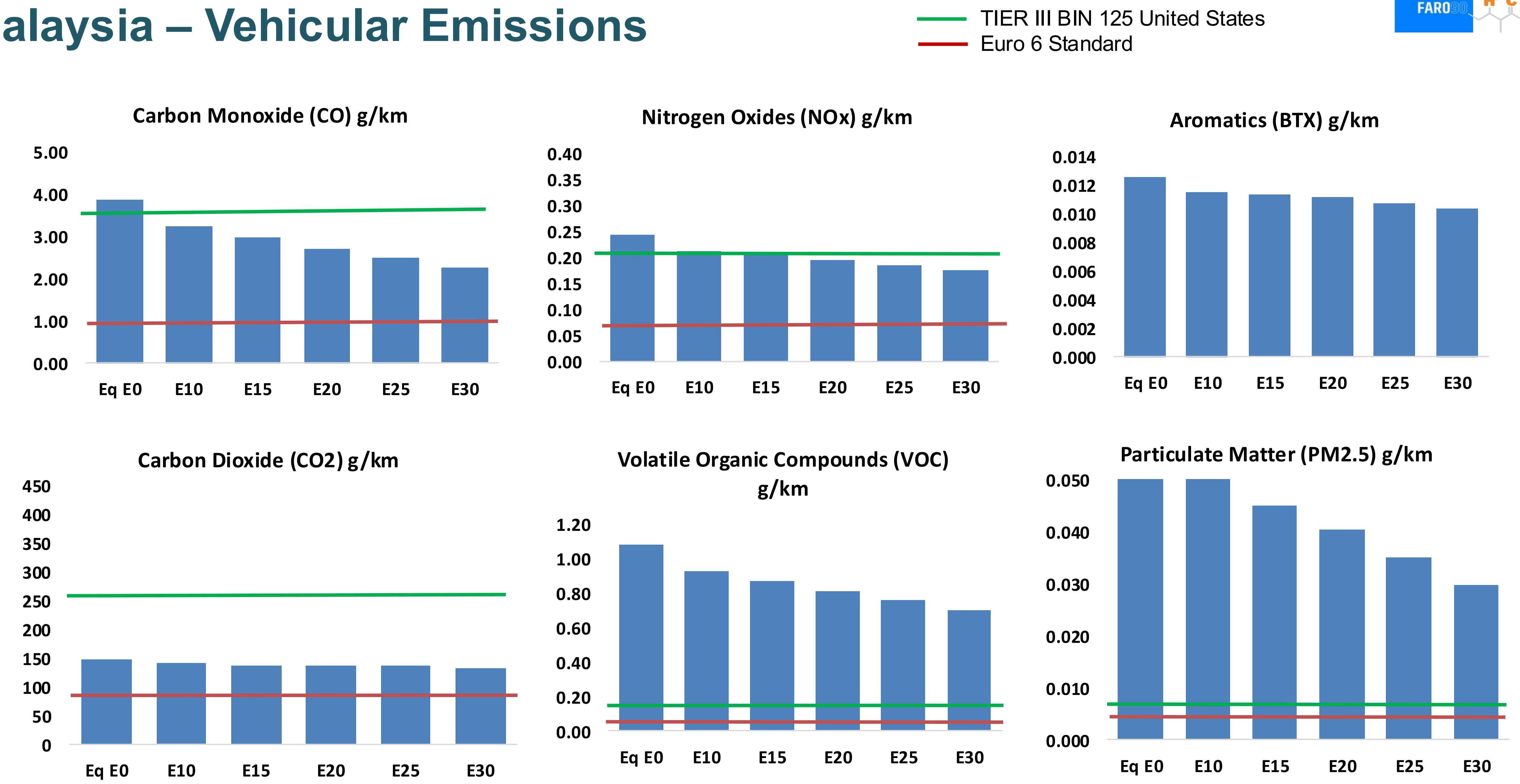
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	1,429
		Single FI	3-Ways		>8 years	62,852
		Multi FI	3-Ways		>8 years	419,881
			Euro III		>8 years	785,206
			Euro IV		>8 years	1,552,675
			Euro V		>8 years	725,323
			Euro 6		>8 years	1,526,338
					4-8 years	1,498,043
					<4 years	2,065,786
			Hybrid		All	105,257
		Electric			All	44,757
	Medium	Carburator	3-Ways	PCV Tank	>8 years	773
		Single FI	3-Ways		>8 years	34,058
		Multi FI	3-Ways		>8 years	228,274
			Euro III		>8 years	428,596
			Euro IV		>8 years	849,702
			Euro V		>8 years	398,170
			Euro 6		>8 years	837,240
					4-8 years	820,478
					<4 years	1,128,102
			Hybrid		All	56,677
		Electric			All	24,100
Sml Engines	Light	4-Cycle	None	PCV	>8 years	460,497
			Euro III		>8 years	847,320
			Euro IV		>8 years	2,172,689
			Euro V		>8 years	1,037,351
			Euro 6		>8 years	1,834,339
					4-8 years	2,339,534
					<4 years	2,655,143
			Hybrid		All	379
		Electric			All	16,529
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	217
		Fuel-Injection	3-Ways		>8 years	59,353
			Euro III		>8 years	96,760
			Euro IV		>8 years	233,350
			Euro V		>8 years	100,773
			Euro 6		>8 years	184,154
					4-8 years	176,765
					<4 years	201,346
			Hybrid		All	0
		Electric			All	9,302

0 500,000 1,000,000 1,500,000 2,000,000 2,500,000 3,000,000

Vehicle Fleet: 26,019,519
Gasoline Vehicle Fleet: 22,285,447

Source: Data Gov Malaysia

Malaysia – Vehicular Emissions



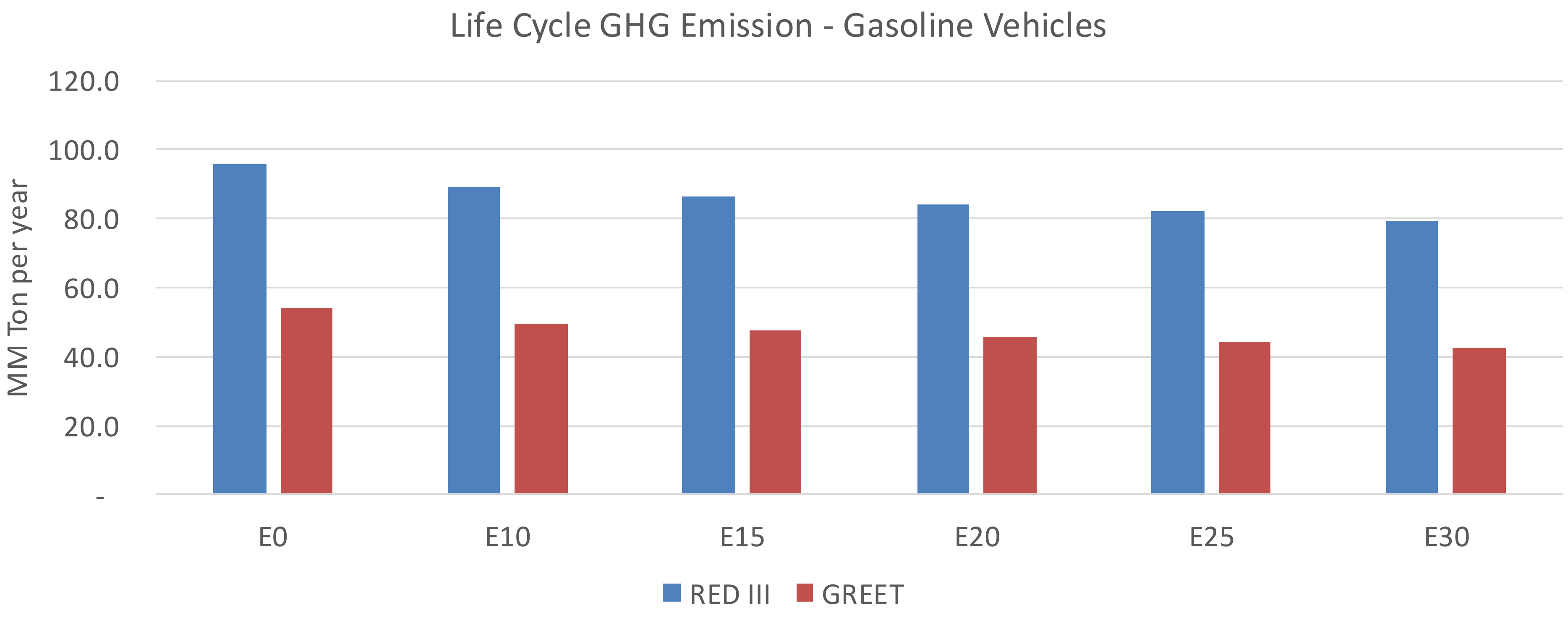
Source: Faro90

Malaysia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,122,769	1,946,988	1,771,207	1,624,705	1,476,146	1,361,571	1,232,569	-17%	-30%
CO2	80,754,602	78,678,055	76,716,872	75,174,459	74,411,212	74,065,375	72,700,137	-5%	-8%
VOC	588,196	547,157	506,117	473,302	440,654	415,663	381,915	-14%	-25%
NOx	133,247	123,031	115,972	112,349	106,542	101,977	96,610	-13%	-20%
SOx	926	856	786	727	670	608	544	-15%	-28%
NH3	35,790	35,436	35,081	35,248	35,644	35,923	36,155	-2%	0%
N2O	1,037	1,032	1,027	1,032	1,054	1,065	1,078	-1%	2%
CH4	130,499	130,499	130,499	132,457	135,328	137,807	140,156	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,748	2,600	2,452	2,343	2,239	2,161	2,050	-11%	-19%
Acetaldehyde	10,170	11,672	17,126	26,080	35,532	40,602	47,975	68%	249%
Formaldehyde	42,474	41,294	48,138	56,524	59,217	65,510	71,315	13%	39%
Benzene	6,902	6,633	6,285	6,185	6,134	5,866	5,677	-9%	-11%
PM2.5	5,392	4,902	4,412	3,980	3,558	3,106	2,632	-18%	-34%
PM10	27,999	25,454	22,908	20,669	18,477	16,128	13,669	-18%	-34%

Source: Faro90

Malaysia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	407.7
Target @ 2035 (MMT/year)	285.4
Delta	122.3

vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.1%	12.4%	15.6%	18.4%	22.1%
RED II/III	0.0%	6.8%	9.5%	12.0%	14.3%	17.1%

Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.1%	5.5%	6.9%	8.2%	9.8%
RED II/III	0.0%	5.3%	7.4%	9.4%	11.2%	13.4%

Source: Greet, RED II/III, climateactiontracker.org, F90

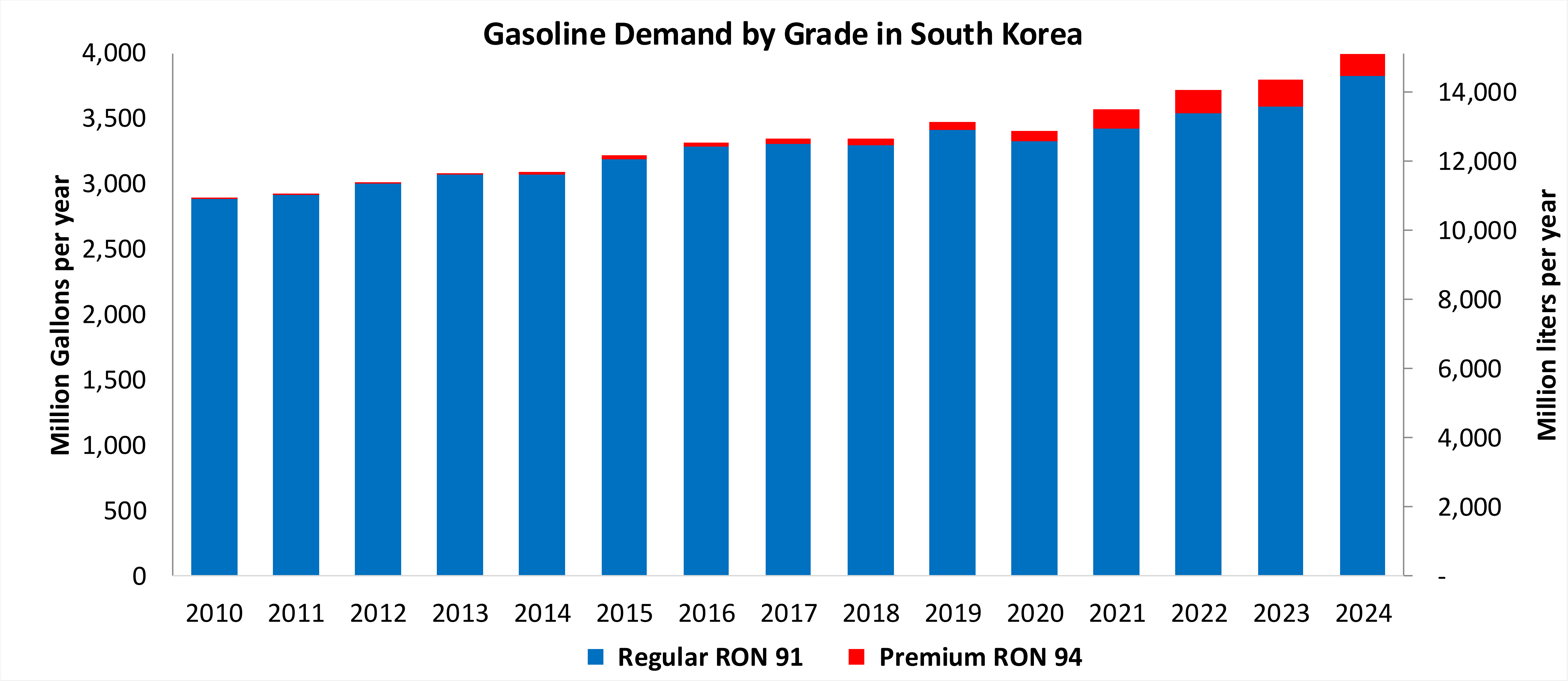
Ethanol Gasoline Blending in South Korea

In 2024, gasoline consumption in South Korea was 15,109 million liters (3,991 million gallons) and growth rate was 2.8%. Gasoline production and net imports were 32,661 and -17,552 million liters (8,628 and -4,637 million gallons) correspondingly. South Korea offers two main grades: RON 91 and RON 94) with an average octane of 91.1 RON.

South Korea has implemented requirements for renewable fuels that are currently met with biodiesel blending. Altough there is an oxygen requirement, ethanol is not currently blended in gasoline. Efforts to implement pilot programs have been stalled because of a lack of consensus and reluctance of different actors in the petroleum industry.

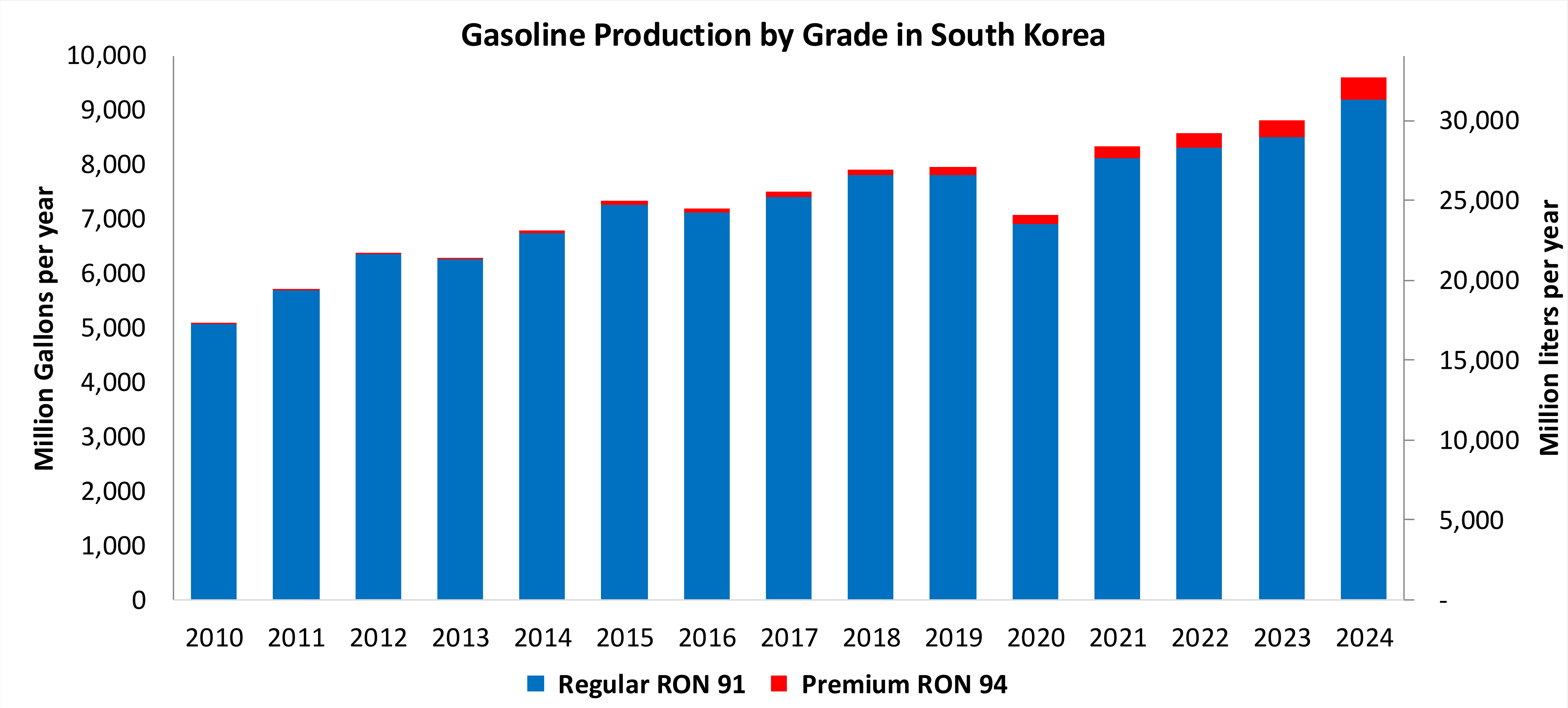
Source: Korean Energy Economics Institute, IEA

Gasoline Demand in South Korea



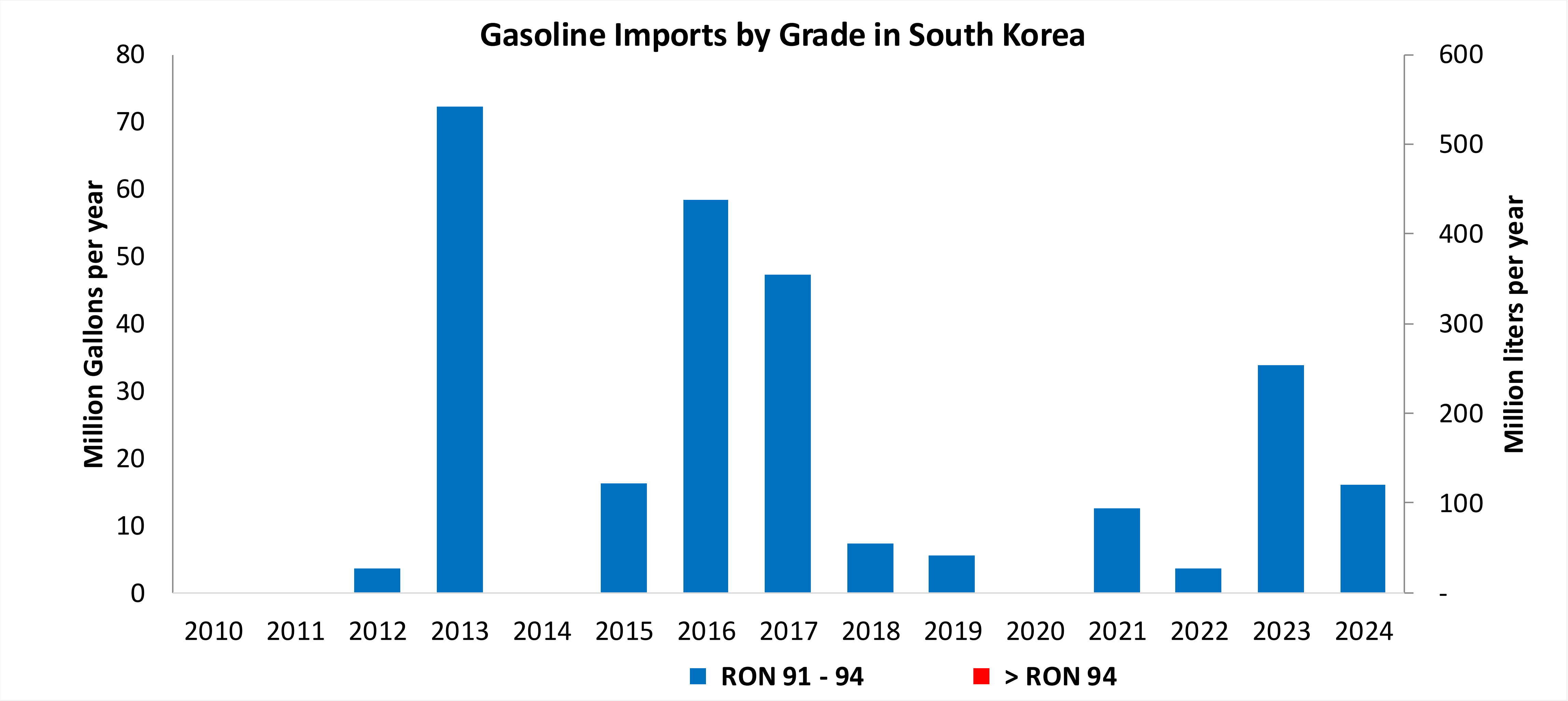
Source: Korean Energy Economics Institute

Gasoline Production in South Korea



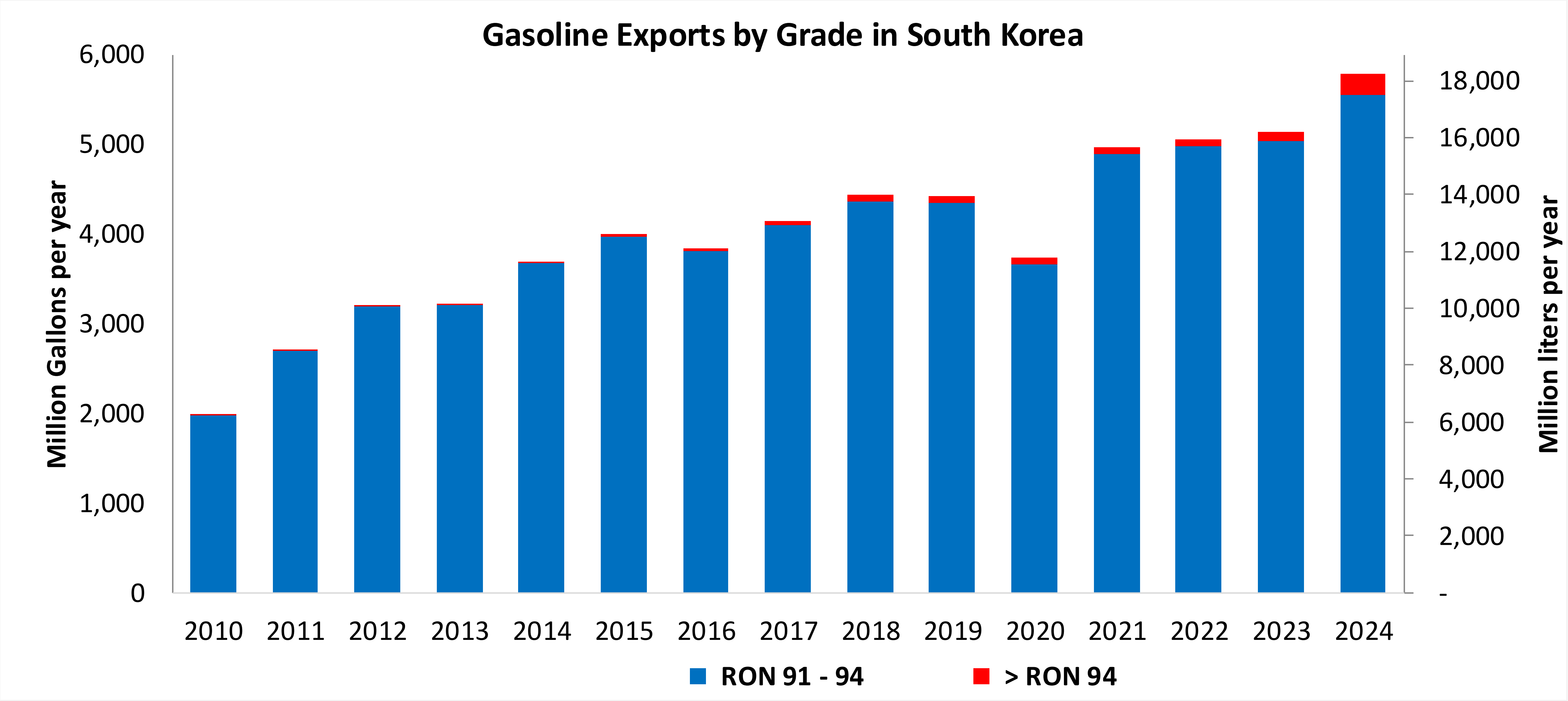
Source: Korean Energy Economics Institute

Gasoline Imports in South Korea



Source: Korean Energy Economics Institute

Gasoline Exports in South Korea



Source: Korean Energy Economics Institute

Ethanol Balance in South Korea

South Korea does not produce or consumes bioethanol in a commercial level for gasoline blending. It imports large quantities of bioethanol for industrial uses.

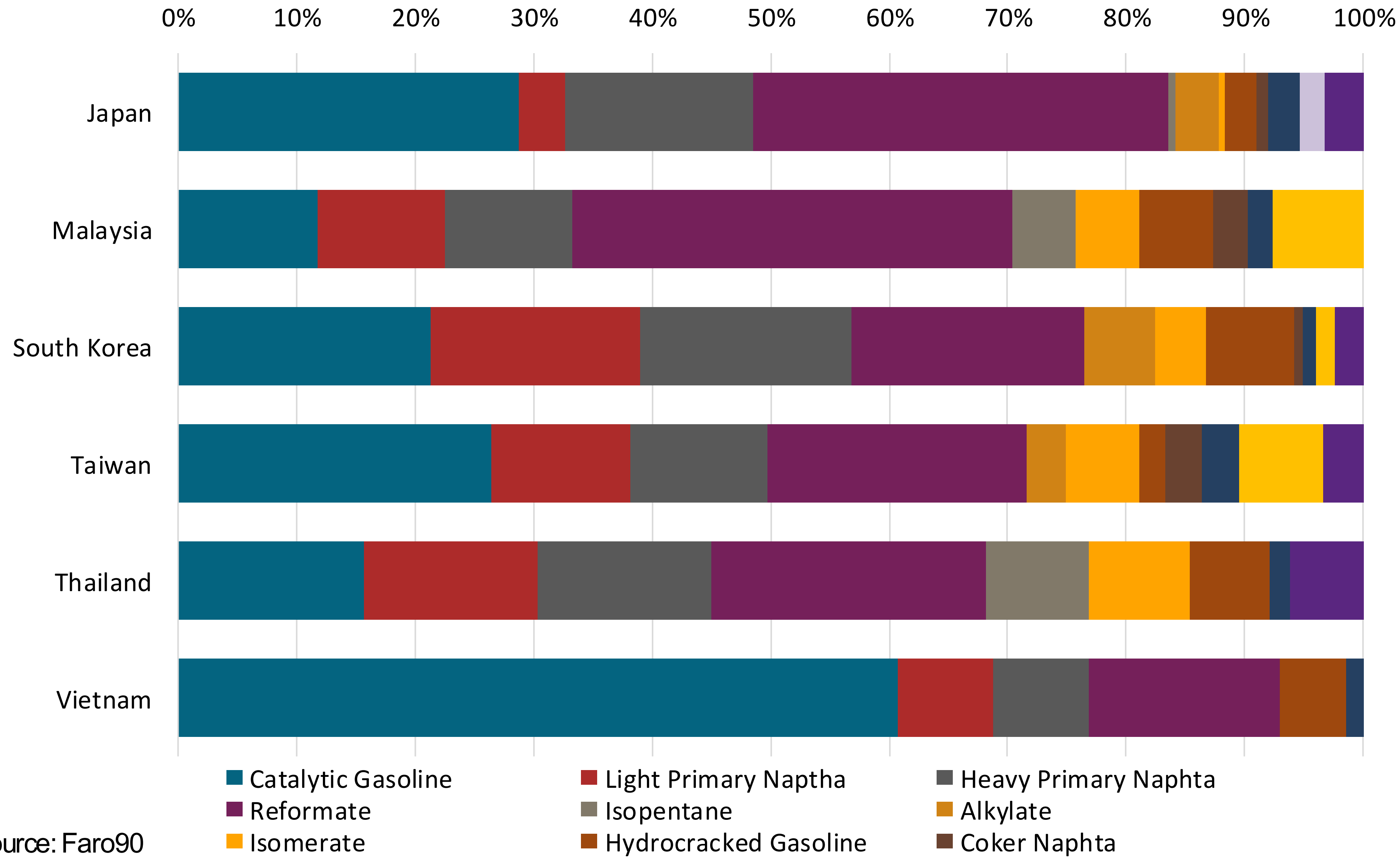
Gasoline Quality in South Korea

Name	Clean Air Conservation Act			
Implementation Year	2009			
Applicability	Nationwide			
Grade	RON 91		RON 94	
	Min	Max	Min	Max
RON	91.0		94.0	
MON				
AKI				
Benzene (%vol)		0.7		0.7
Aromatics (% vol)		24.0		24.0
Olefins (%vol)		18.0		18.0
Lead (g/L)		0.013		0.013
Manganese (mg/L)		-		-
Sulfur (ppm)		10		10
Oxygen (%v)	0.5	2.3	0.5	2.3
Ethanol (%vol)				
MTBE (%vol)				
RVP (psi)		8.7		8.7

Gasoline Component Blending in Asia

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naptha
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

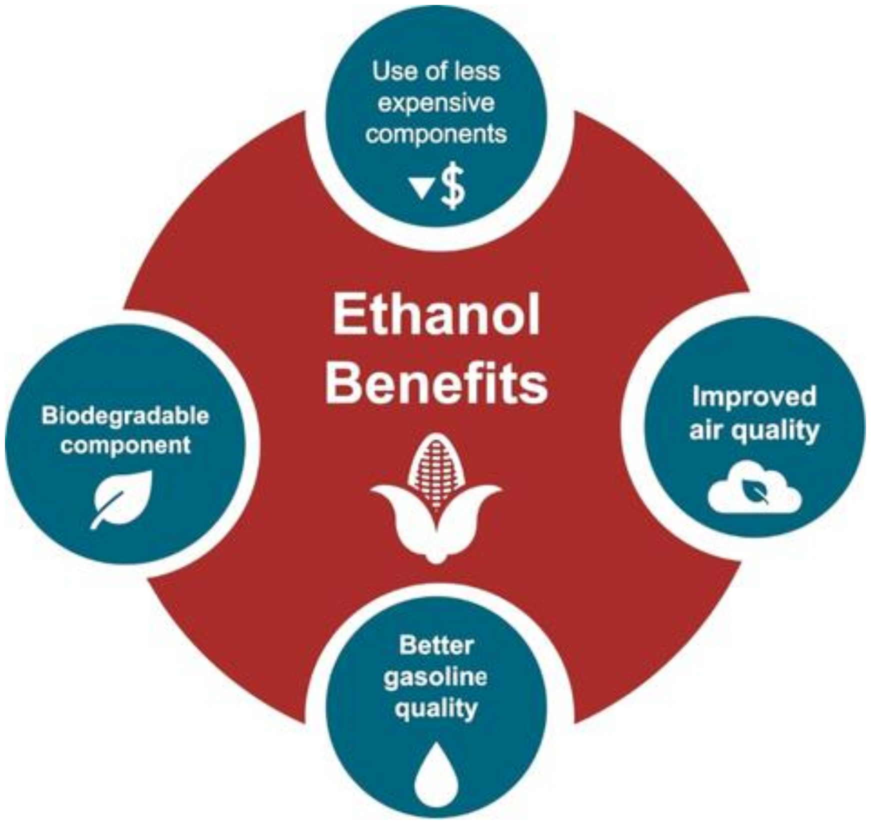
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

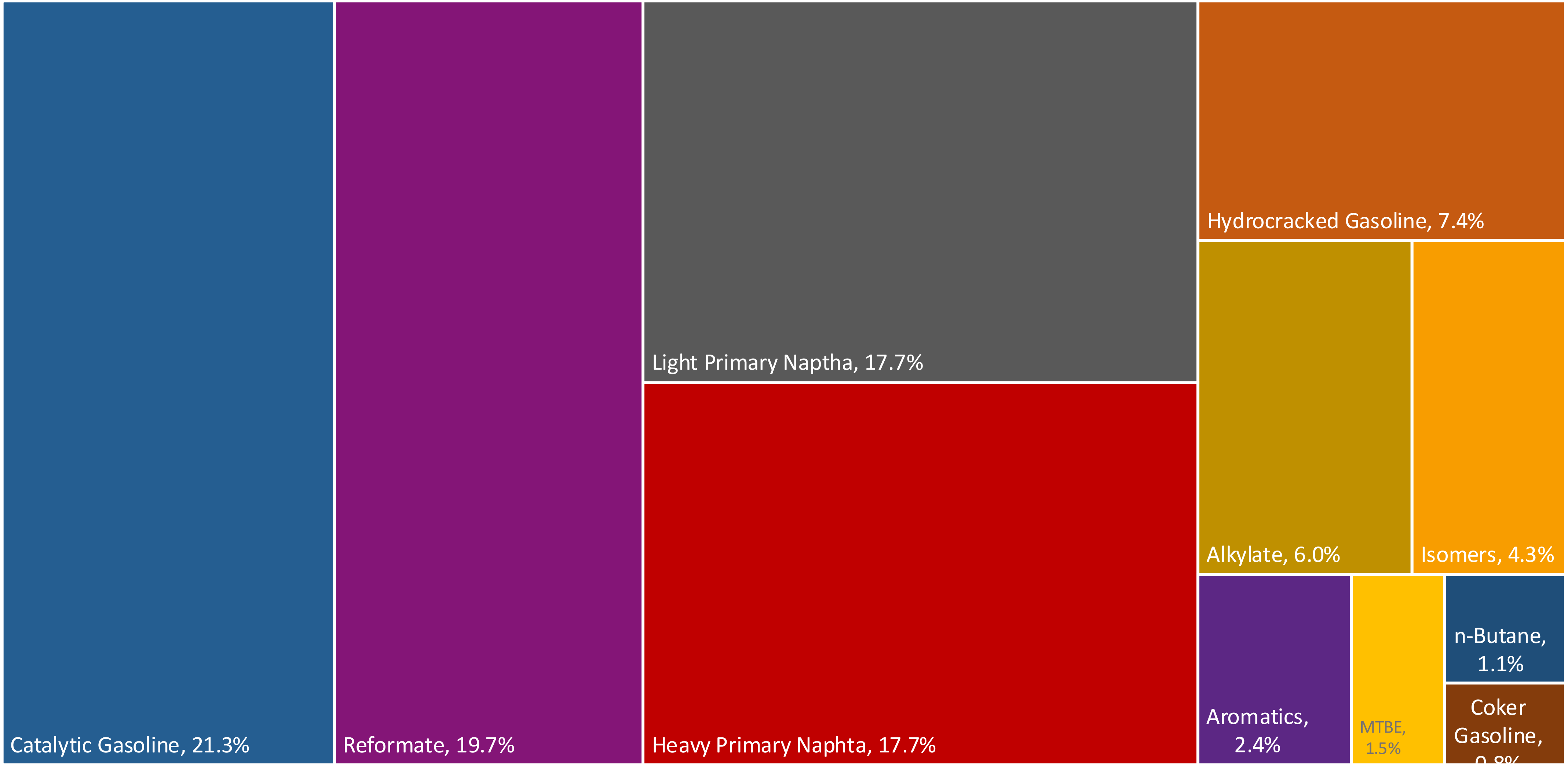
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

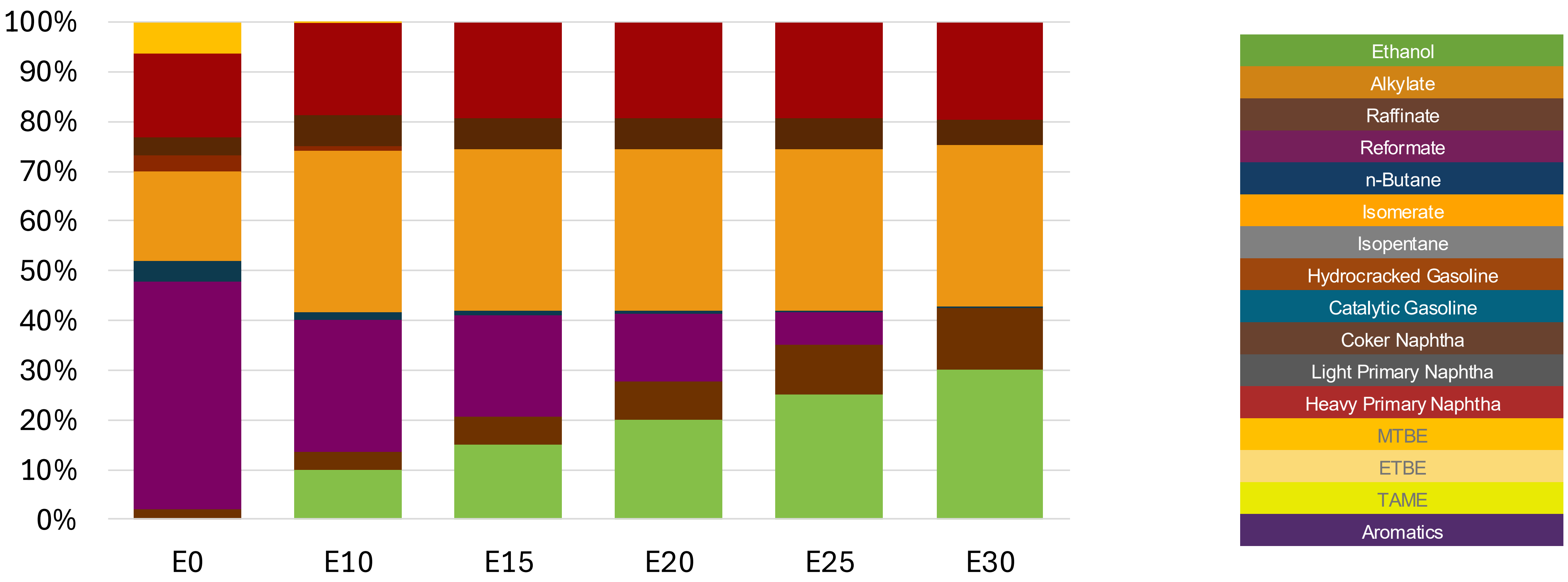


South Korea Available Blending Components



Source: Faro90

South Korea – Gasoline 91 – Constant Octane

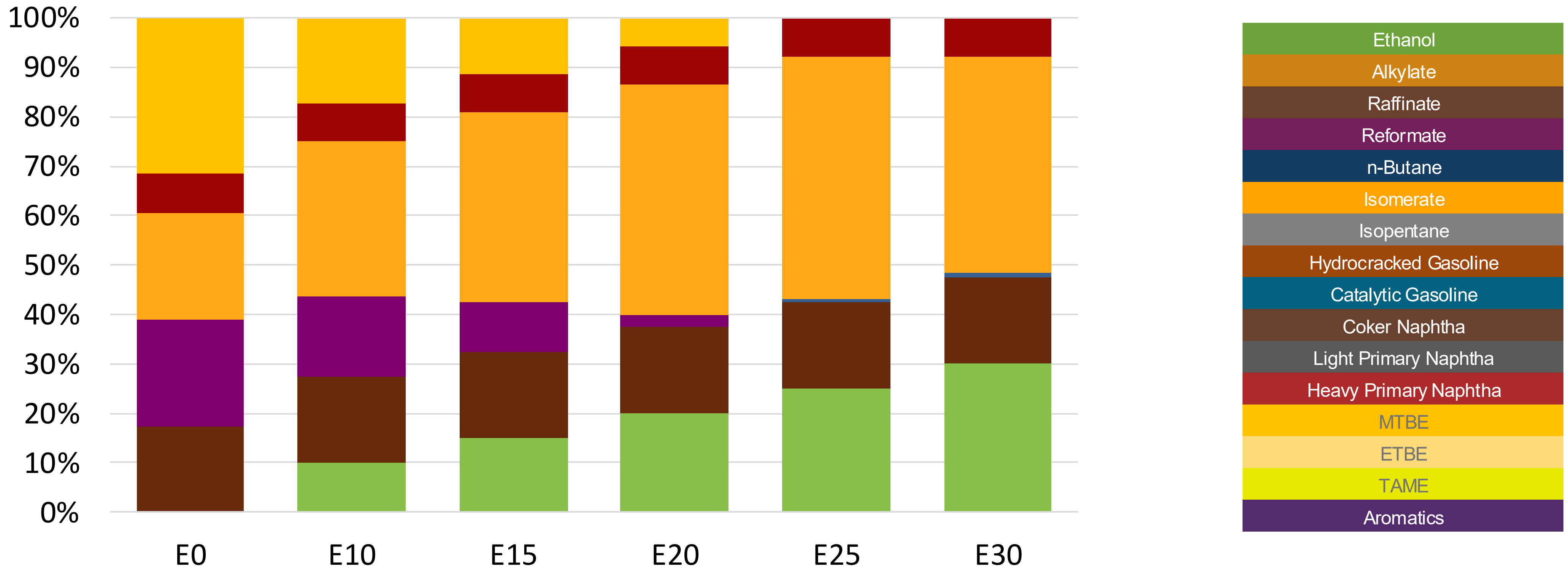


Octane (RON)	91.0	91.0	92.1	93.2	94.3	95.3
Price (US$/gal)	$ 2.257	$ 2.137	$ 2.094	$ 2.050	$ 2.001	$ 1.949

Average CIF 2024 Prices

Source: Faro90

South Korea - Gasoline 94 - Constant Octane

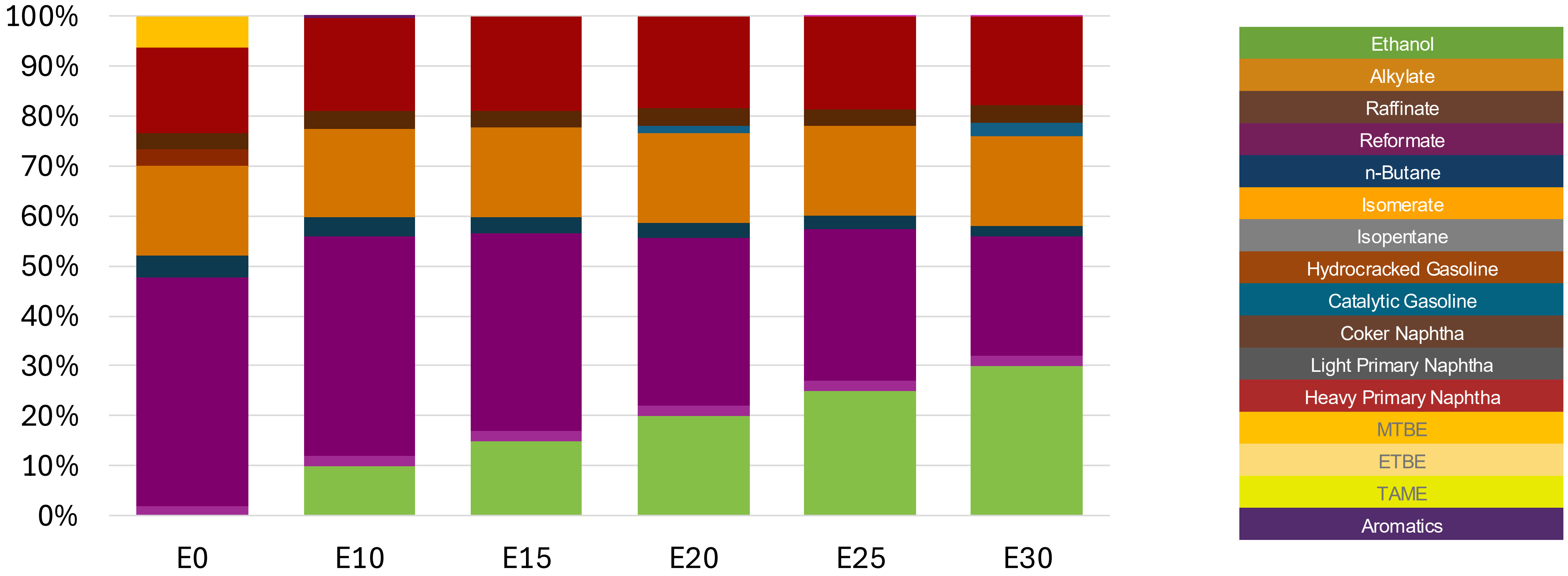


Octane (RON)	94.0	94.0	94.0	94.0	94.7	96.9
Price (US$/gal)	$ 2.240	$ 2.185	$ 2.140	$ 2.086	$ 2.060	$ 2.080

Average CIF 2024 Prices

Source: Faro90

South Korea – Gasoline 91 – Increased Octane

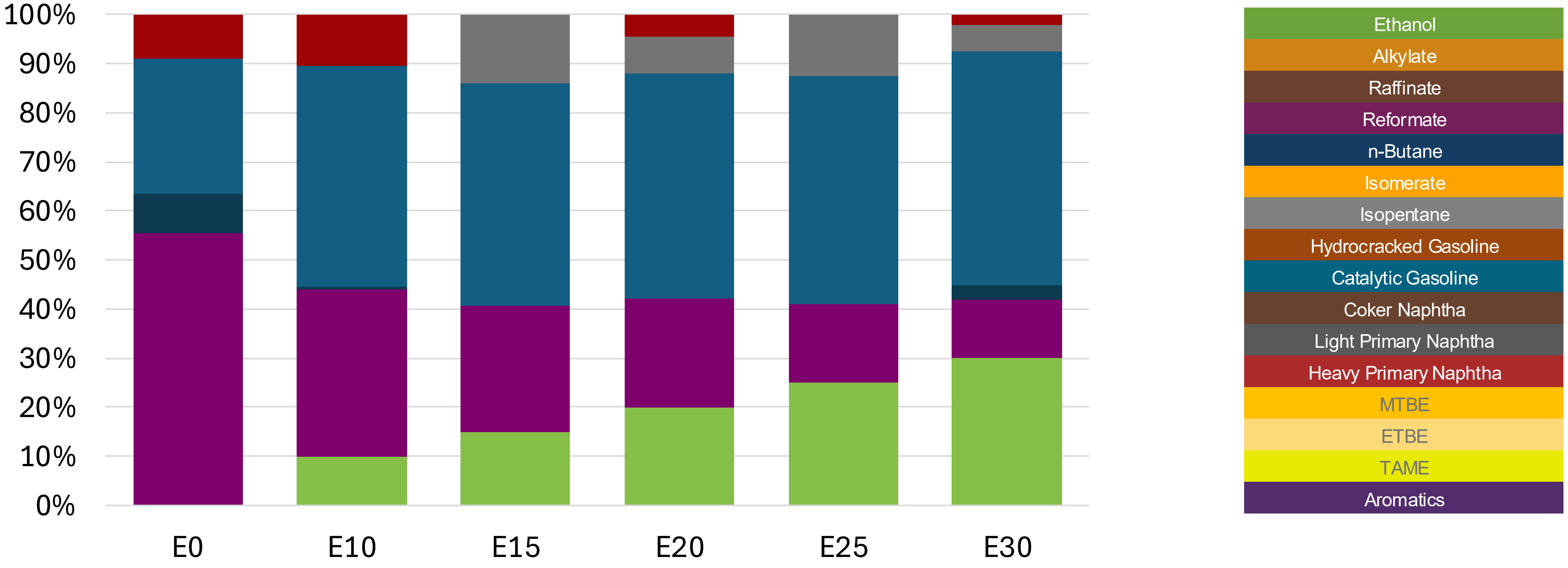


Octane (RON)	91.0	93.7	95.4	97.1	98.8	100.3
Price (US$/gal)	$ 2.257	$ 2.257	$ 2.257	$ 2.257	$ 2.257	$ 2.257

Average CIF 2024 Prices

Source: Faro90

South Korea – Gasoline 94 – Increased Octane



Octane (RON)	94.0	94.0	94.2	95.6	97.3	99.5
Price (US$/gal)	$ 2.240	$ 2.240	$ 2.240	$ 2.240	$ 2.240	$ 2.240

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

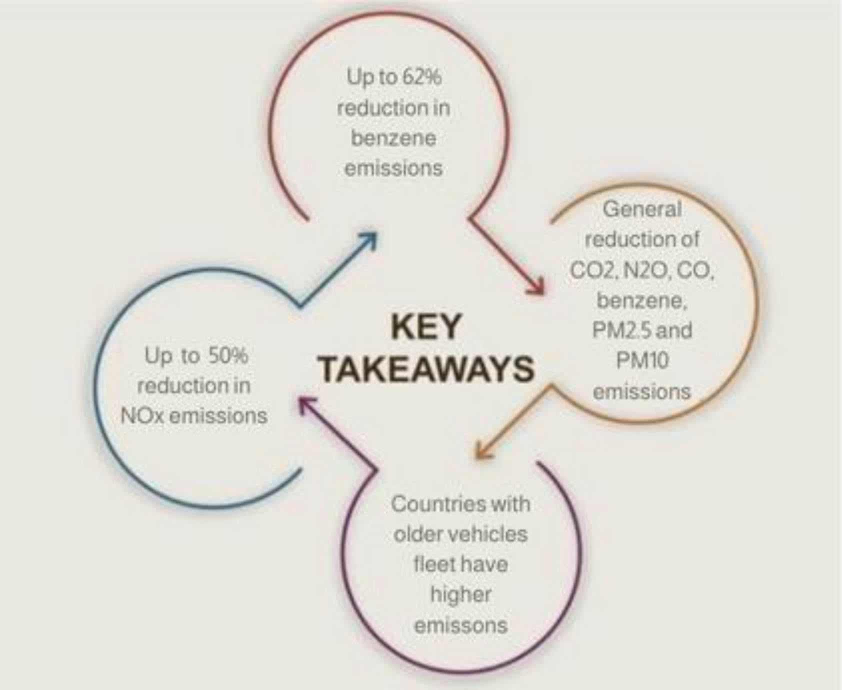
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - South Korea

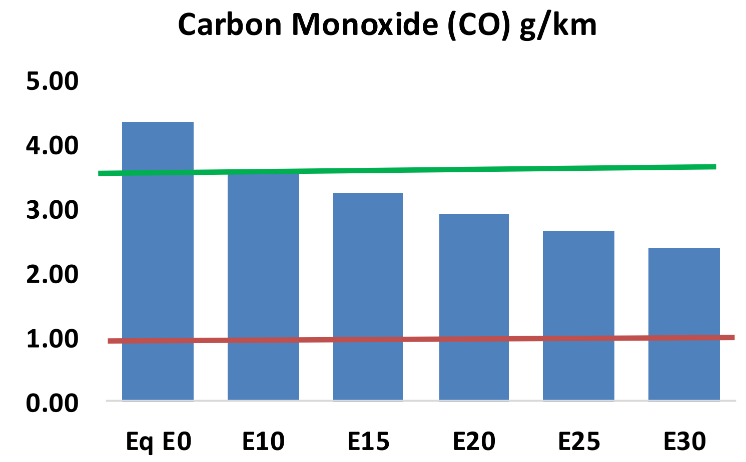
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	70,131
		Single FI	3-Ways		>8 years	350,948
		Multi FI	3-Ways		>8 years	1,028,517
			Euro III		>8 years	1,158,130
			Euro IV		>8 years	1,992,869
			Euro V		>8 years	684,400
			Euro 6		>8 years	1,669,245
					4-8 years	1,962,533
					<4 years	1,588,508
			Other		All	1,272,314
			Hybrid		All	1,245,121
		Electric			All	420,832
	Medium	Carburator	3-Ways	PCV Tank	>8 years	44,838
		Single FI	3-Ways		>8 years	224,377
		Multi FI	3-Ways		>8 years	657,576
			Euro III		>8 years	740,444
			Euro IV		>8 years	1,274,129
			Euro V		>8 years	437,567
			Euro 6		>8 years	1,067,222
					4-8 years	1,254,734
					<4 years	1,015,603
			Other		All	813,447
			Hybrid		All	796,061
		Electric			All	269,057
Sml Engines	Light	4-Cycle	None	PCV	>8 years	132,701
			Euro III		>8 years	193,096
			Euro IV		>8 years	492,528
			Euro V		>8 years	155,242
			Euro 6		>8 years	391,891
					4-8 years	387,364
					<4 years	513,935
			Hybrid		All	0
		Electric			All	0
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	50,383
		Fuel-Injection	3-Ways		>8 years	600,152
			Euro III		>8 years	151,964
			Euro IV		>8 years	380,908
			Euro V		>8 years	183,380
			Euro 6		>8 years	799,757
					4-8 years	1,222,361
					<4 years	1,087,366
			Hybrid		All	0
		Electric			All	0

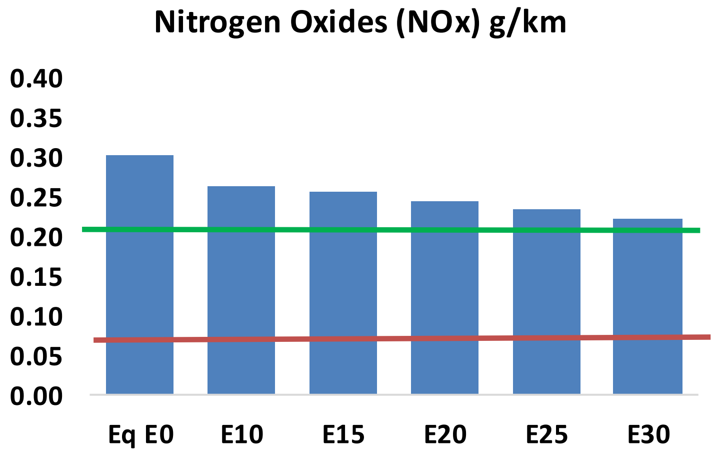
Vehicle Fleet: 28,781,630
Gasoline Vehicle Fleet: 16,830,060

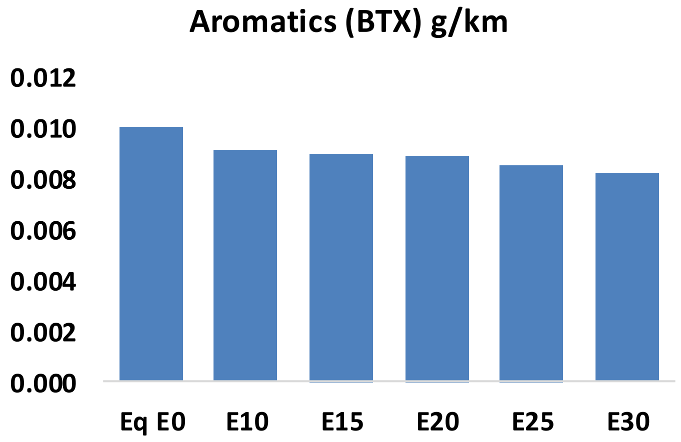
Source: MOLIT

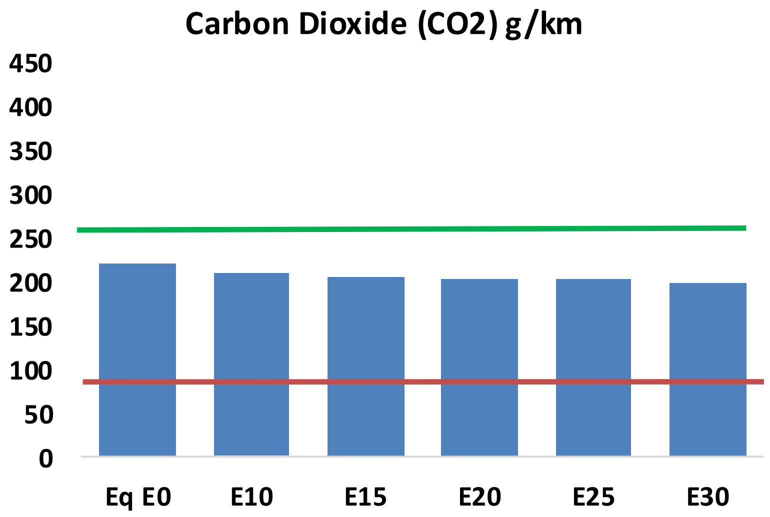
South Korea – Vehicular Emissions

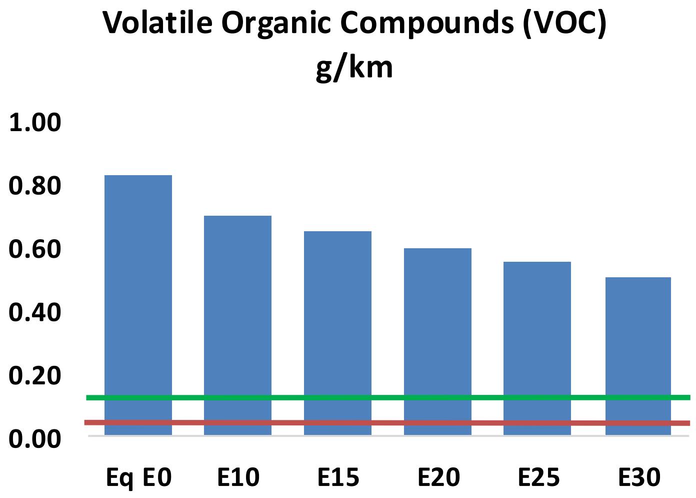
TIER III BIN 125 United States
Euro 6 Standard

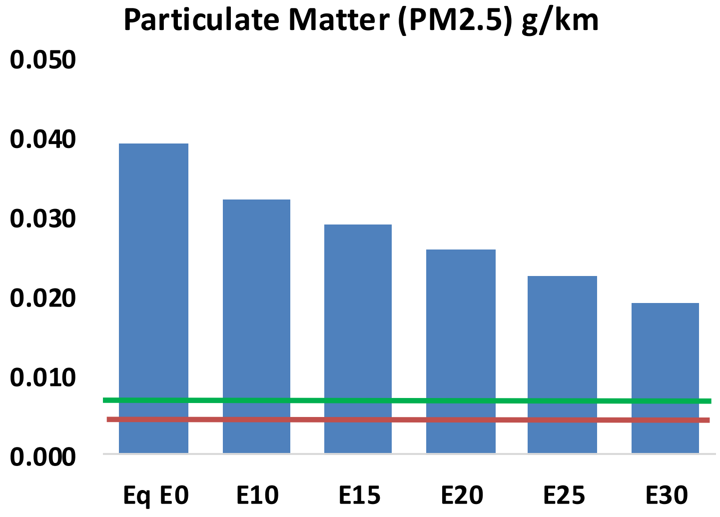












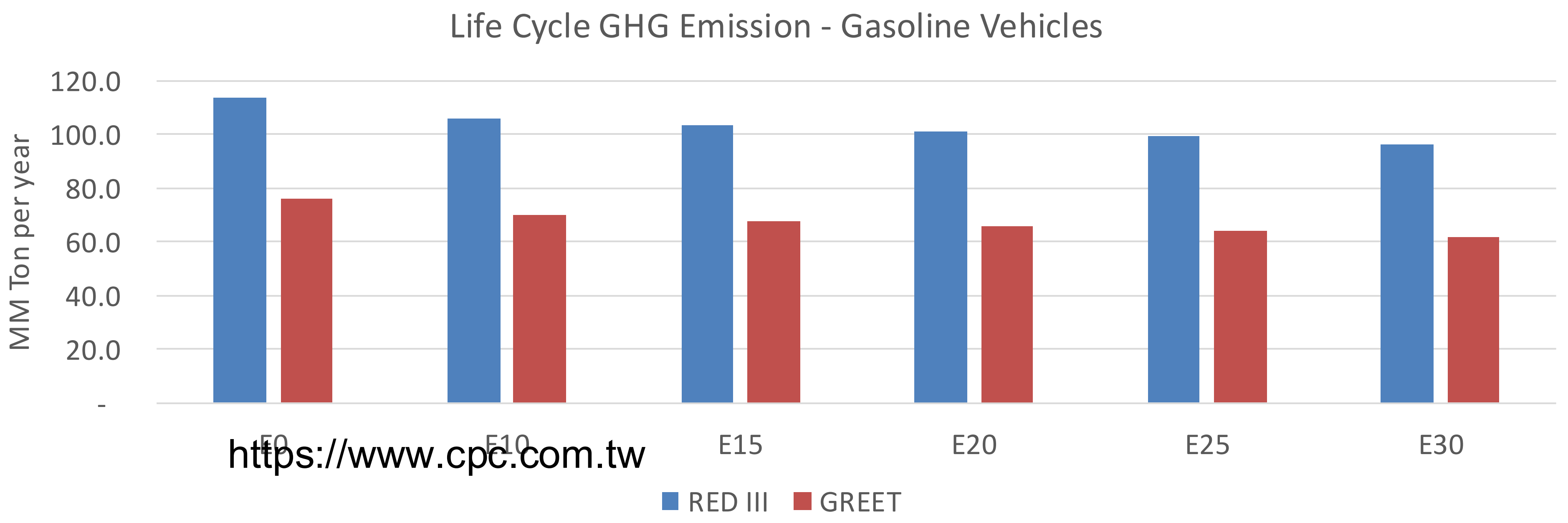
Source: Faro90

South Korea – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,591,317	2,359,846	2,128,374	1,932,397	1,732,283	1,577,533	1,414,428	-18%	-33%
CO2	130,279,254	126,929,216	123,765,291	121,276,958	120,045,633	119,459,096	117,257,122	-5%	-8%
VOC	489,471	452,129	414,788	384,244	353,532	329,925	298,394	-15%	-28%
NOx	180,005	166,205	156,882	152,209	144,597	138,669	131,664	-13%	-20%
SOx	1,520	1,405	1,290	1,193	1,100	999	892	-15%	-28%
NH3	46,891	46,427	45,962	46,181	46,700	47,065	47,369	-2%	0%
N2O	2,724	2,709	2,698	2,712	2,768	2,798	2,830	-1%	2%
CH4	101,103	101,103	101,103	102,619	104,844	106,765	108,584	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,005	1,895	1,785	1,704	1,626	1,567	1,484	-11%	-19%
Acetaldehyde	6,582	7,554	11,084	16,879	22,996	26,277	31,049	68%	249%
Formaldehyde	25,164	24,465	28,519	33,488	35,083	38,812	42,251	13%	39%
Benzene	5,947	5,715	5,415	5,330	5,285	5,055	4,891	-9%	-11%
PM2.5	7,324	6,658	5,992	5,406	4,833	4,218	3,575	-18%	-34%
PM10	15,949	14,499	13,049	11,773	10,525	9,186	7,786	-18%	-34%

Source: Faro90

South Korea – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	682.3
Target @ 2035 (MMT/year)	332.3
Delta	349.9

vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.3%	11.1%	13.8%	16.0%	19.1%
RED II/III	0.0%	6.4%	8.7%	10.9%	12.6%	15.1%

Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	1.8%	2.4%	3.0%	3.5%	4.2%
RED II/III	0.0%	2.1%	2.8%	3.5%	4.1%	4.9%

Source: Greet, RED II/III, climateactiontracker.org, F90

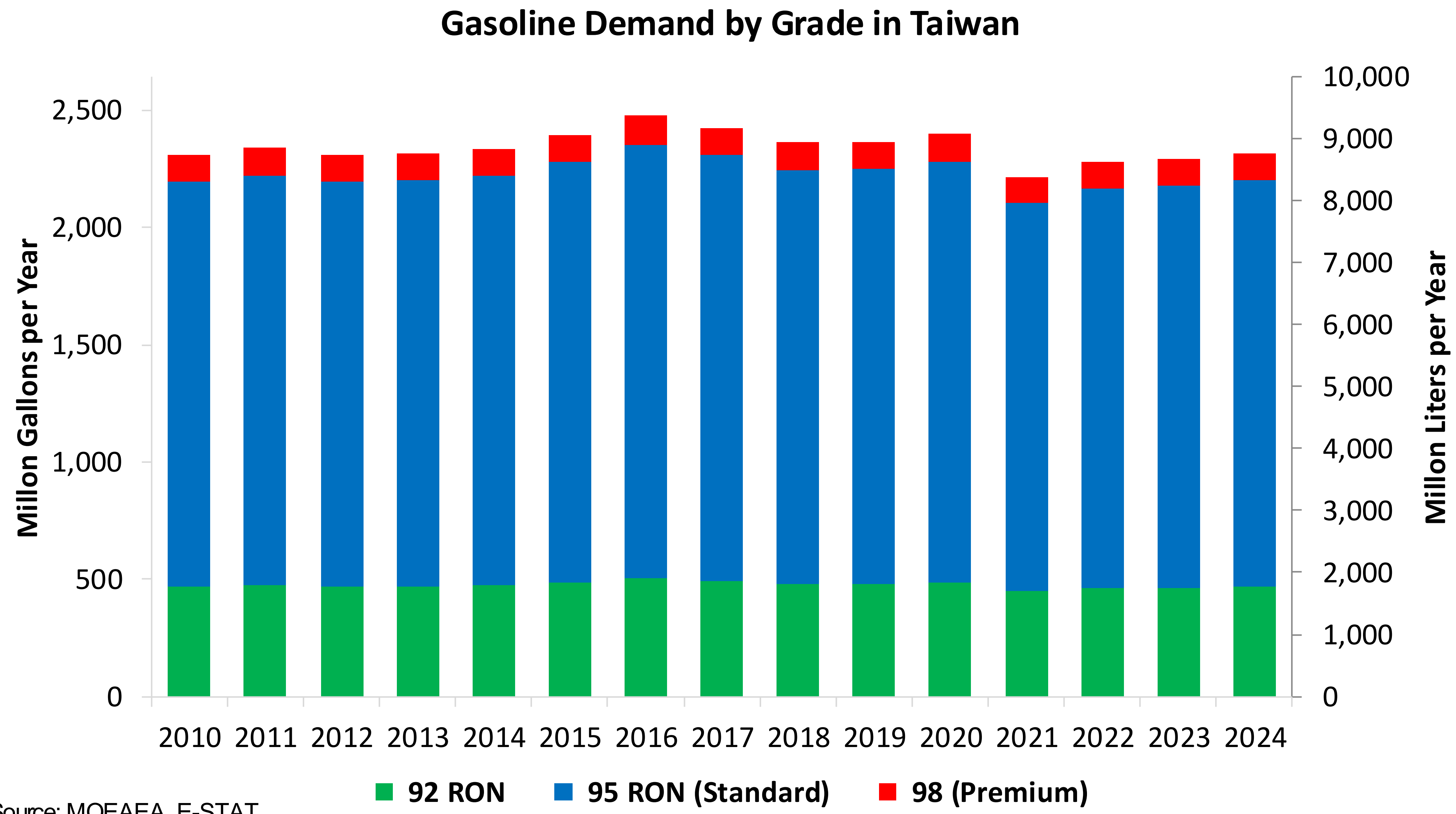
Ethanol Gasoline Blending in Taiwan

In 2024, gasoline consumption in Taiwan was 8,761 million liters (2,314 million gallons) and growth rate was -0.4%. Gasoline production and net imports were 12,450 and -3,689 million liters (3,289 and -975 million gallons) correspondingly. Taiwan offers three main grades: RON 92, RON 95 and RON 98) with an average octane of 94.5 RON.

Taiwan has policies to promote bioethanol in transportation to reduce imported fuel dependency and lower carbon emissions, using it as an oxygenate additive. While there was a limited trial of E3 gasohol in the mid-2000s, the lack of domestic production and insufficient incentives led to its discontinuation. Only a few gas stations offer it. Ethanol is supplied by imports and corresponds to an average content of 0.2%. CPC's 2026 budget indicates that the company plans to procure 96,000 liters of ethanol for E3 blends.
Future bioethanol use depends on developing profitable cellulosic ethanol production from non-food waste like rice straw and food waste, rather than imported feedstocks.

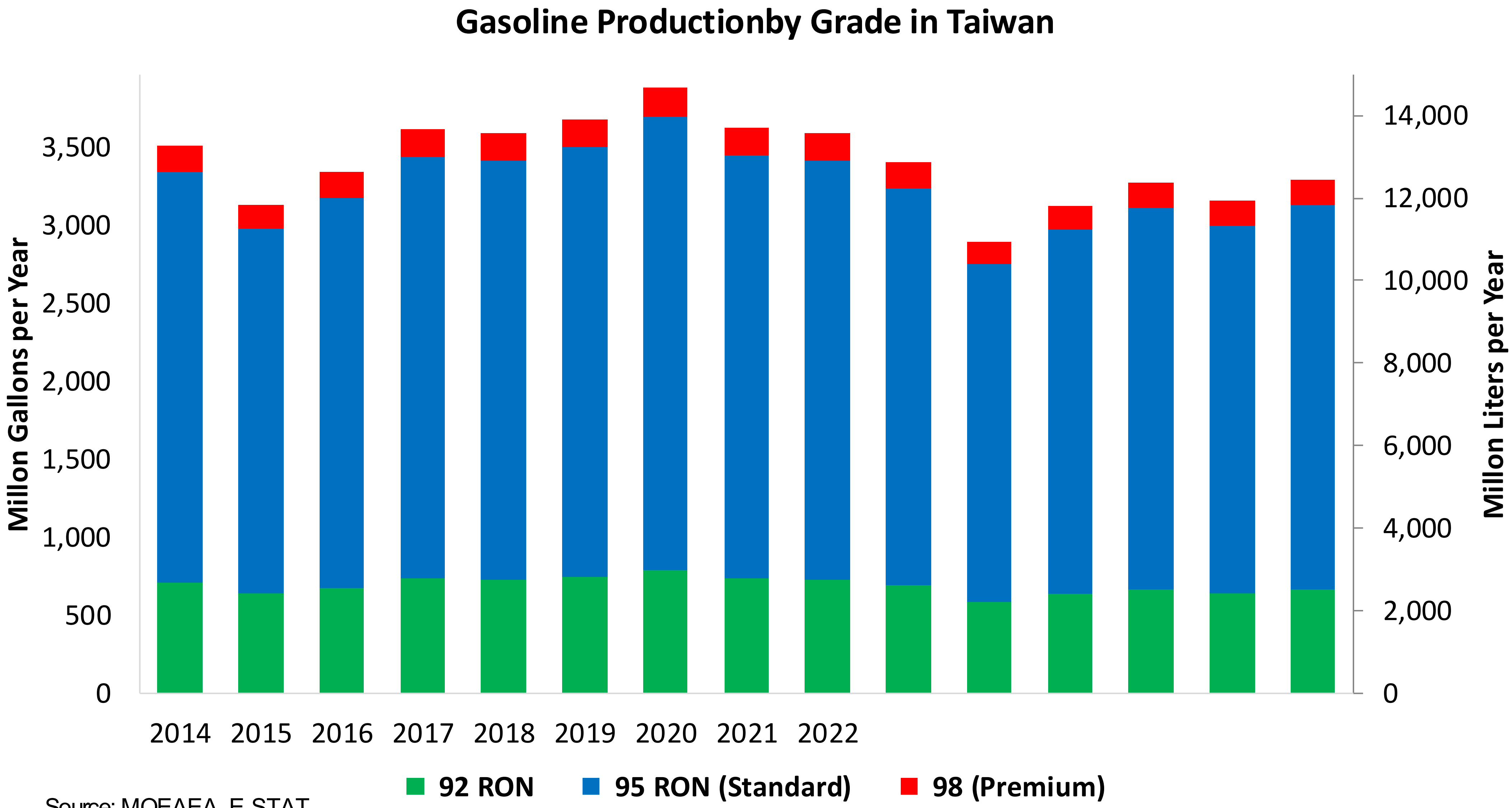
Source: MOEAEA, E-STAT, MOENV, USDA

Gasoline Demand in Taiwan



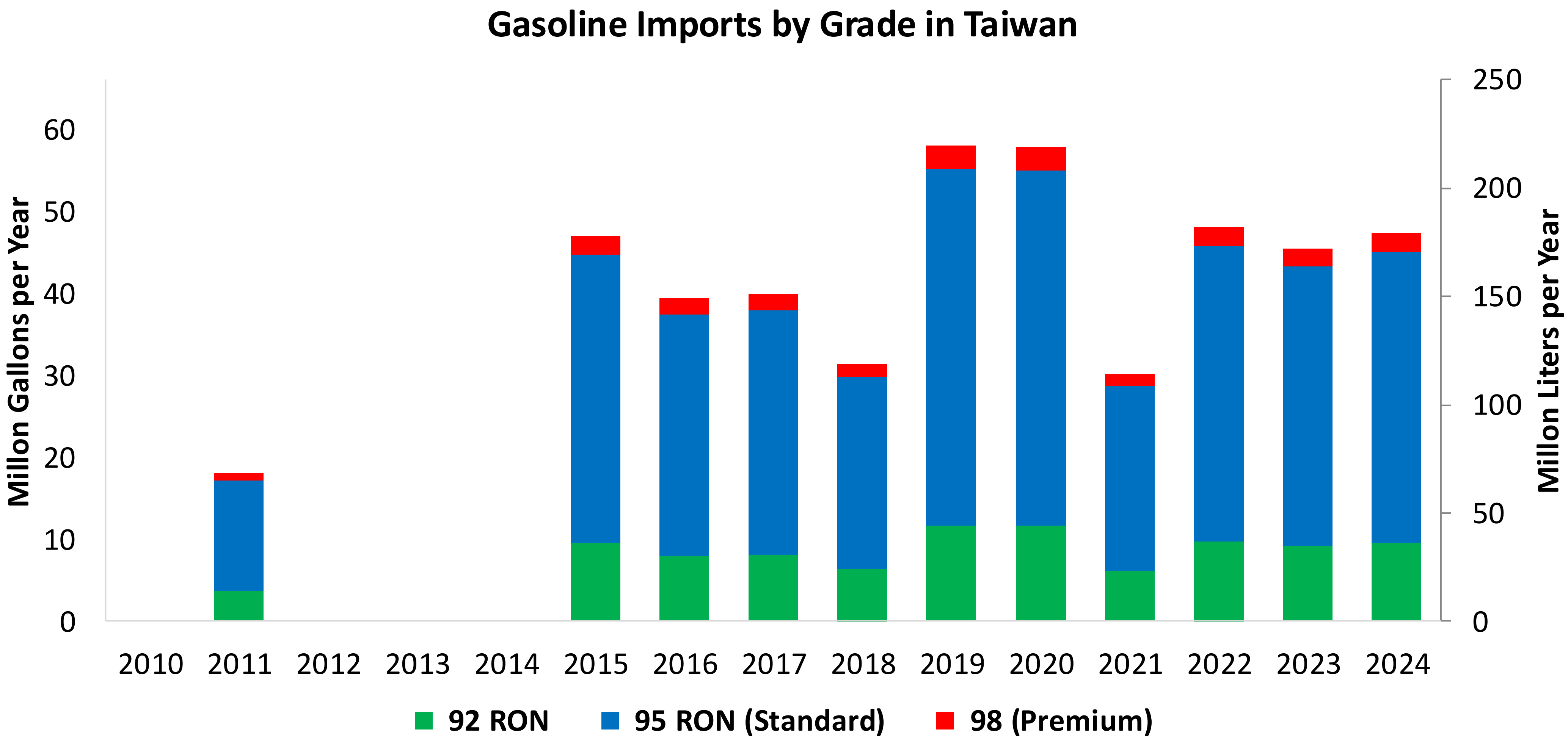
Source: MOEAEA, E-STAT

Gasoline Production in Taiwan



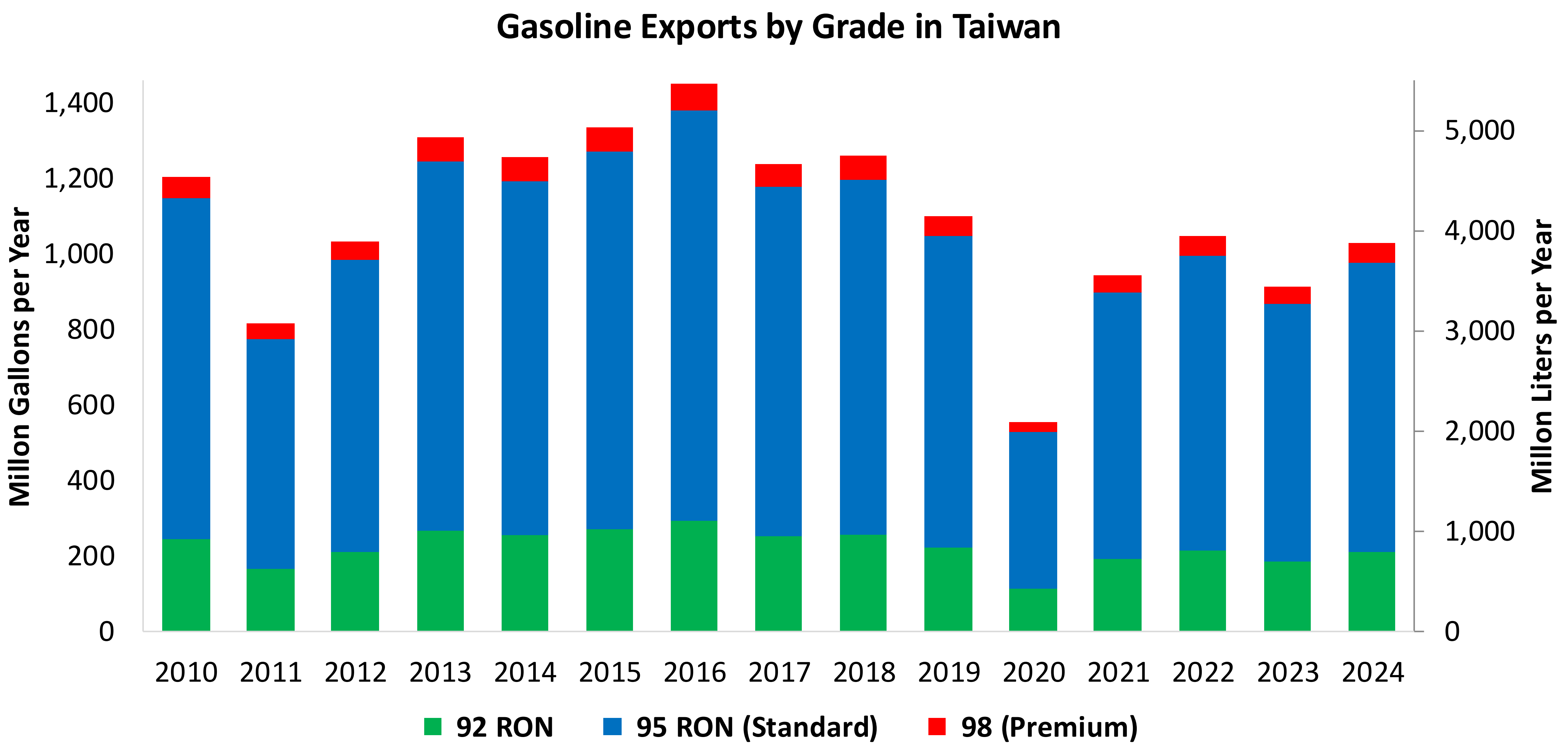
Source: MOEAEA, E-STAT

Gasoline Imports in Taiwan



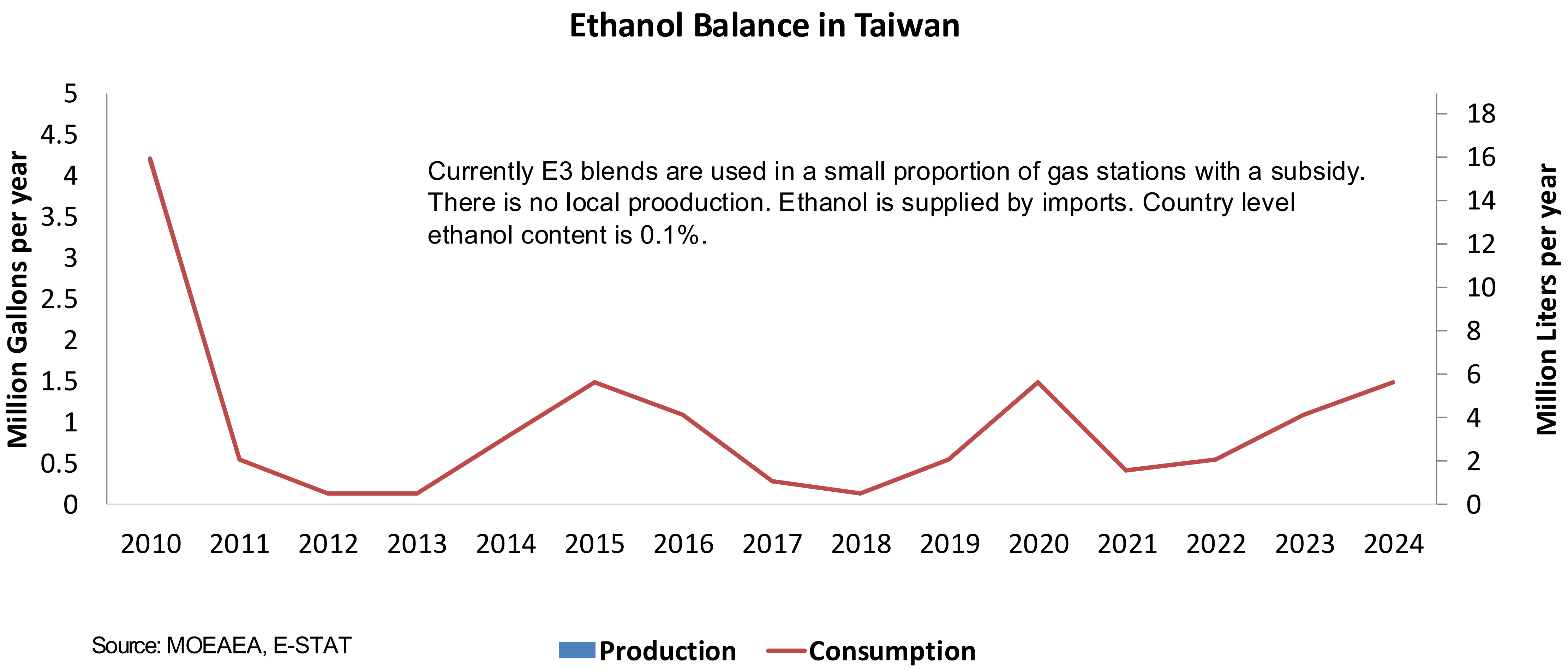
Source: MOEAEA, E-STAT

Gasoline Exports in Taiwan



Source: MOEAEA, E-STAT

Ethanol Balance in Taiwan



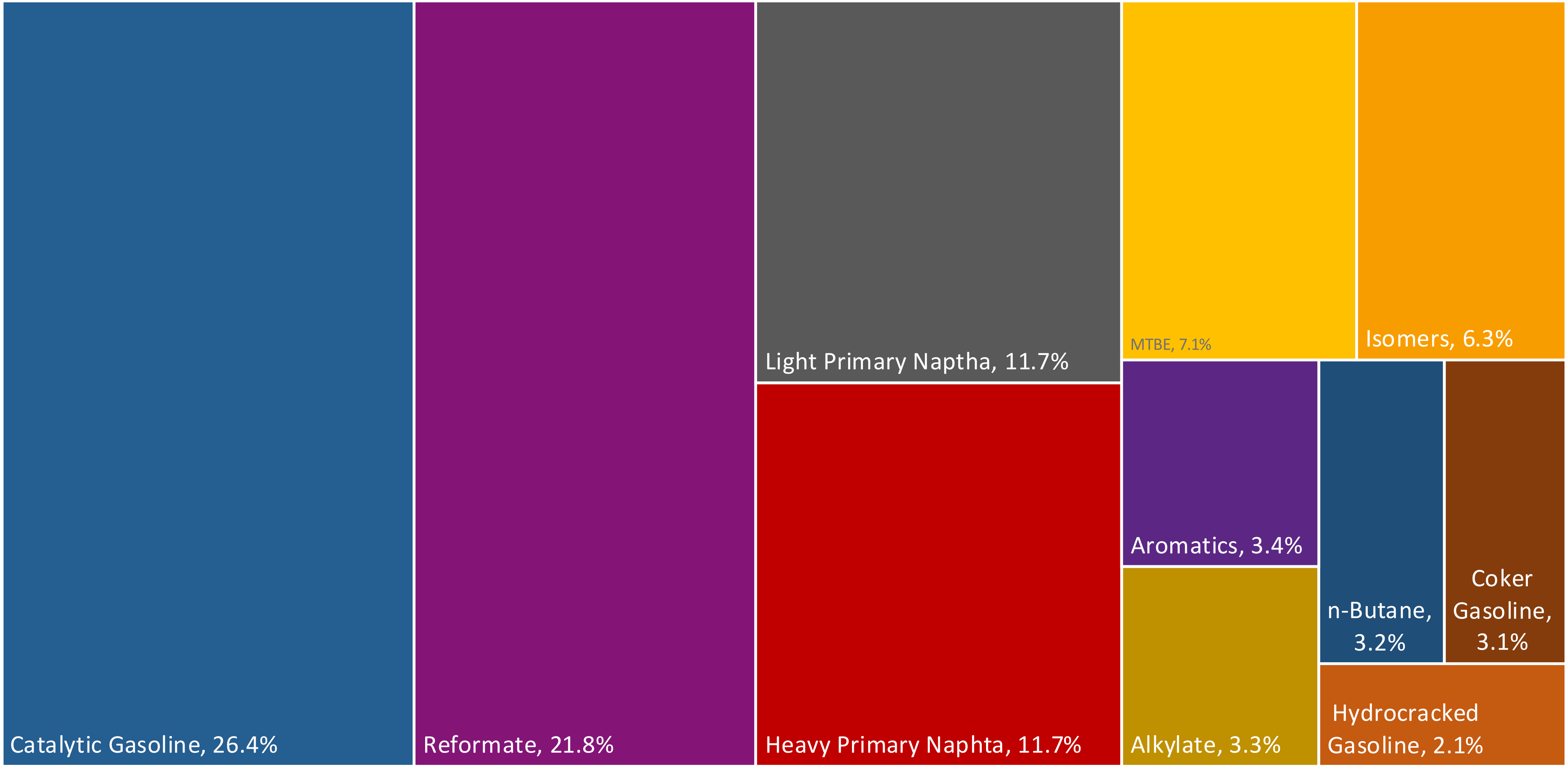
Source: MOEAEA, E-STAT

Gasoline Quality in Taiwan

Name	Mobile Pollution Source Fuel Composition Control Standard Regulation					
Implementation Year	2024					
Applicability	Nationwide					
Grade	92 RON		95 RON (Standard)		98 RON (Premium)	
	Min	Max	Min	Max	Min	Max
RON	92.0		95.0		98.0	
MON						
AKI						
Benzene (%vol)		0.8		0.8		0.8
Aromatics (% vol)		35.0		35.0		35.0
Olefins (%vol)		18.0		18.0		18.0
Lead (g/L)						
Manganese (mg/L)						
Sulfur (ppm)		10		10		10
Oxygen (%v)		3.2		3.2		3.2
Ethanol (%vol)		10.0		10.0		10.0
MTBE (%vol)						
RVP (kPa)	45.0	66.9	45.0	66.9	45.0	66.9

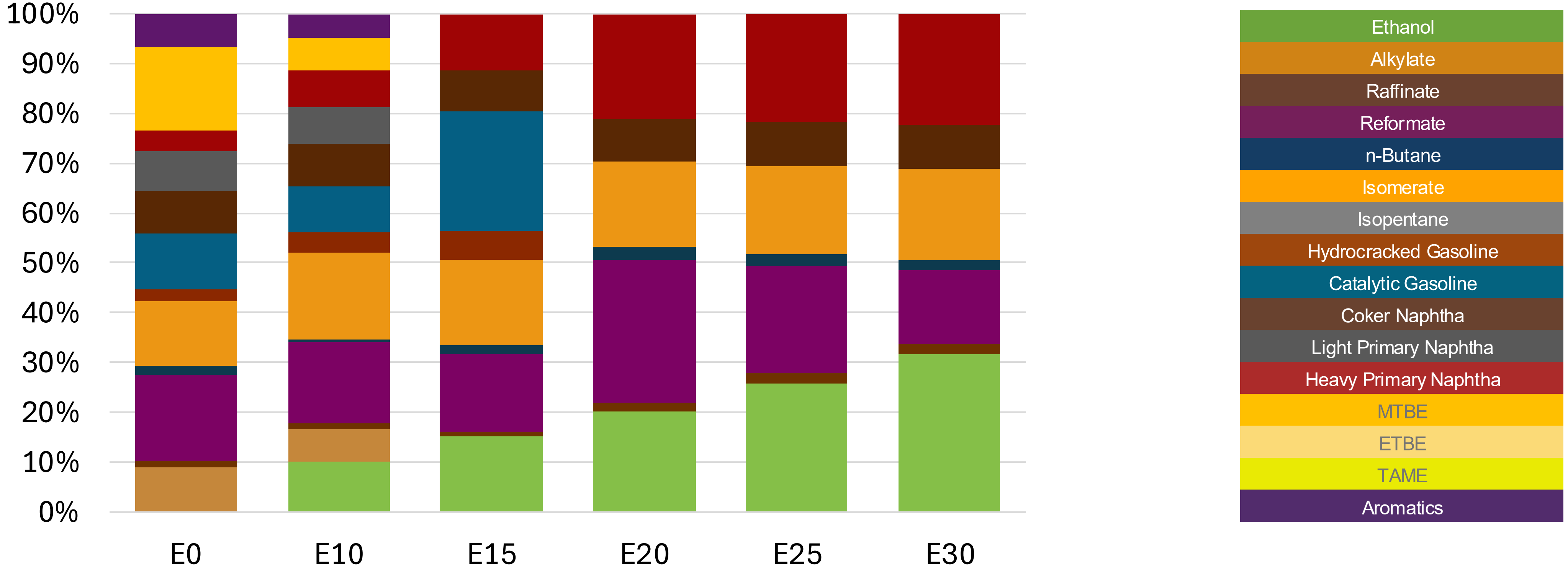
Source: MOEAEA, MOENV, Formosa Petrochemical

Taiwan Available Blending Components



Source: Faro90

Taiwan – Gasoline 95 – Constant Octane

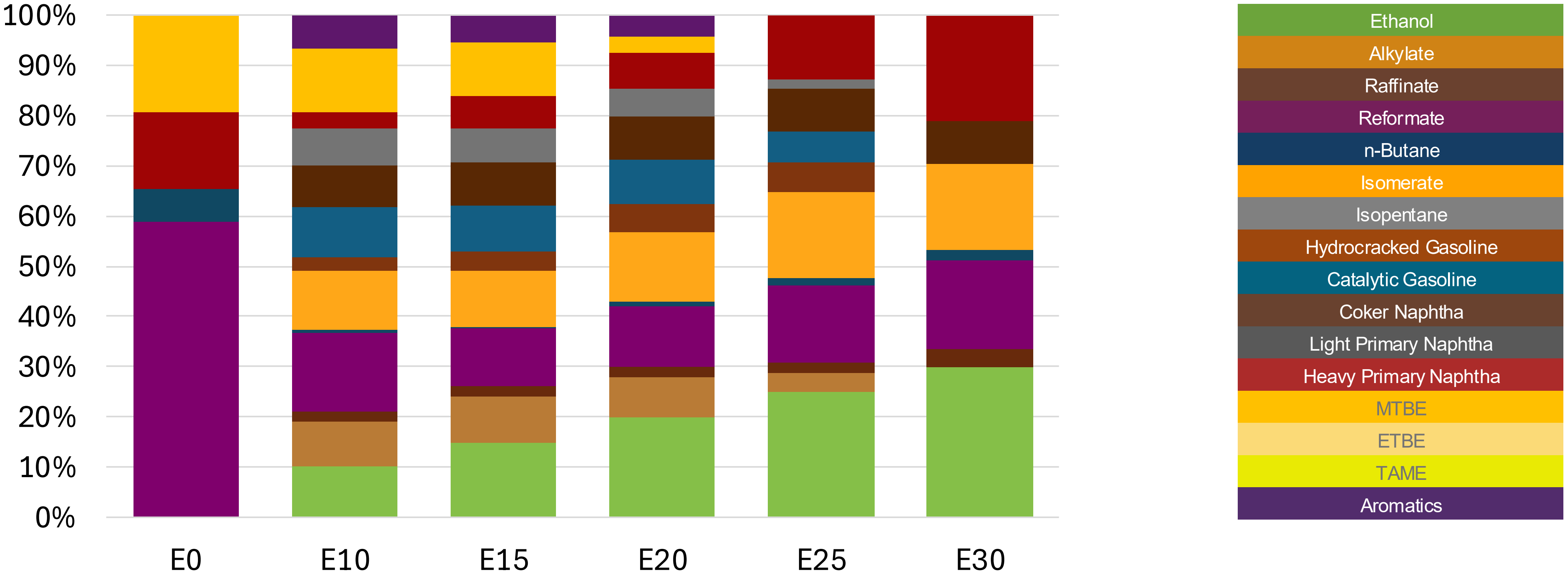


	E0	E10	E15	E20	E25	E30
Octane (RON)	95.0	95.0	95.0	95.8	95.0	95.0
Price (US$/gal)	$ 2.388	$ 2.307	$ 2.223	$ 2.149	$ 2.061	$ 1.995

Average CIF 2024 Prices

Source: Faro90

Taiwan - Gasoline 98 - Constant Octane

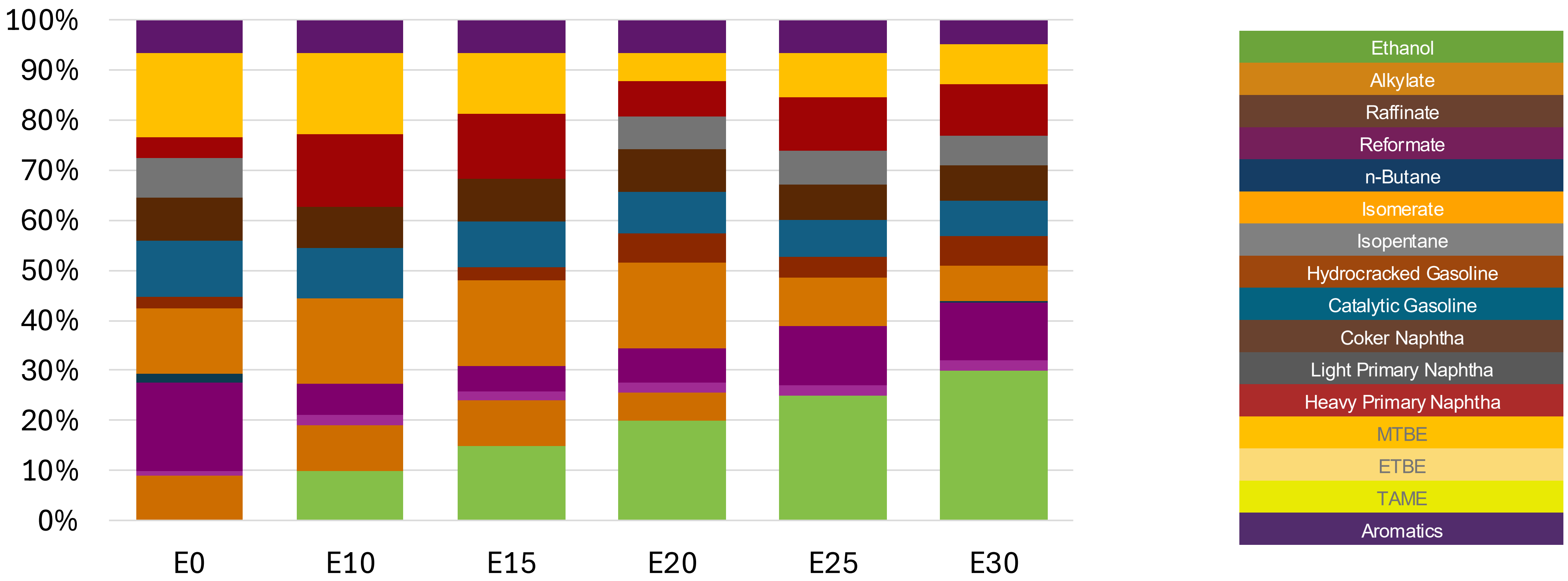


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.6
Price (US$/gal)	$ 2.445	$ 2.418	$ 2.346	$ 2.305	$ 2.210	$ 2.117

Average CIF 2024 Prices

Source: Faro90

Taiwan – Gasoline 95 – Increased Octane

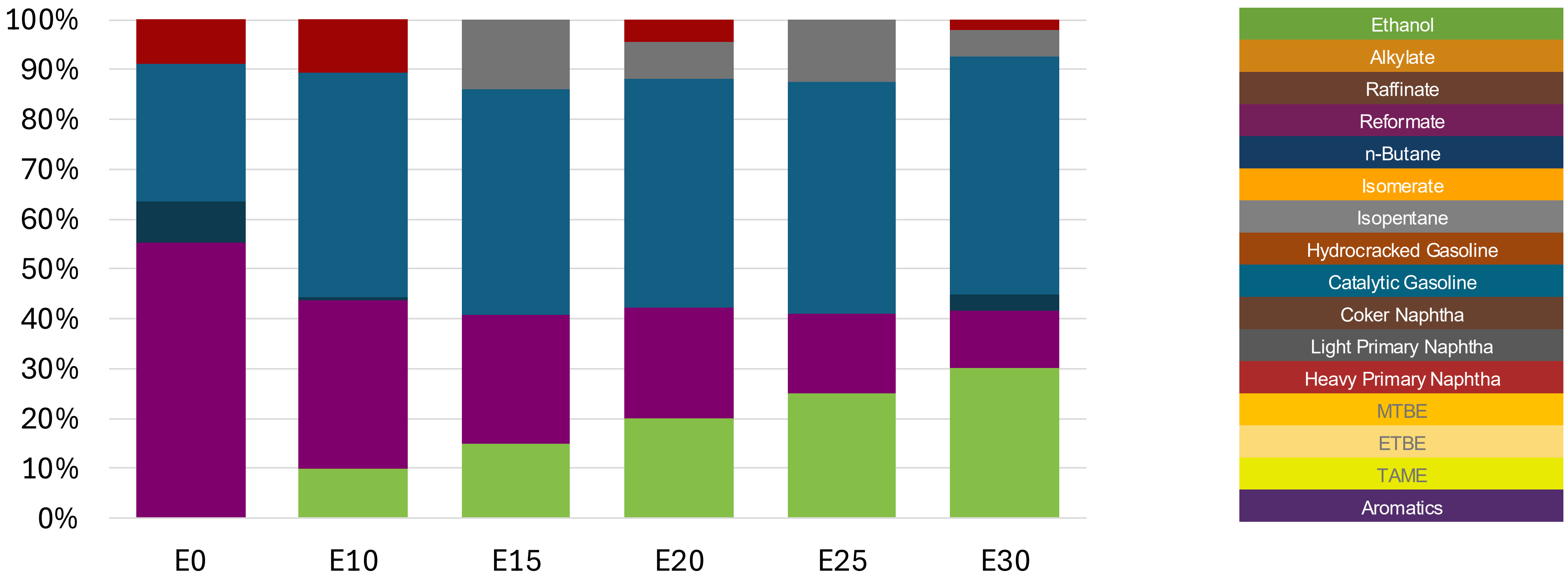


Octane (RON)	95.0	96.4	97.4	98.1	100.6	102.0
Price (US$/gal)	$ 2.388	$ 2.388	$ 2.388	$ 2.388	$ 2.388	$ 2.388

Average CIF 2024 Prices

Source: Faro90

Taiwan – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.8	99.9	101.3	102.6	103.9
Price (US$/gal)	$ 2.445	$ 2.445	$ 2.445	$ 2.445	$ 2.445	$ 2.445

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Taiwan

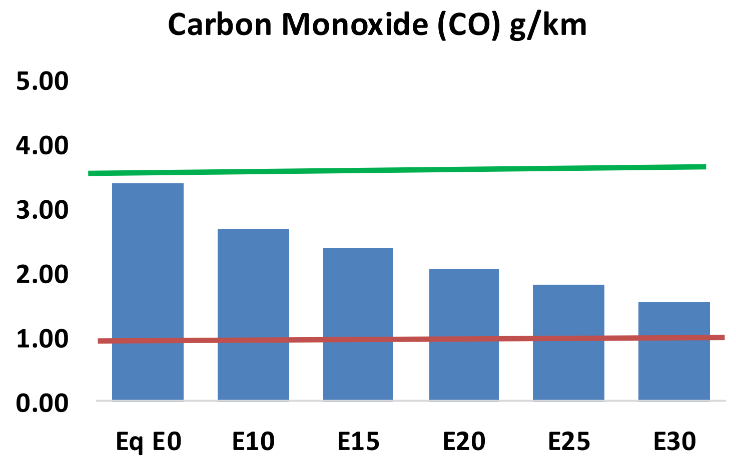
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	9,555
		Single FI	3-Ways		>8 years	277,656
		Multi FI	3-Ways		>8 years	781,852
			Euro III		>8 years	713,122
			Euro IV		>8 years	943,717
			Euro V		>8 years	1,415,819
			Euro 6		>8 years	1,156,342
					4-8 years	981,907
					<4 years	960,692
			Hybrid		All	143,305
		Electric			All	48,801
	Medium	Carburator	3-Ways	PCV Tank	>8 years	17,020
		Single FI	3-Ways		>8 years	67,268
		Multi FI	3-Ways		>8 years	226,812
			Euro III		>8 years	117,564
			Euro IV		>8 years	177,121
			Euro V		>8 years	88,574
			Euro 6		>8 years	99,272
					4-8 years	94,150
					<4 years	109,767
			Hybrid		All	3,244
		Electric			All	974
Sml Engines	Light	4-Cycle	None	PCV	>8 years	884,870
			Euro III		>8 years	2,061,596
			Euro IV		>8 years	3,370,098
			Euro V		>8 years	3,408,923
			Euro 6		>8 years	203,066
					4-8 years	1,706,126
					<4 years	2,057,361
			Hybrid		All	0
		Electric			All	808,708
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	16
		Fuel-Injection	3-Ways		>8 years	10,919
			Euro III		>8 years	16,174
			Euro IV		>8 years	53,747
			Euro V		>8 years	30,133
			Euro 6		>8 years	61,123
					4-8 years	57,191
					<4 years	50,710
			Hybrid		All	11,543
		Electric			All	12,195

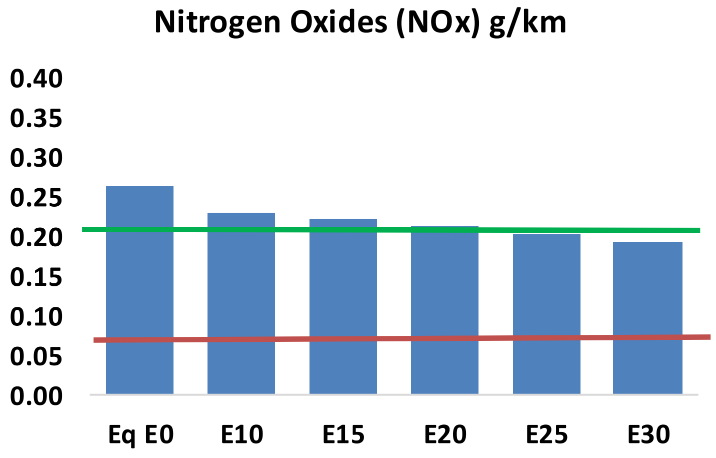
Vehicle Fleet: 23,239,031
Gasoline Vehicle Fleet: 20,069,210

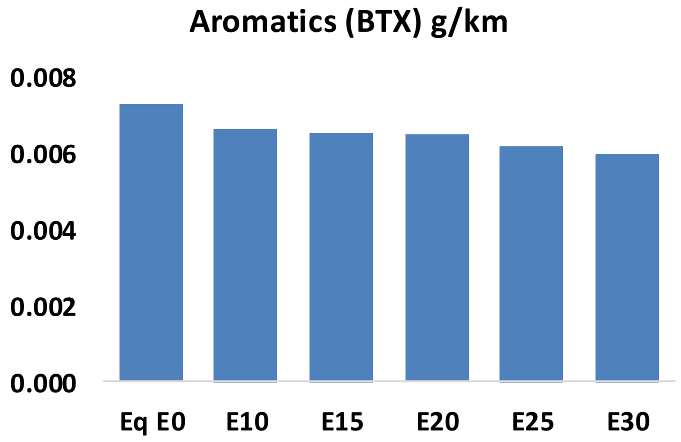
Source: DATA-GOV.tw, CIECData

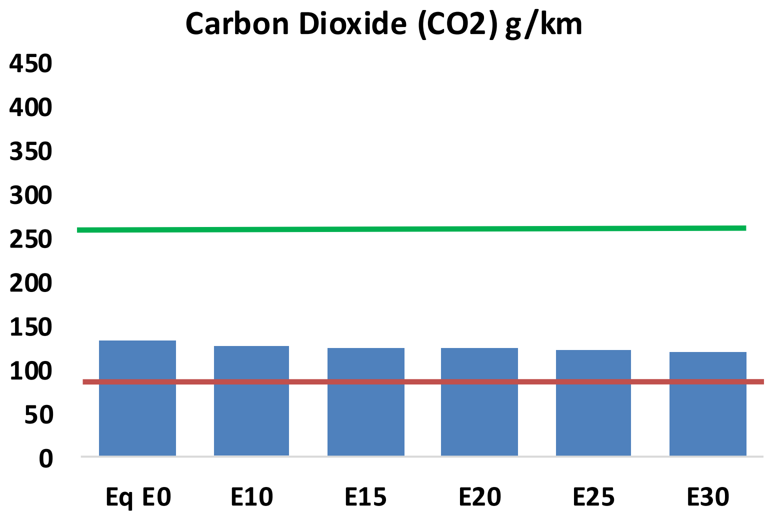
Taiwan – Vehicular Emissions

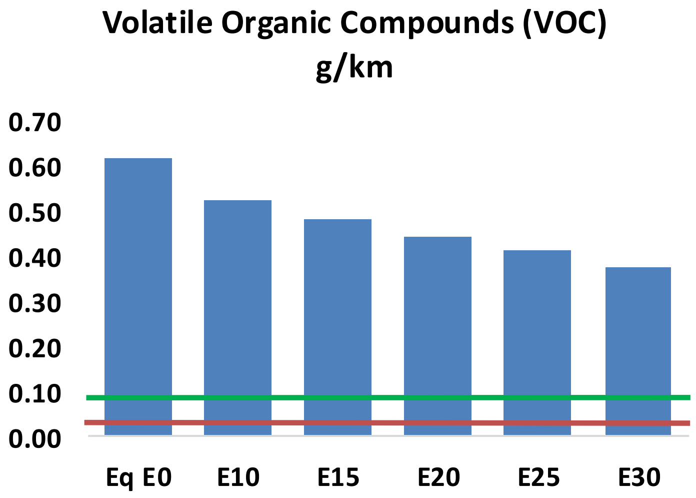
TIER III BIN 125 United States
Euro 6 Standard

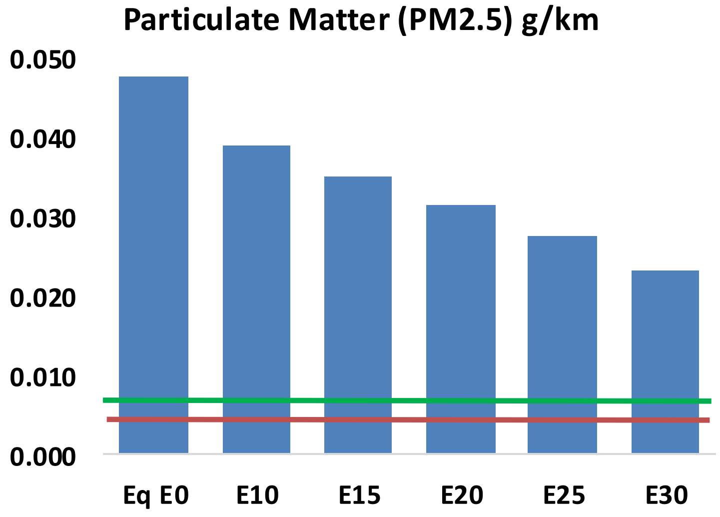












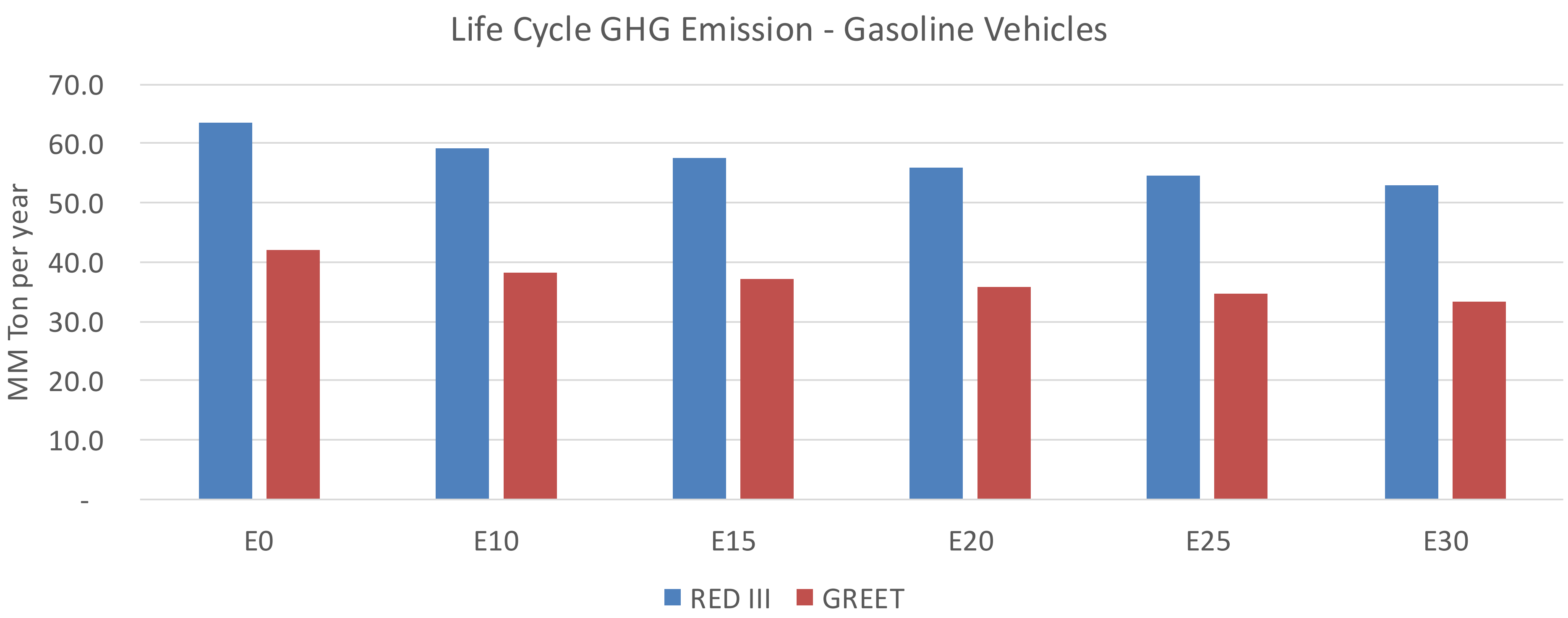
Source: Faro90

Taiwan – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,603,124	1,435,644	1,268,164	1,122,290	971,520	854,387	729,007	-21%	-39%
CO2	62,635,536	61,024,908	59,503,759	58,307,420	57,715,425	57,451,104	56,392,115	-5%	-8%
VOC	291,022	268,550	246,078	227,644	209,084	194,810	175,771	-15%	-28%
NOx	124,311	114,781	108,287	105,003	99,686	95,530	90,629	-13%	-20%
SOx	686	634	582	538	496	451	403	-15%	-28%
NH3	26,238	25,978	25,718	25,840	26,131	26,335	26,505	-2%	0%
N2O	766	762	759	763	779	787	796	-1%	2%
CH4	79,981	79,981	79,981	81,180	82,940	84,459	85,899	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,061	1,917	1,773	1,659	1,545	1,457	1,339	-14%	-25%
Acetaldehyde	6,412	7,359	10,797	16,443	22,402	25,598	30,247	68%	249%
Formaldehyde	25,834	25,116	29,278	34,379	36,017	39,844	43,375	13%	39%
Benzene	3,435	3,301	3,128	3,078	3,053	2,920	2,825	-9%	-11%
PM2.5	4,384	3,986	3,587	3,236	2,893	2,525	2,140	-18%	-34%
PM10	18,103	16,457	14,812	13,363	11,946	10,427	8,838	-18%	-34%

Source: Faro90

Taiwan – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	291.7
Target @ 2035 (MMT/year)	204.2
Delta	87.5

vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.9%	11.8%	14.8%	17.2%	20.5%
RED II/III	0.0%	7.0%	9.5%	12.0%	14.1%	16.7%

Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.3%	5.7%	7.1%	8.2%	9.8%
RED II/III	0.0%	5.1%	6.9%	8.7%	10.2%	12.2%

Source: Greet, RED II/III, climateactiontracker.org, F90

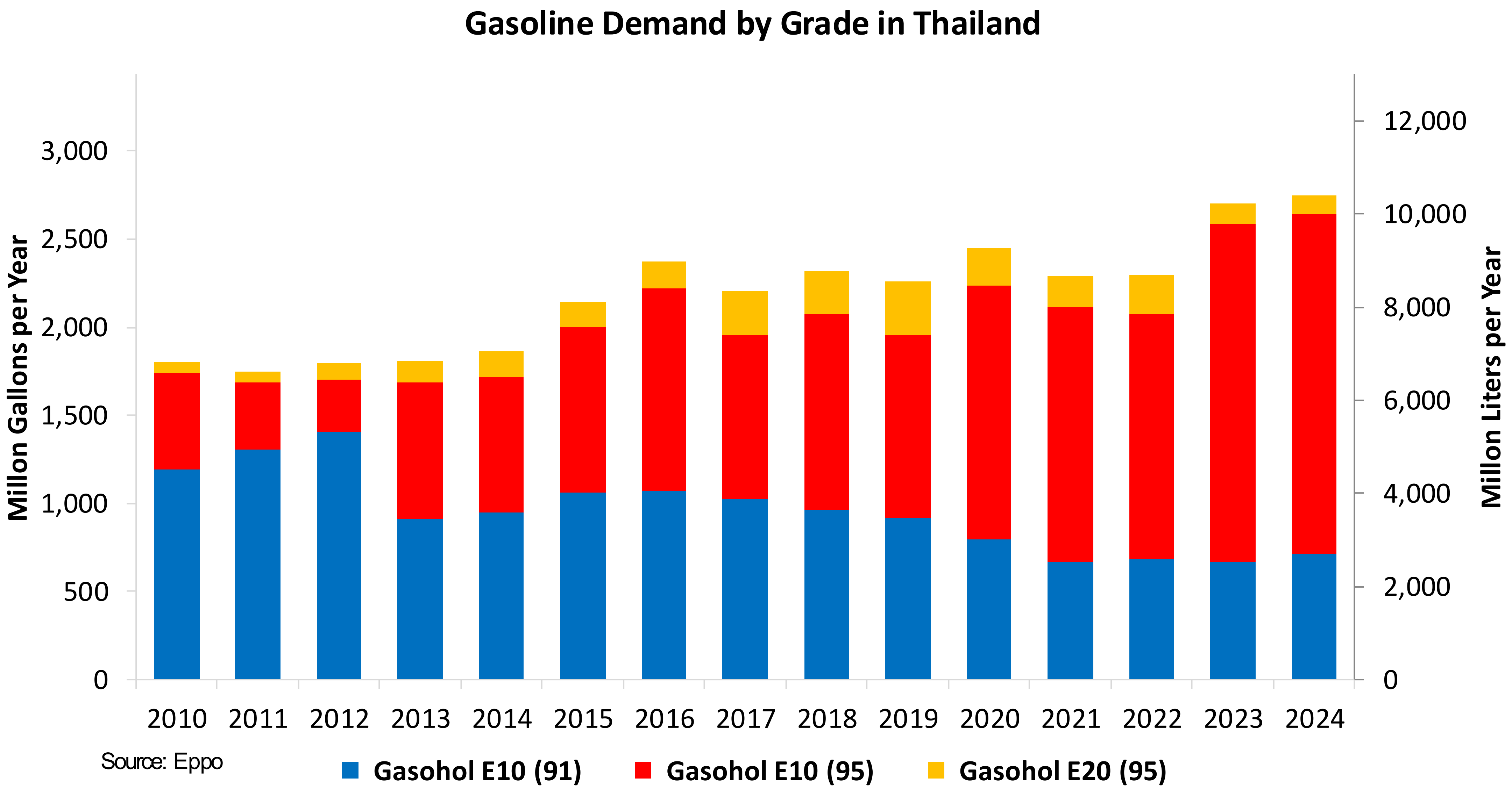
Ethanol Gasoline Blending in Thailand

In 2024, gasoline consumption in Thailand was 11,452 million liters (3,025 million gallons) and growth rate was -0.5%. Gasoline production and net imports were 13,337 and -1,885 million liters (3,523 and -498 million gallons) correspondingly. Thailand offers two main grades: RON 91 (E10) and RON 95 (E10 and E20) with an average octane of 94.1 RON.

Thailand's bioethanol production primarily uses sugarcane and cassava as feedstocks, with approximately 30 facilities producing over 1 billion liters annually. The promotion of bio-based products like ethanol is part of Thailand's bioeconomy policy, supporting both the energy sector and domestic economic growth. In 2024, ethanol consumption supplied by local production in Thailand was 1,289 million liters (341 million gallons) representing 11.3% of gasoline demand.

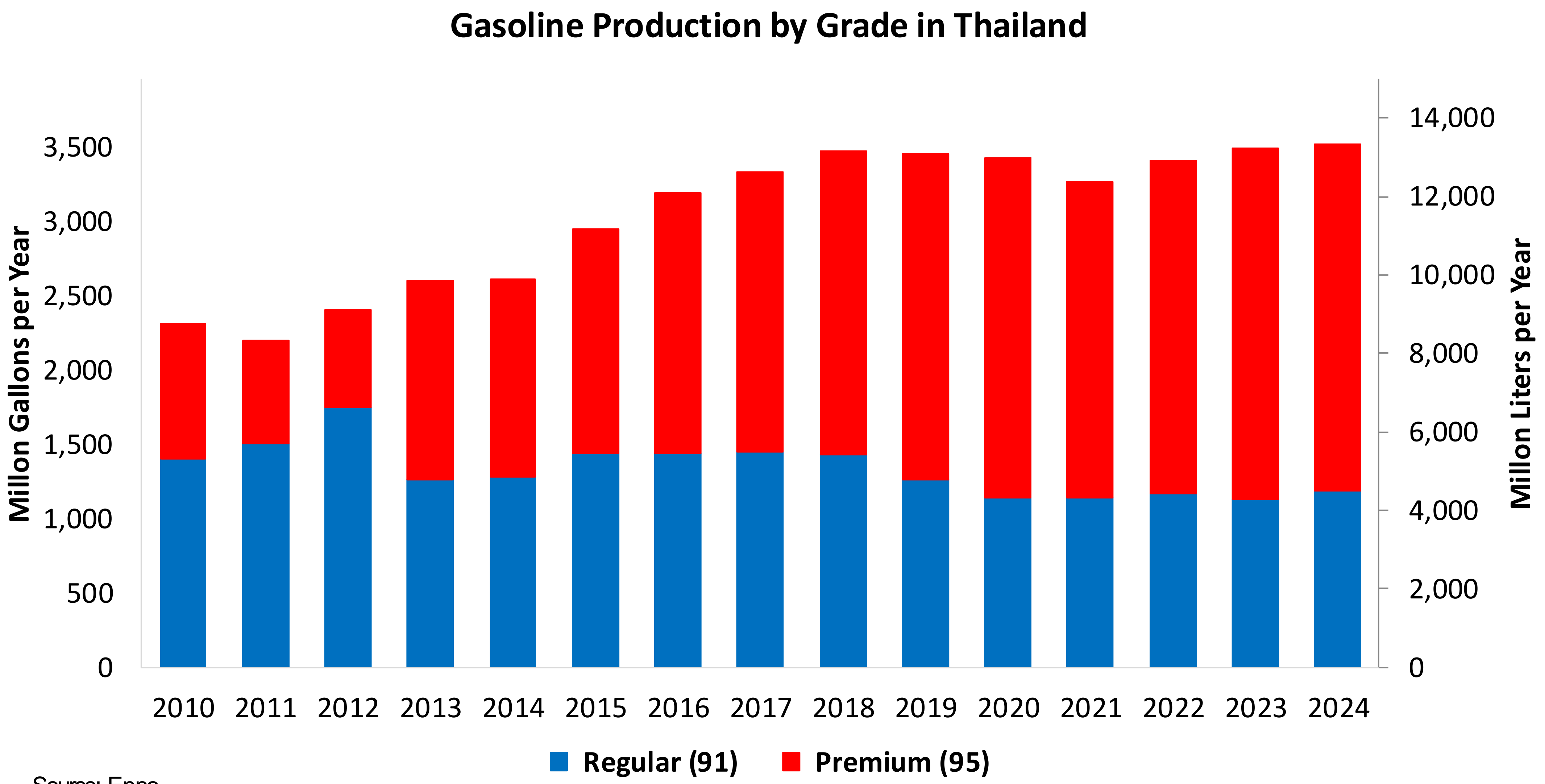
Source: Eppo, USDA, Department of Energy Buisiness

Gasoline Demand in Thailand



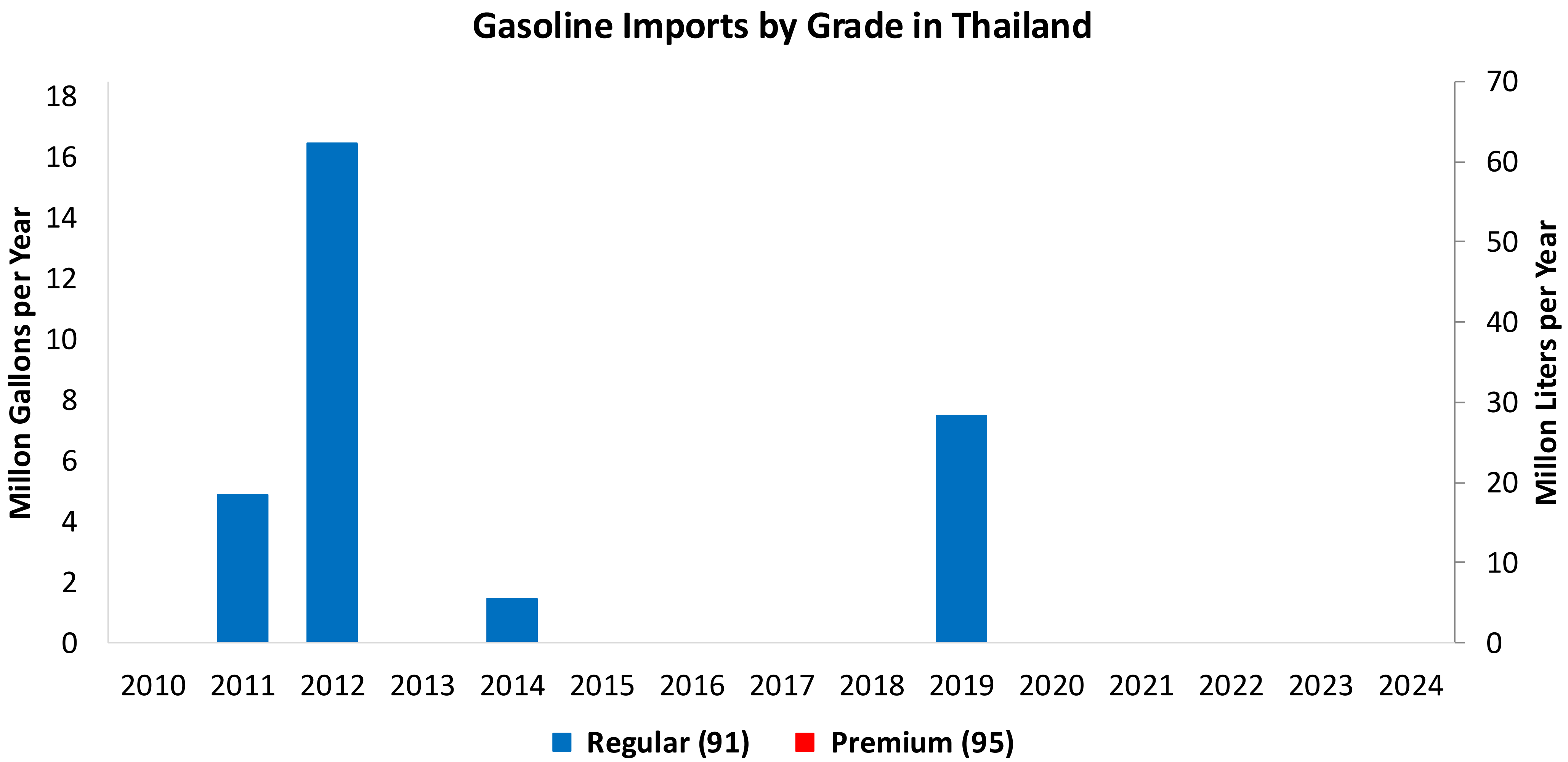
Source: Eppo

Gasoline Production in Thailand



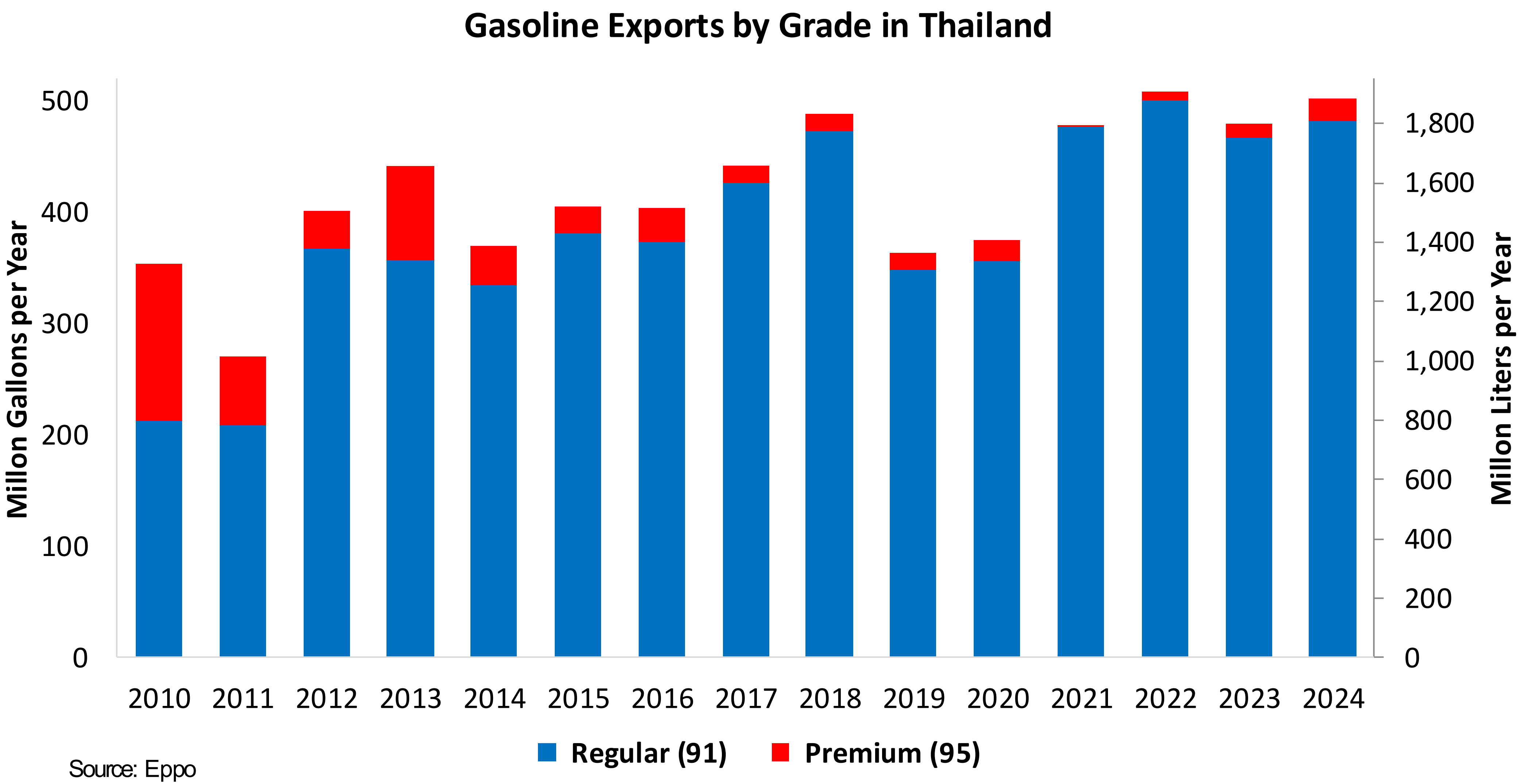
Source: Eppo

Gasoline Imports in Thailand



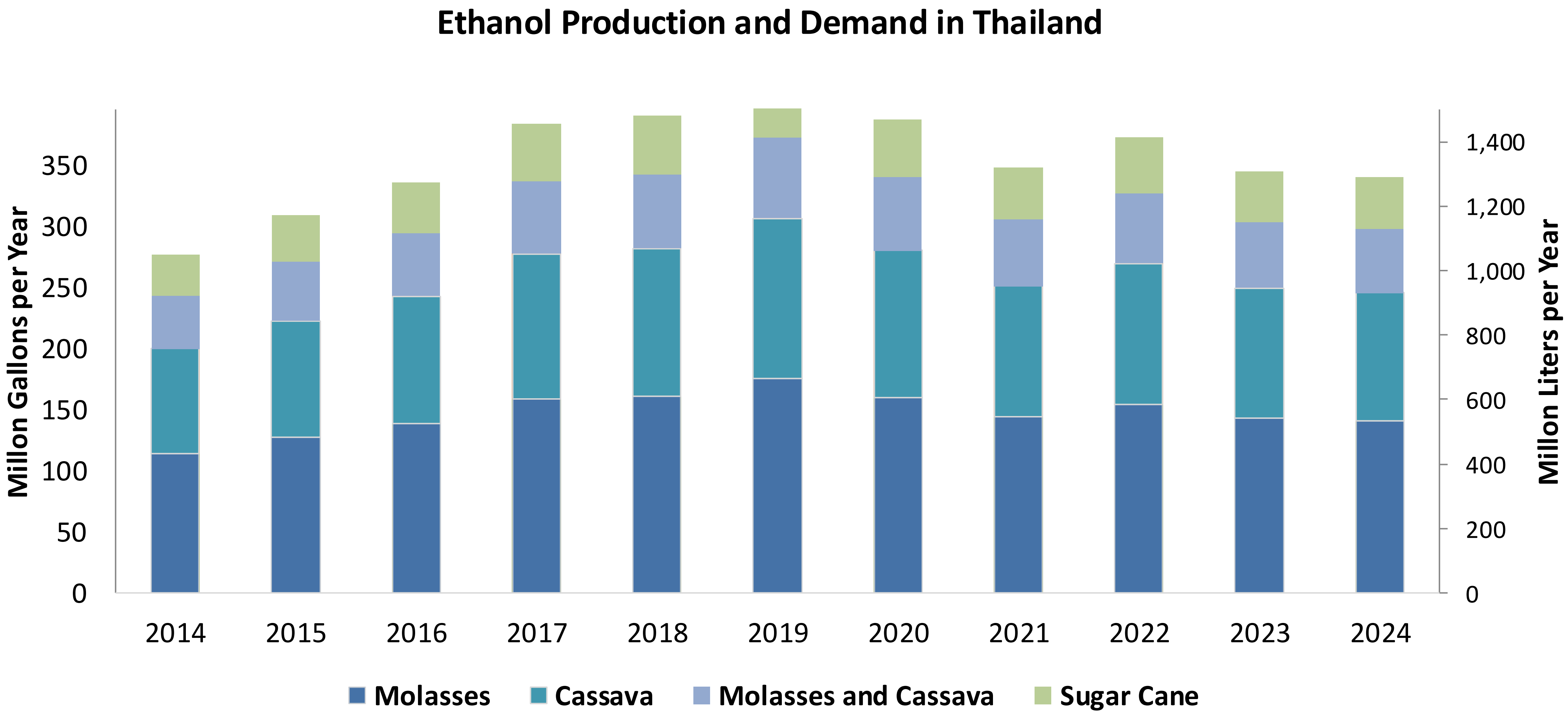
Source: Eppo

Gasoline Exports in Thailand



Source: Eppo

Ethanol Balance in Thailand



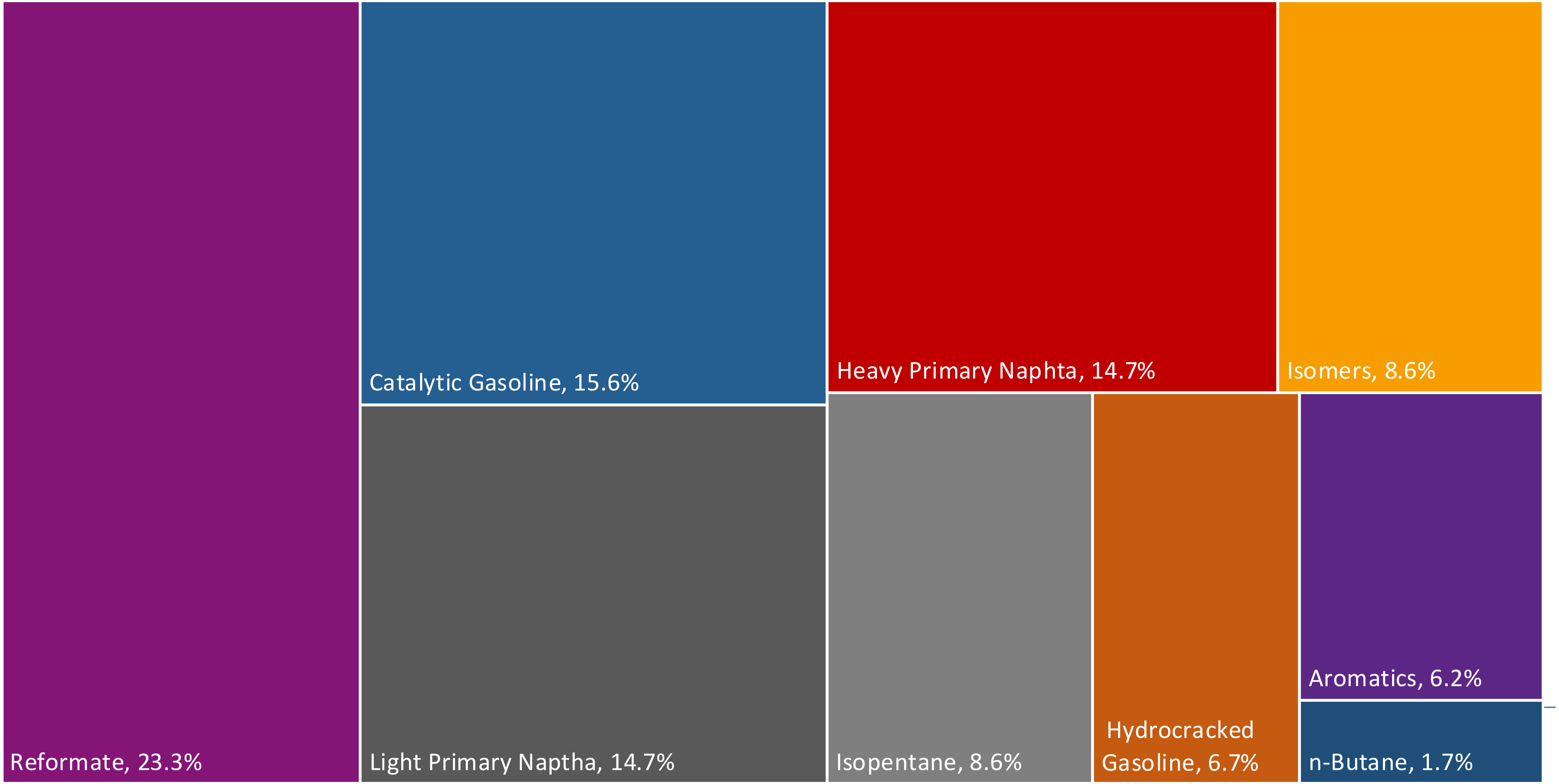
Source: Eppo, USDA

Gasoline Quality in Thailand

Name	B.E. 2557 (2014)					
Implementation Date	2014					
Applicability	Nationwide					
Grade	Gasohol E10 - 91		Gasohol E10 - 95		Gasohol E20	
	Min	Max	Min	Max	Min	Max
RON	91.0		95.0		95.0	
MON	80.0		84.0		84.0	
AKI						
Benzene (%vol)		1.0		1.0		1.0
Aromatics (% vol)		35.0		35.0		35.0
Olefins (%vol)		18.0		18.0		18.0
Lead (g/L)		0.005		0.005		0.005
Phosporous (g/L)		0.013		0.013		0.013
Sulfur (ppm)		50		50		50
Oxygen (%w)						
Ethanol (%vol)		10.0		10.0		20.0
Other Oxygenates	The approval of the Director General of the Department of Energy Business					
PVR 37.8°C (kPa)		62		62		62

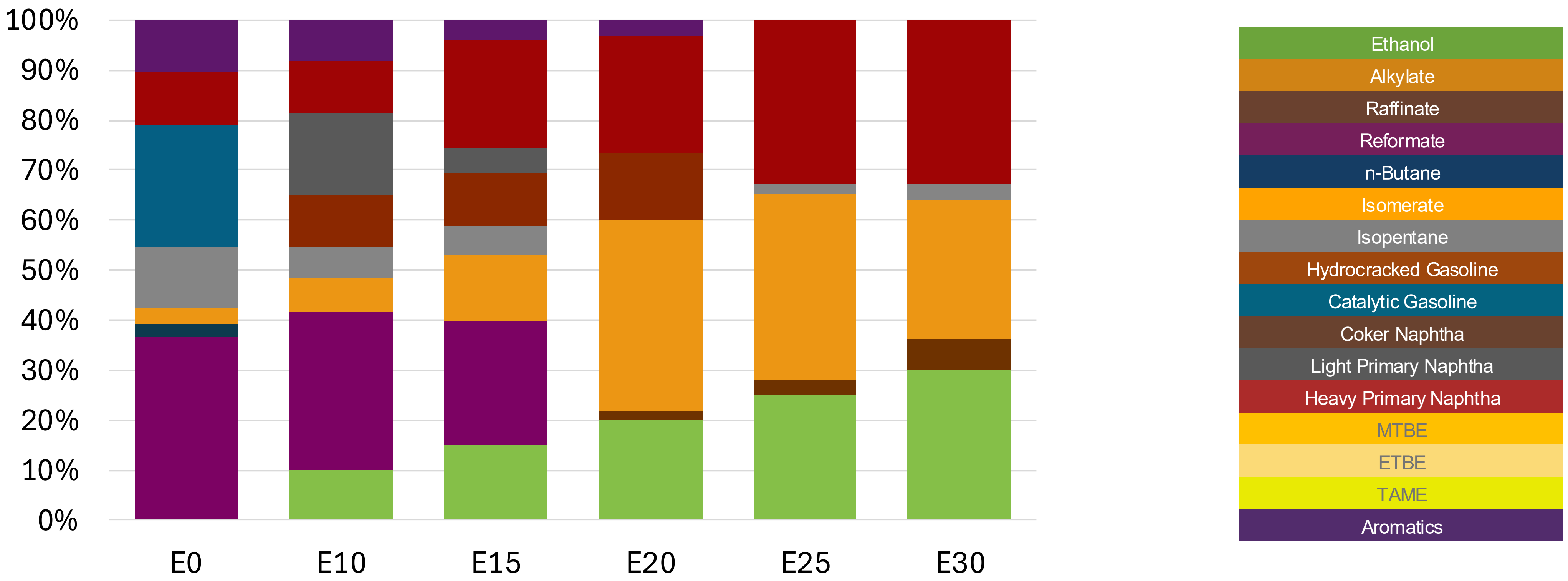
Source: Department of Energy Buisiness

Thailand Available Blending Components



Source: Faro90

Thailand – Gasoline 91 – Constant Octane

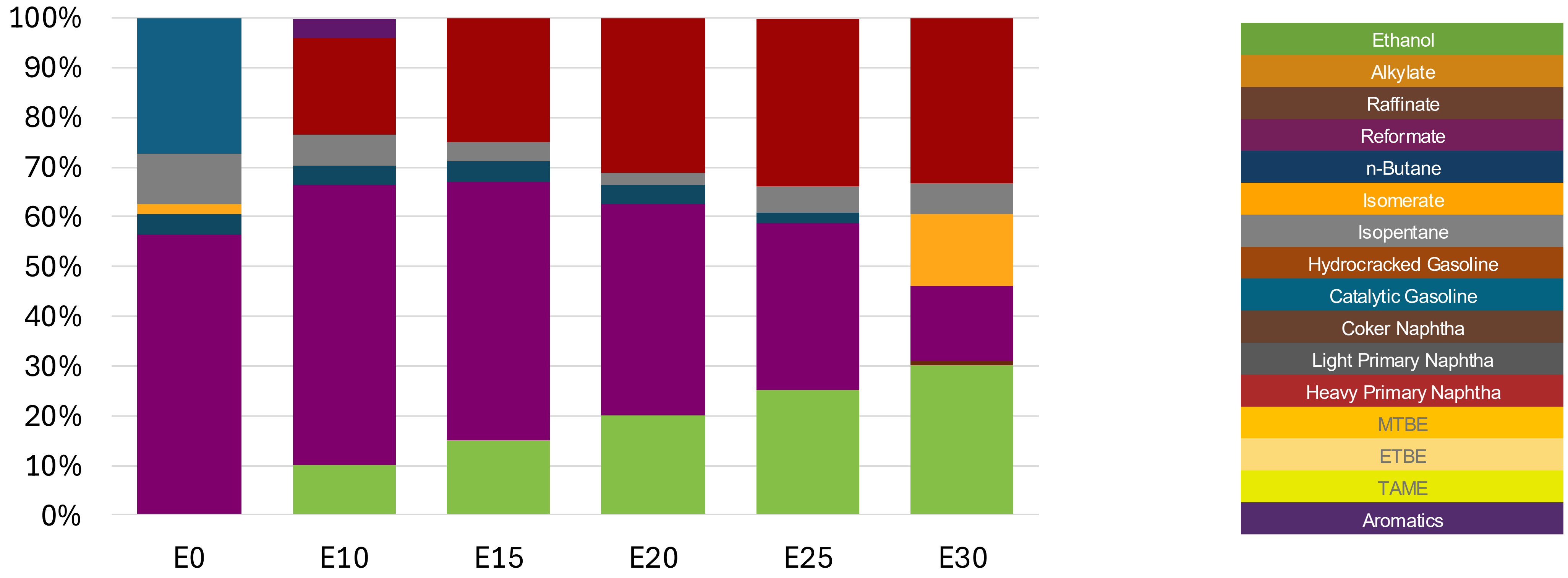


Octane (RON)	91.0	91.0	91.0	91.4	91.6	93.1
Price (US$/gal)	$ 2.256	$ 2.123	$ 2.007	$ 1.947	$ 1.808	$ 1.766

Average CIF 2024 Prices

Source: Faro90

Thailand - Gasoline 95 - Constant Octane

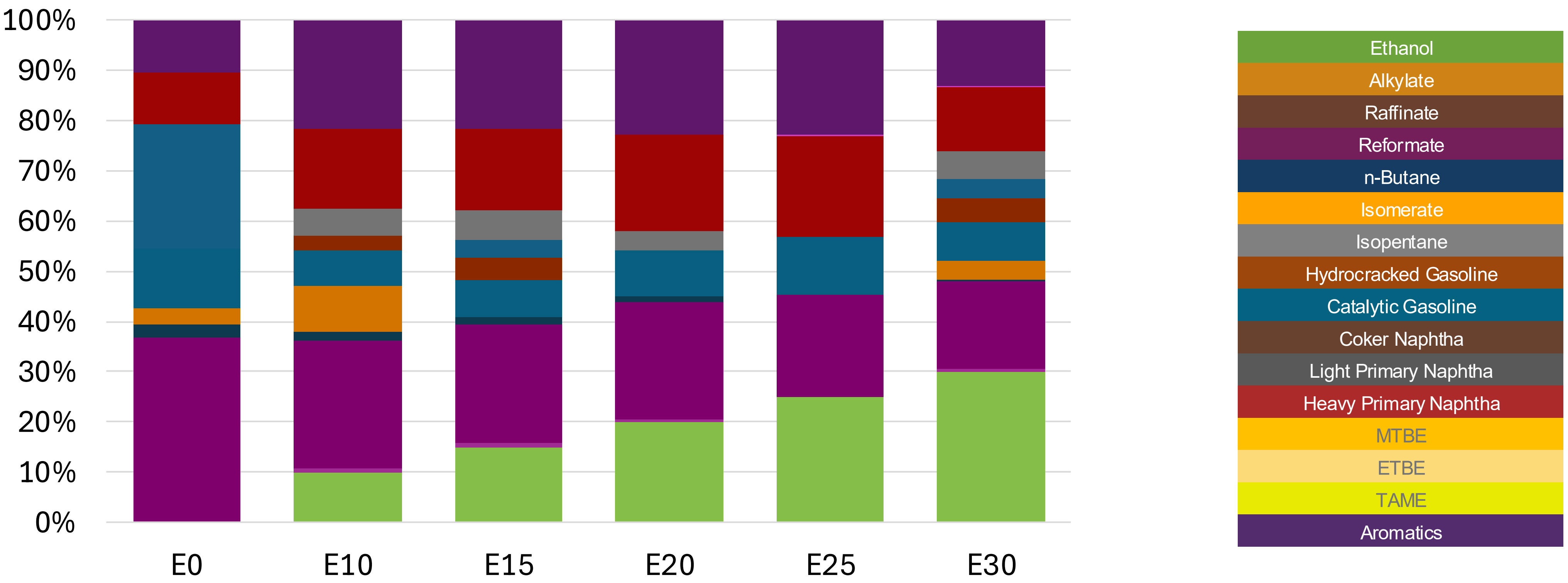


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US$/gal)	$ 2.505	$ 2.276	$ 2.179	$ 2.075	$ 1.968	$ 1.873

Average CIF 2024 Prices

Source: Faro90

Thailand – Gasoline 91 – Increased Octane

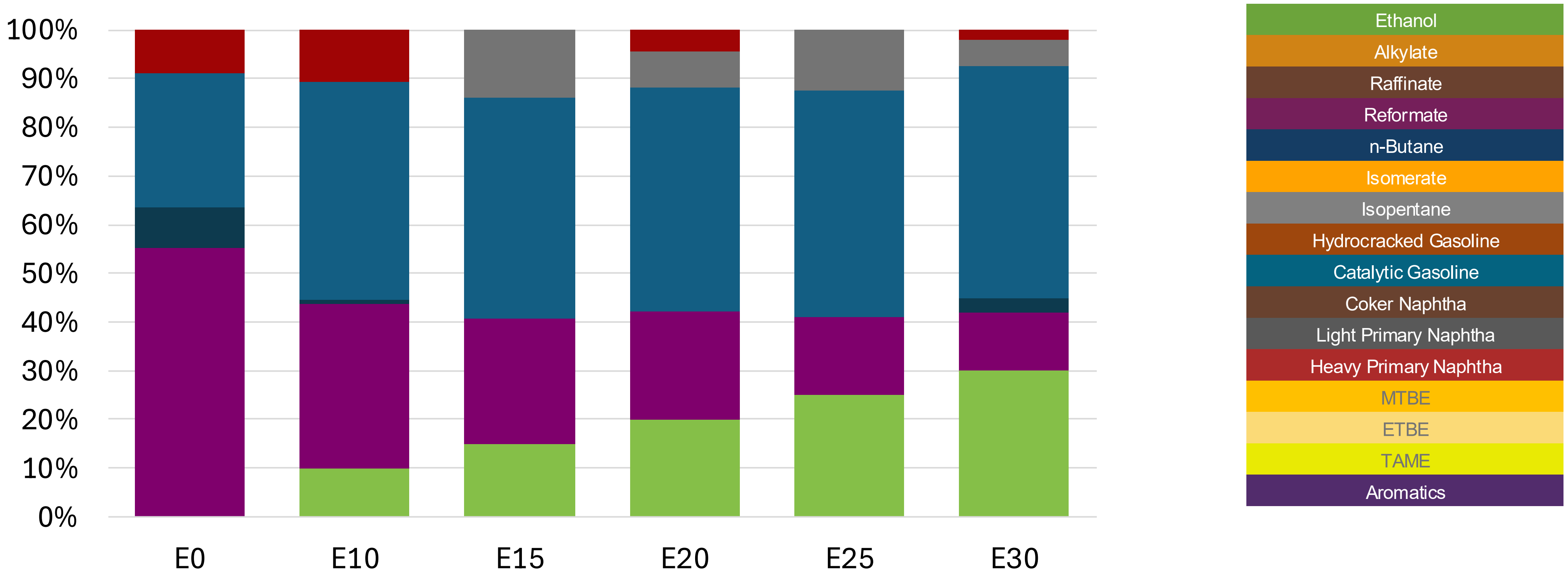


Octane (RON)	91.0	93.8	95.7	97.5	99.2	100.7
Price (US$/gal)	$ 2.256	$ 2.256	$ 2.256	$ 2.256	$ 2.256	$ 2.256

Average CIF 2024 Prices

Source: Faro90

Thailand – Gasoline 95 – Increased Octane



Octane (RON)	95.0	97.9	99.7	101.2	102.8	104.2
Price (US$/gal)	$ 2.505	$ 2.505	$ 2.505	$ 2.505	$ 2.505	$ 2.505

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Thailand

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	2,702
		Single FI	3-Ways		>8 years	120,633
		Multi FI	3-Ways		>8 years	1,065,474
			Euro III		>8 years	910,601
			Euro IV		>8 years	1,904,335
			Euro V		>8 years	1,379,562
			Euro 6		>8 years	2,508,335
					4-8 years	2,572,348
					<4 years	2,313,467
			Hybrid		All	376,495
		Electric			All	103,755
	Medium	Carburator	3-Ways	PCV Tank	>8 years	4,105
		Single FI	3-Ways		>8 years	134,931
		Multi FI	3-Ways		>8 years	1,259,587
			Euro III		>8 years	683,553
			Euro IV		>8 years	1,662,576
			Euro V		>8 years	705,347
			Euro 6		>8 years	1,330,395
					4-8 years	[illegible]
					<4 years	[illegible]
			Hybrid		All	230,643
		Electric			All	63,561
Sml Engines	Light	4-Cycle	None	PCV	>8 years	1,308,247
			Euro III		>8 years	39,712
			Euro IV		>8 years	5,456,632
			Euro V		>8 years	2,984,729
			Euro 6		>8 years	4,375,067
					4-8 years	4,464,037
					<4 years	4,414,259
			Hybrid		All	9,788
		Electric			All	56,423
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	2,938
		Fuel-Injection	3-Ways		>8 years	366,599
			Euro III		>8 years	144,896
			Euro IV		>8 years	355,572
			Euro V		>8 years	191,361
			Euro 6		>8 years	349,214
					4-8 years	293,575
					<4 years	280,065
			Hybrid		All	2
		Electric			All	2,961

30,005,903

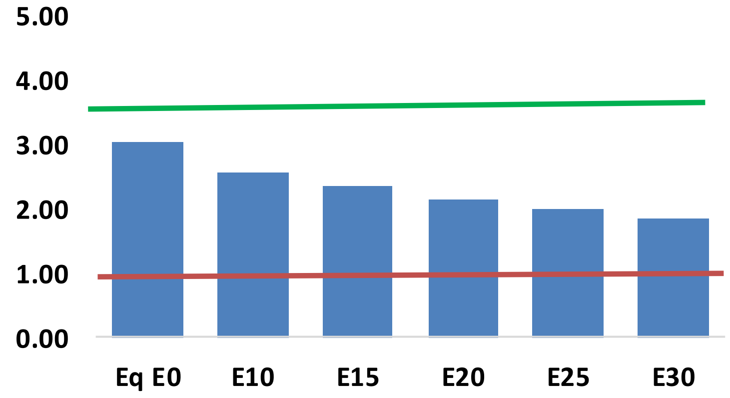
Vehicle Fleet: 46,475,523
Gasoline Vehicle Fleet: 30,005,903

Source: datagov.mot.go.th

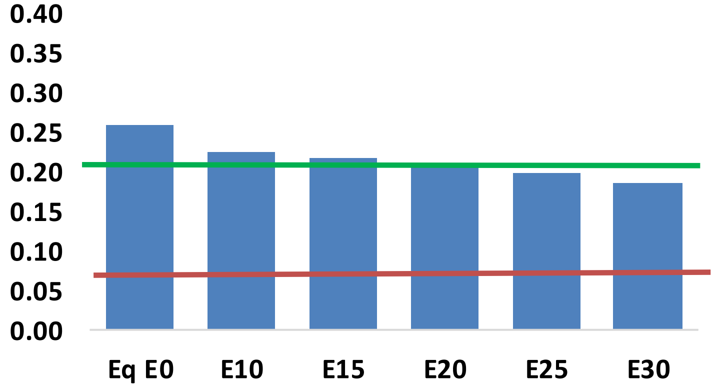
Thailand – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard

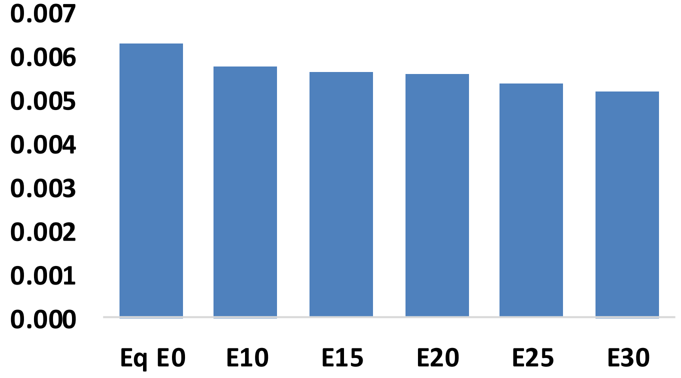
Carbon Monoxide (CO) g/km



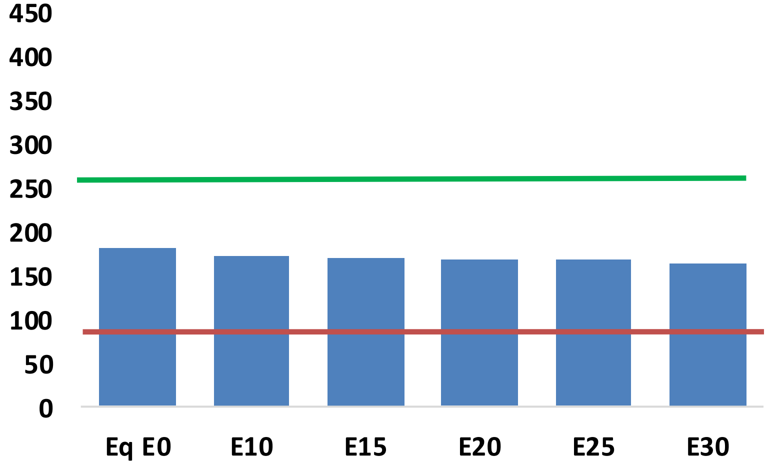
Nitrogen Oxides (NOx) g/km



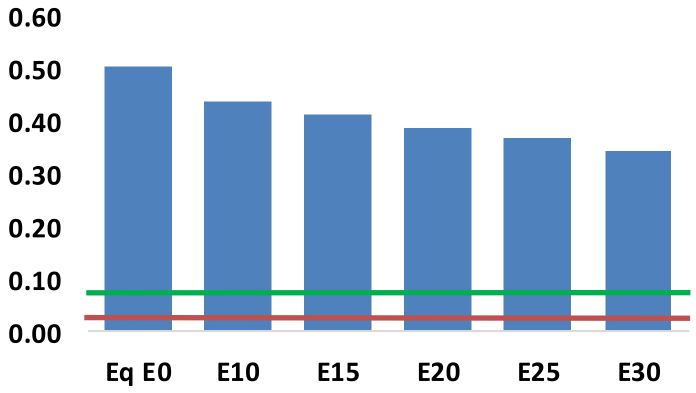
Aromatics (BTX) g/km



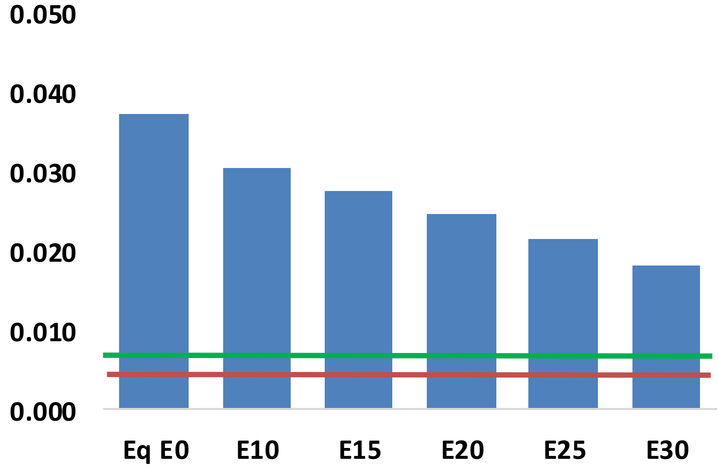
Carbon Dioxide (CO2) g/km



Volatile Organic Compounds (VOC) g/km



Particulate Matter (PM2.5) g/km



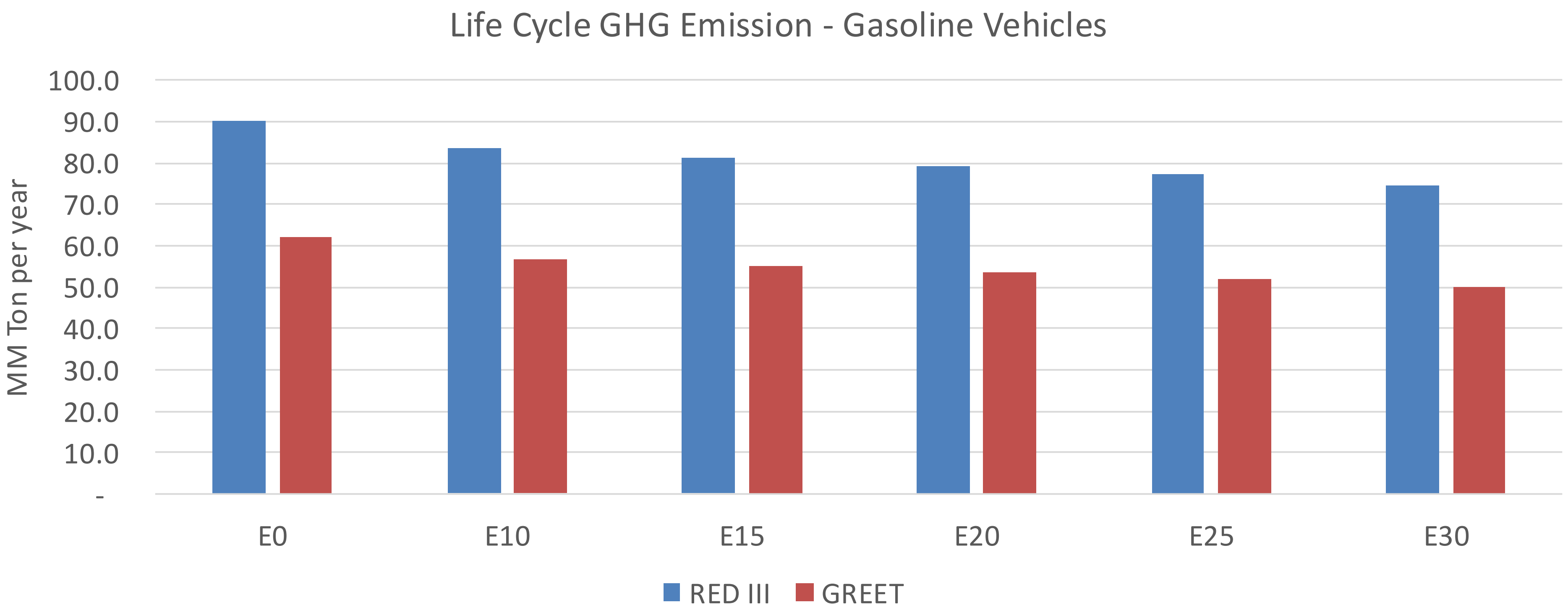
Source: Faro90

Thailand – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,771,402	1,631,957	1,492,513	1,377,607	1,261,681	1,172,451	1,075,545	-16%	-29%
CO2	105,993,888	103,268,331	100,694,194	98,669,711	97,667,917	97,249,062	95,456,482	-5%	-8%
VOC	294,251	275,158	256,065	241,107	226,372	215,137	199,807	-13%	-23%
NOx	150,376	138,848	130,903	126,837	120,308	115,182	109,151	-13%	-20%
SOx	1,202	1,111	1,020	943	870	790	706	-15%	-28%
NH3	40,484	40,083	39,682	39,871	40,319	40,635	40,897	-2%	0%
N2O	1,710	1,701	1,694	1,703	1,738	1,757	1,777	-1%	2%
CH4	70,833	70,833	70,833	71,896	73,454	74,800	76,075	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,944	1,844	1,745	1,673	1,605	1,554	1,482	-10%	-17%
Acetaldehyde	5,608	6,437	9,444	14,383	19,595	22,391	26,457	68%	249%
Formaldehyde	21,725	21,121	24,621	28,911	30,288	33,507	36,476	13%	39%
Benzene	3,690	3,546	3,360	3,306	3,279	3,136	3,035	-9%	-11%
PM2.5	6,597	5,998	5,398	4,870	4,354	3,800	3,221	-18%	-34%
PM10	15,141	13,765	12,388	11,177	9,992	8,721	7,392	-18%	-34%

Source: Faro90

Thailand – Gasoline Vehicle Fleet Life Cycle GHG Emissions



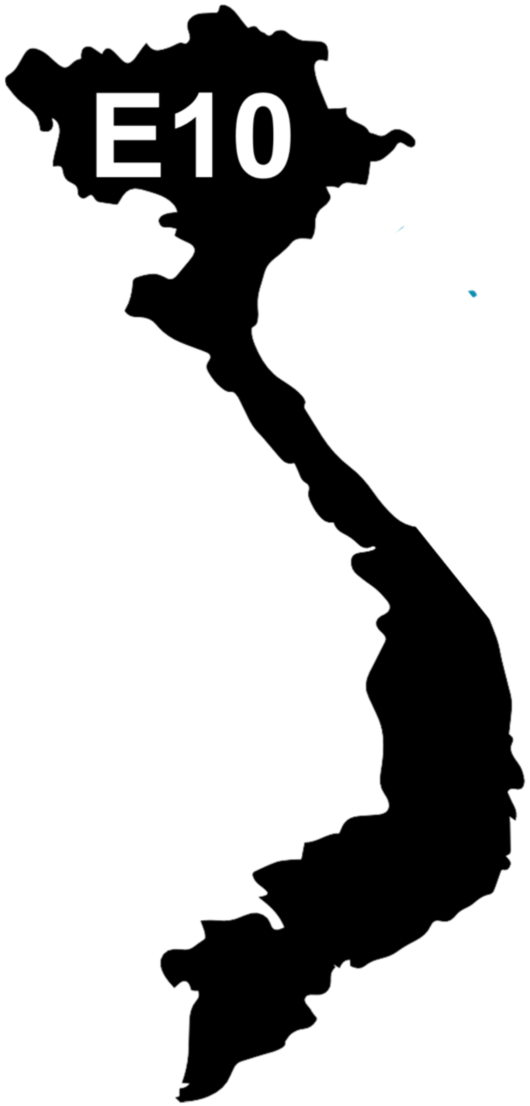
Country Emissions 2024 (MMT/año)	356.2
Target @ 2035 (MMT/year)	211.6
Delta	144.7

vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.3%	11.1%	13.8%	15.9%	19.1%
RED II/III	0.0%	7.2%	9.8%	12.3%	14.4%	17.2%

Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.6%	4.8%	5.9%	6.8%	8.2%
RED II/III	0.0%	4.5%	6.1%	7.7%	9.0%	10.7%

Source: Greet, RED II/III, climateactiontracker.org, F90

Ethanol Gasoline Blending in Vietnam

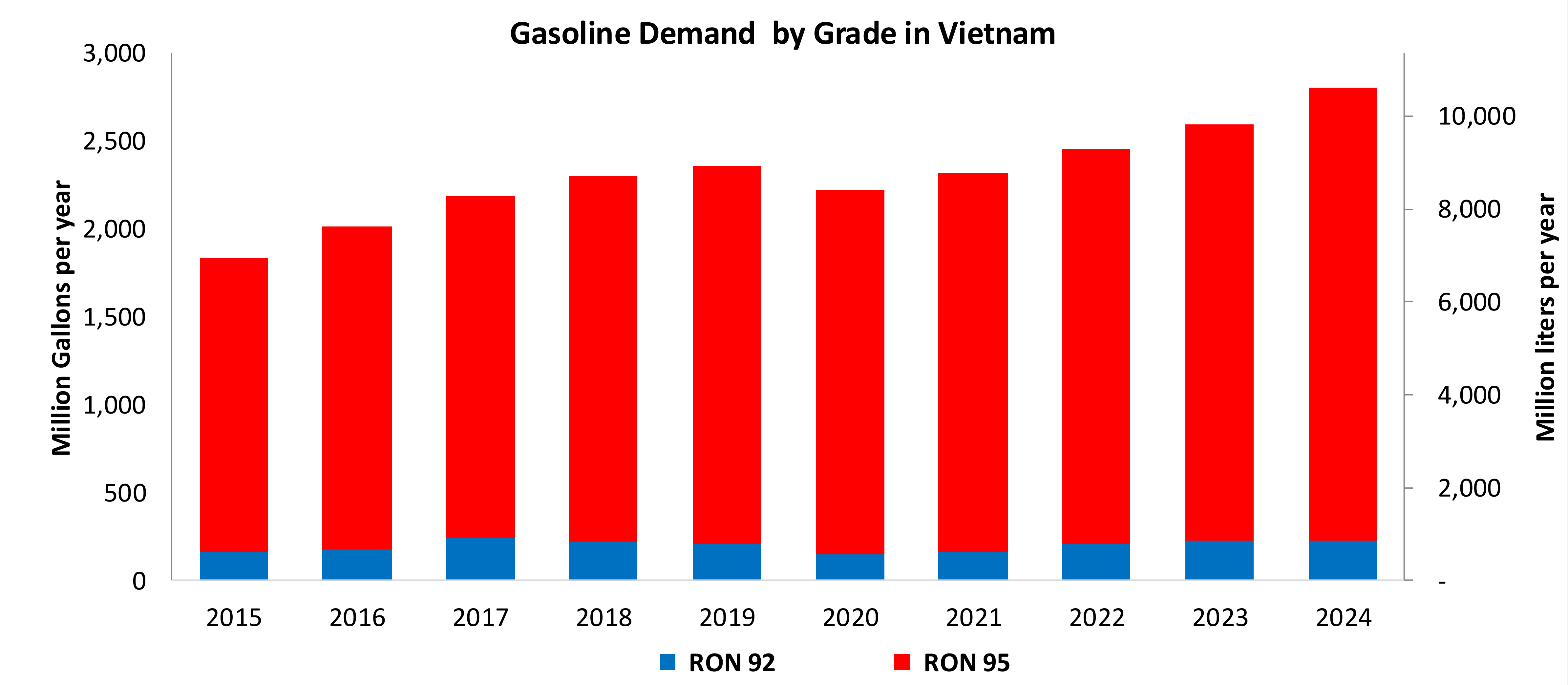


In 2024, gasoline consumption in Vietnam was 10,600 million liters (2,800 million gallons) and growth rate was 3.8%. Gasoline production and net imports were 7,870 and 2,730 million liters (2,079 and 721 million gallons) correspondingly. Vietnam complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 94.8 RON.

Vietnam's bioethanol demand is growing rapidly, driven by government policies like a planned nationwide E10 gasoline mandate set to begin in 2026. Ethanol consumption in Vietnam was 236 million liters (62 million gallons) representing 2.5% of gasoline demand. Ethanol production and Net Imports were 200 and 36 million liters (62 and 9 million gallons) correspondingly.

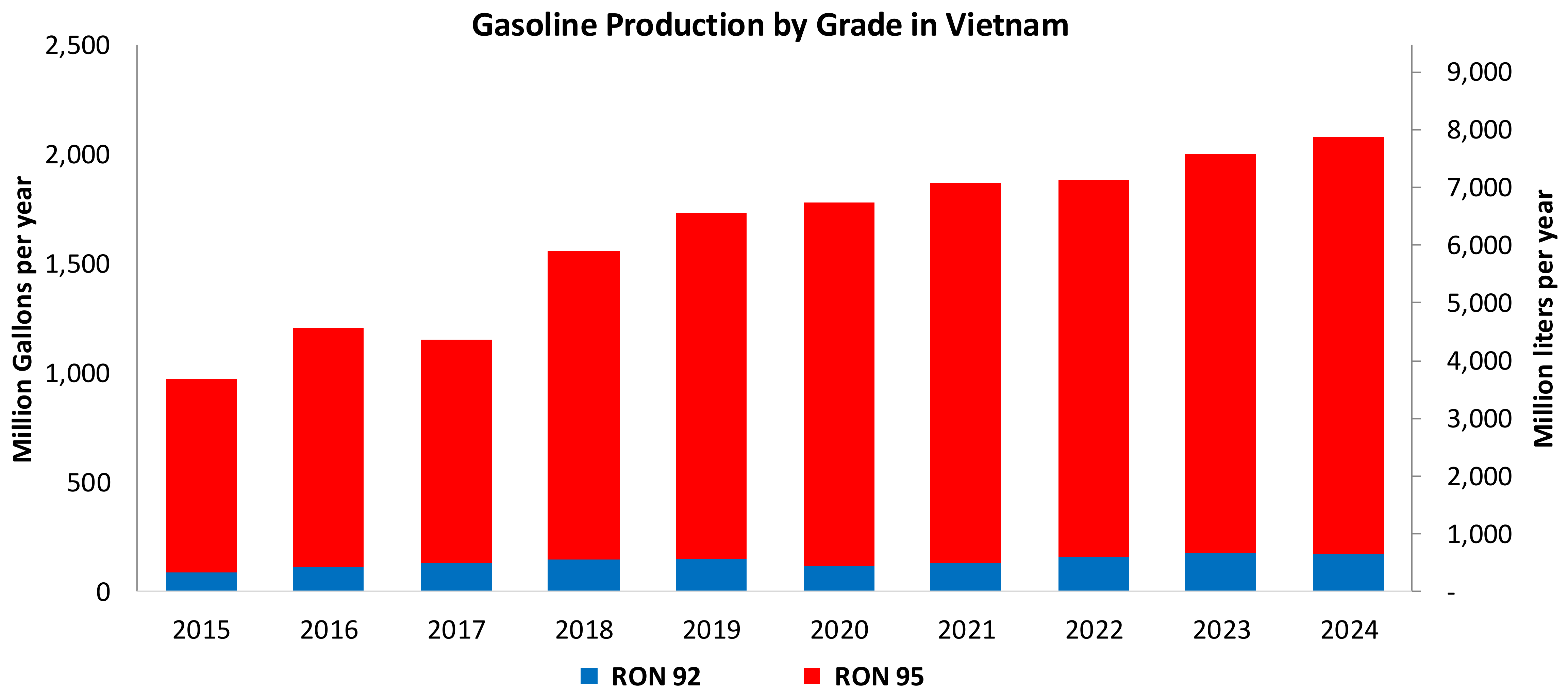
Source: Petro Vietnam, EIA,, EITdata, USDA

Gasoline Demand in Vietnam



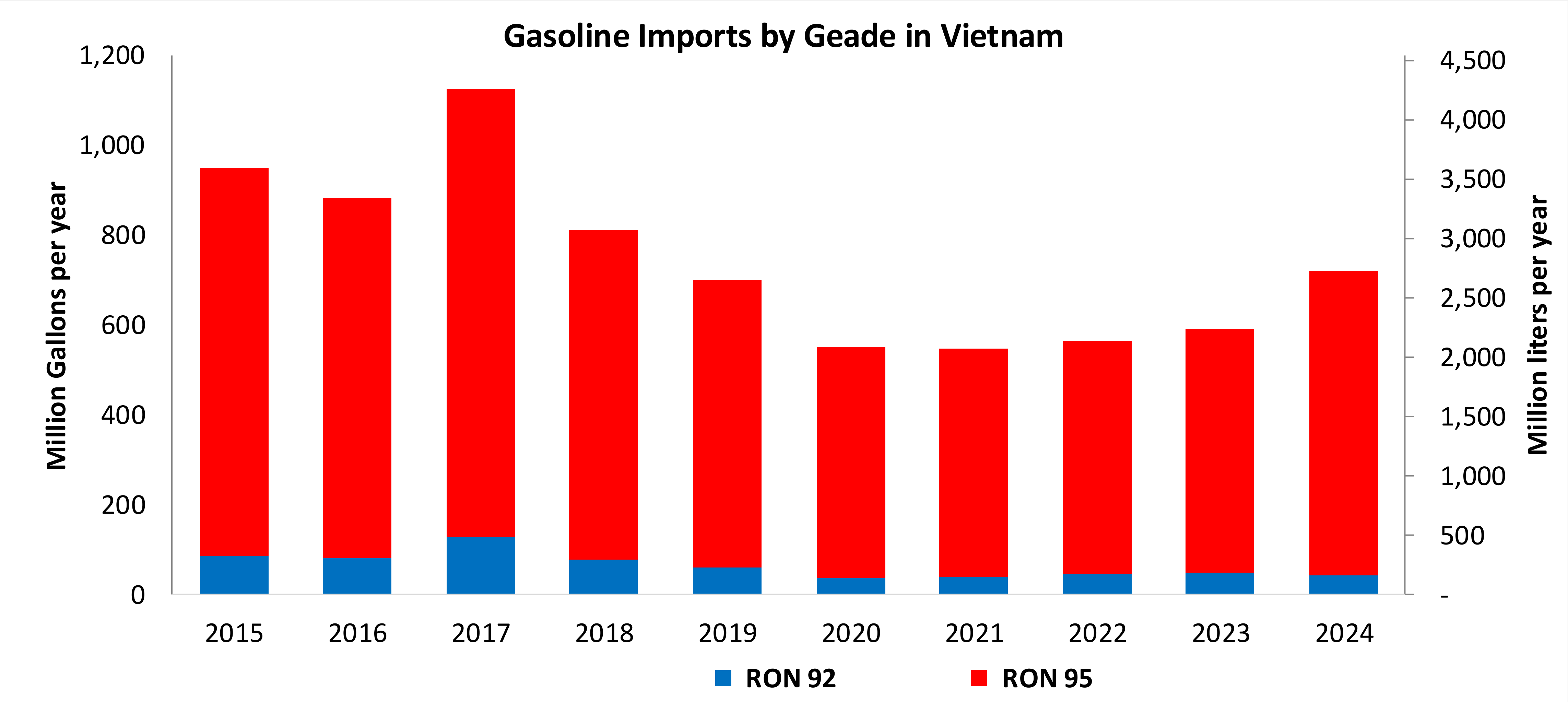
Source: Petro Vietnam, EIA,, EITdata

Gasoline Production in Vietnam



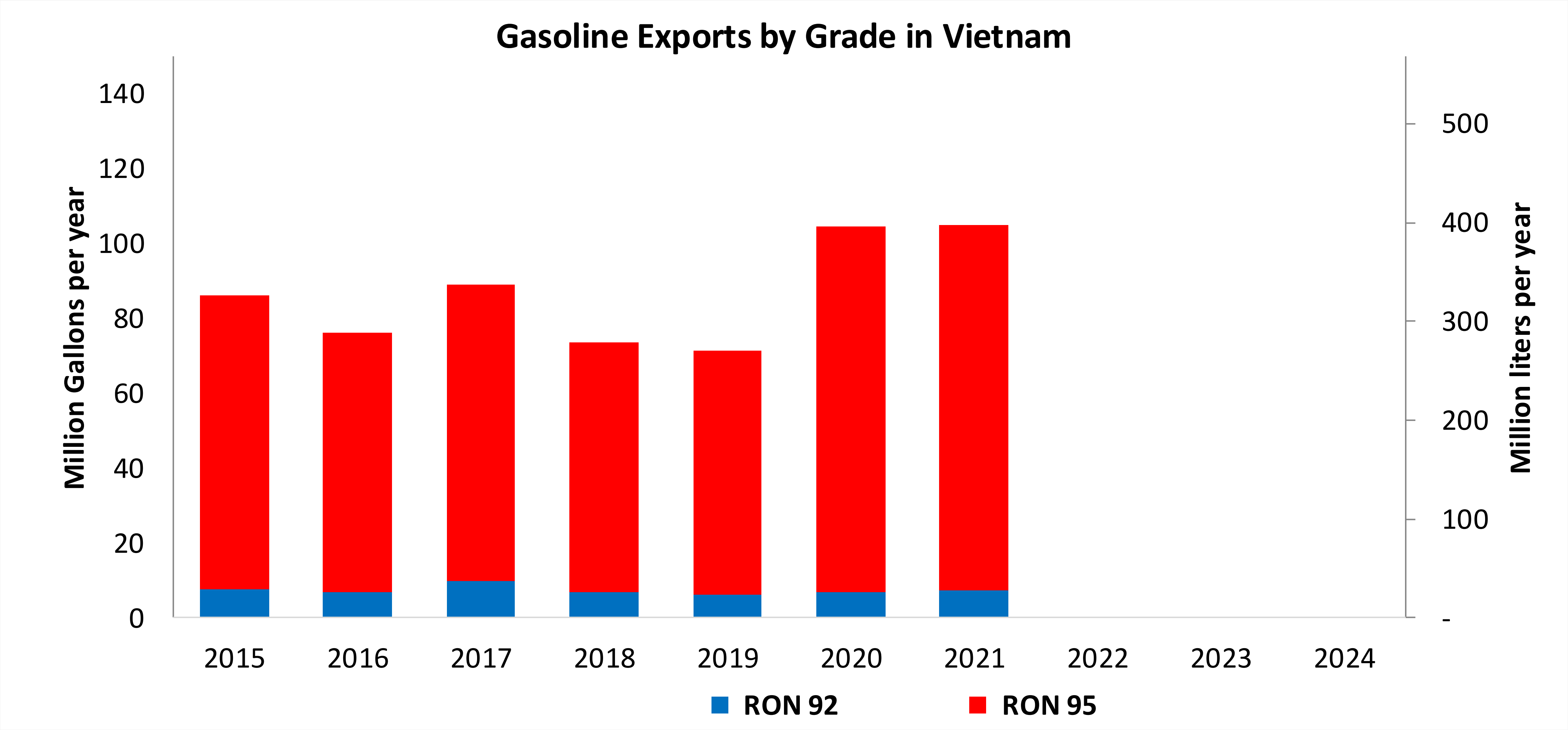
Source: Petro Vietnam, EIA,, EITdata

Gasoline Imports in Vietnam



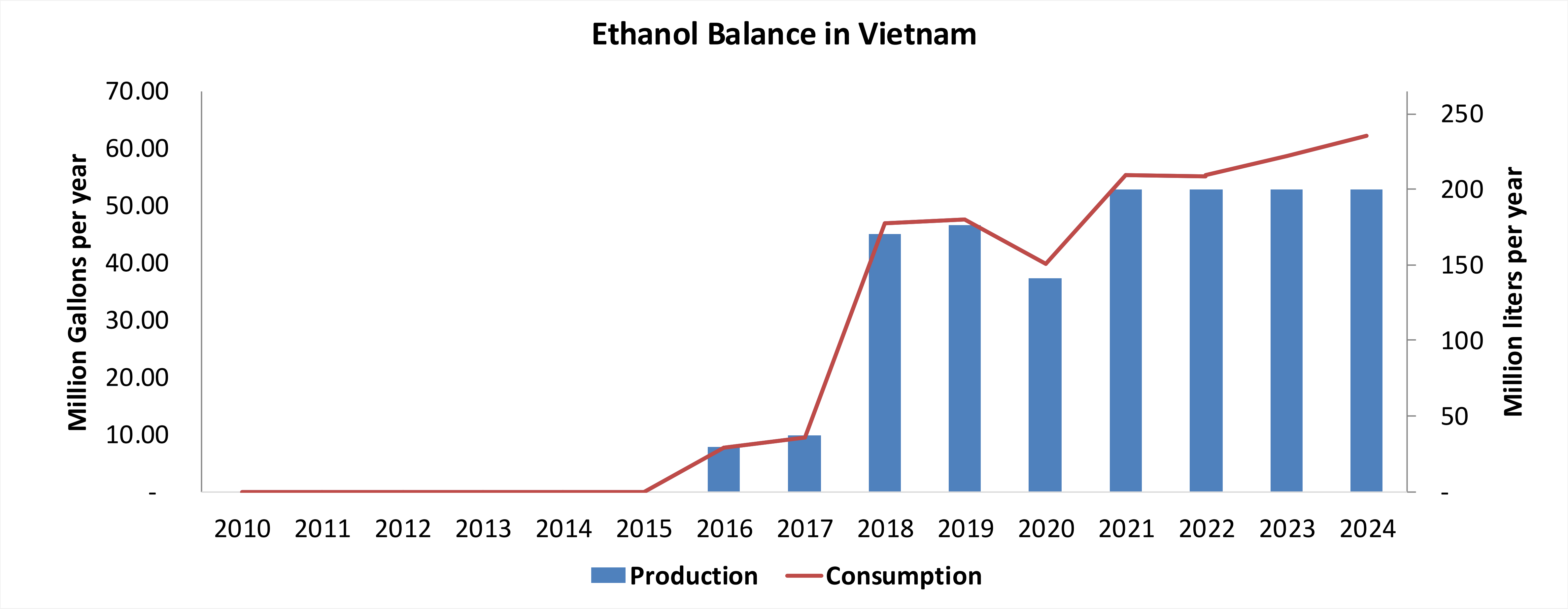
Source: Petro Vietnam, EIA,, EITdata

Gasoline Exports in Vietnam



Source: Petro Vietnam, EIA,, EITdata

Ethanol Balance in Vietnam



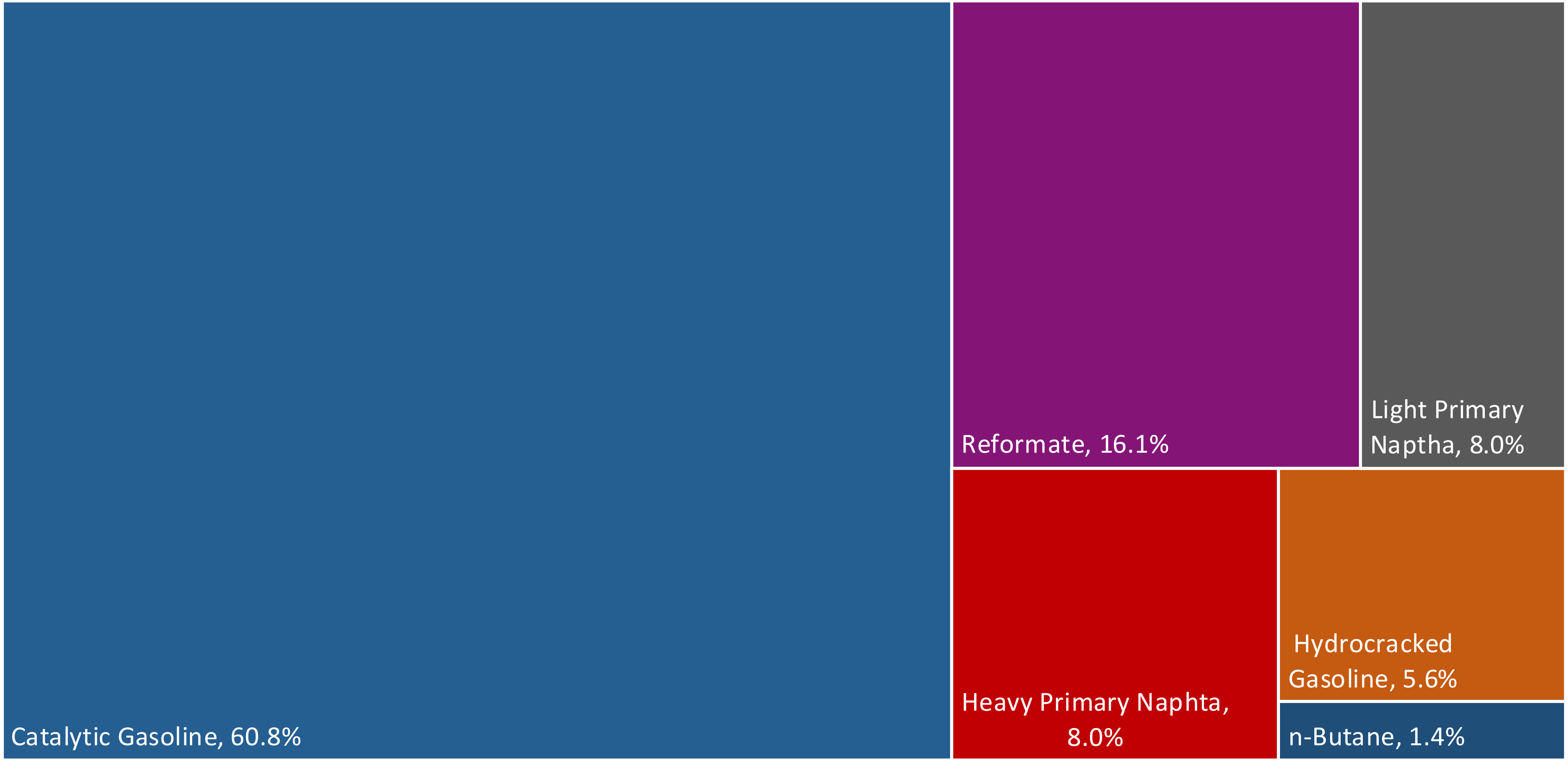
Source: Petro Vietnam, USDA

Gasoline Quality in Vietnam

Name	QCVN 01:2022/BKHCN					
Implementation Year	2023					
Applicability	All territory					
Grade	RON 92		RON 95		RON 97	
	Min	Max	Min	Max	Min	Max
RON	92.0		95.0		97.0	
MON	-		-		-	
AKI	-		-		-	
Benzene (%vol)		1.0		1.0		1.0
Aromatics (% vol)		35.0		35.0		35.0
Olefins (%vol)		18.0		18.0		18.0
Lead (g/L)		0.005		0.005		0.005
Manganese (mg/L)		5.0		5.0		5.0
Sulfur (ppm)		10		10		10
Oxygen (%w)		2.7		2.7		2.7
Ethanol (%vol)		-		-		-
MTBE (%vol)		-		-		-
RVP (psi)		8.7		8.7		8.7

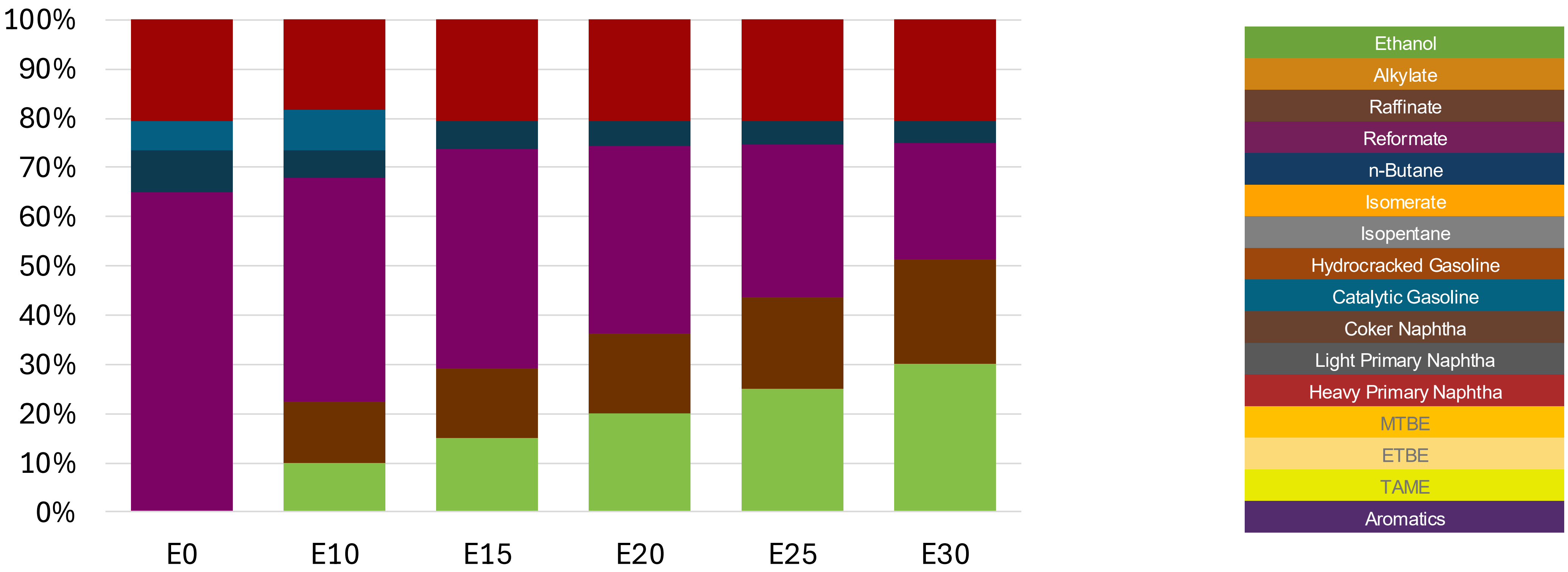
Source: Caselaw

Vietnam Available Blending Components



Source: Faro90

Vietnam – Gasoline 92 – Constant Octane

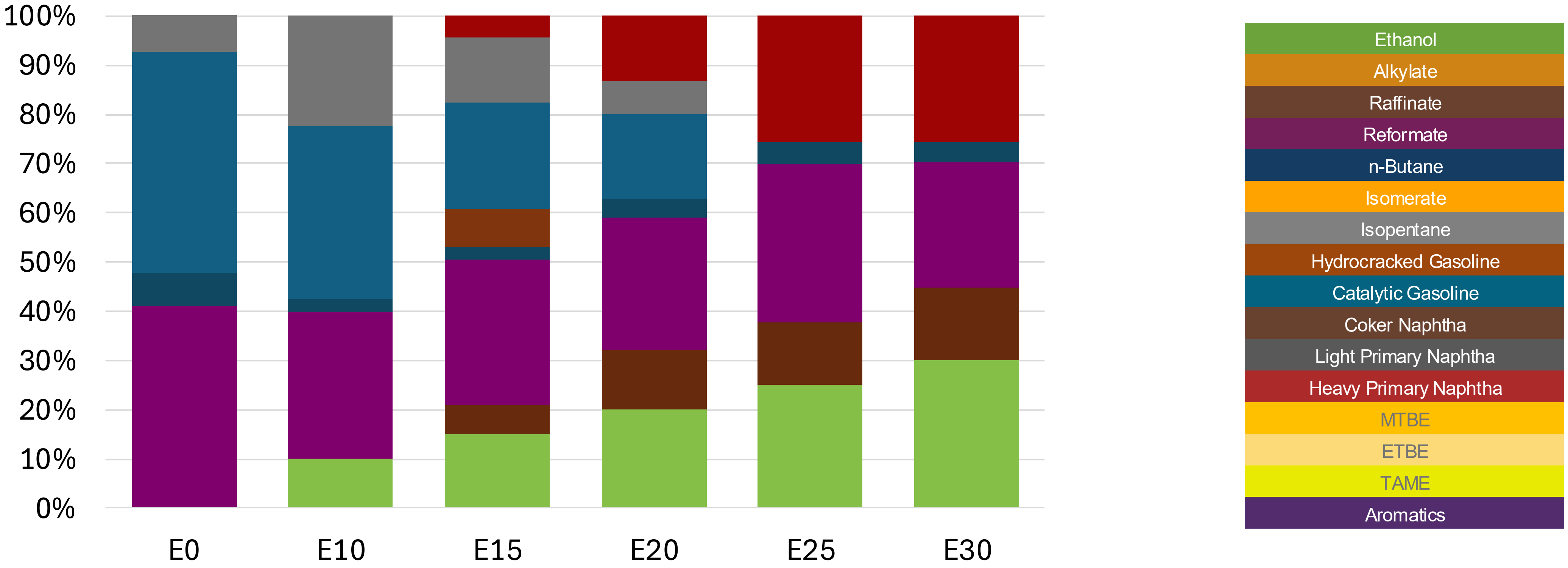


Octane (RON)	92.0	92.0	92.9	94.0	95.0	96.0
Price (US$/gal)	$ 2.280	$ 2.113	$ 2.063	$ 2.020	$ 1.972	$ 1.921

Average CIF 2024 Prices

Source: Faro90

Vietnam – Gasoline 95 – Constant Octane

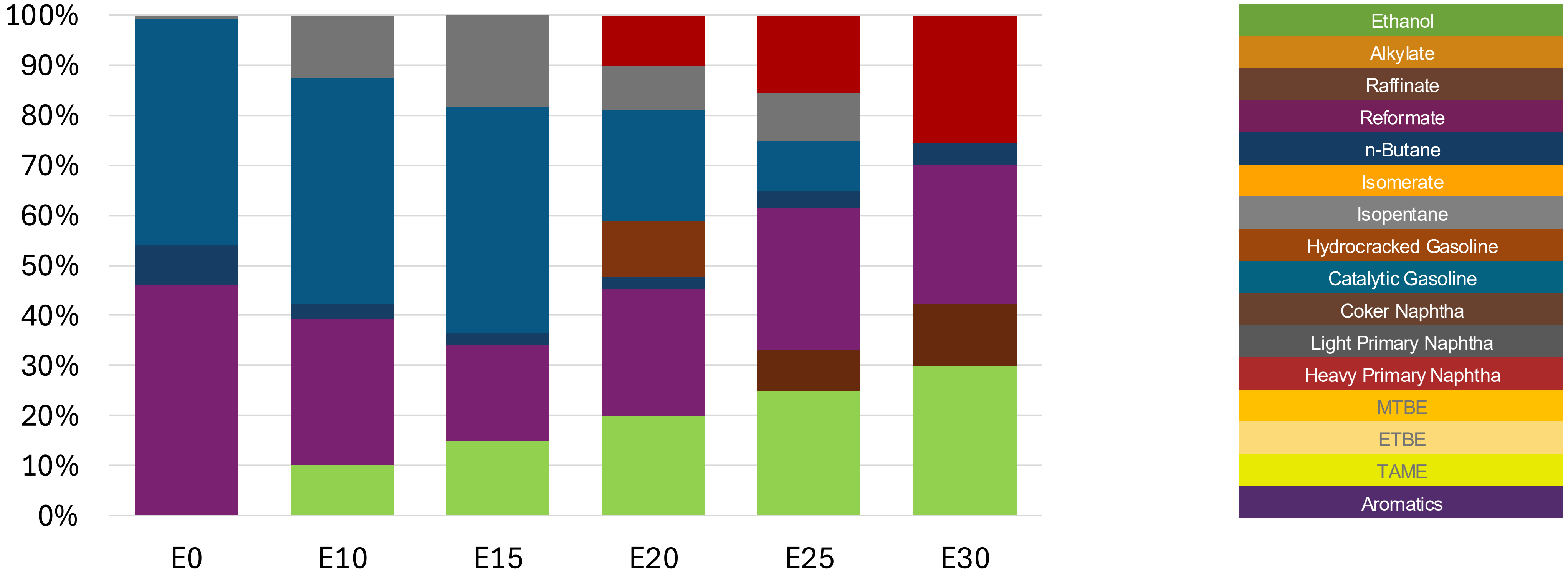


Octane (RON)	95.0	95.0	95.0	95.0	95.2	96.3
Price (US$/gal)	$ 2.436	$ 2.305	$ 2.214	$ 2.090	$ 1.981	$ 1.933

Average CIF 2024 Prices

Source: Faro90

Vietnam - Gasoline 97 - Constant Octane

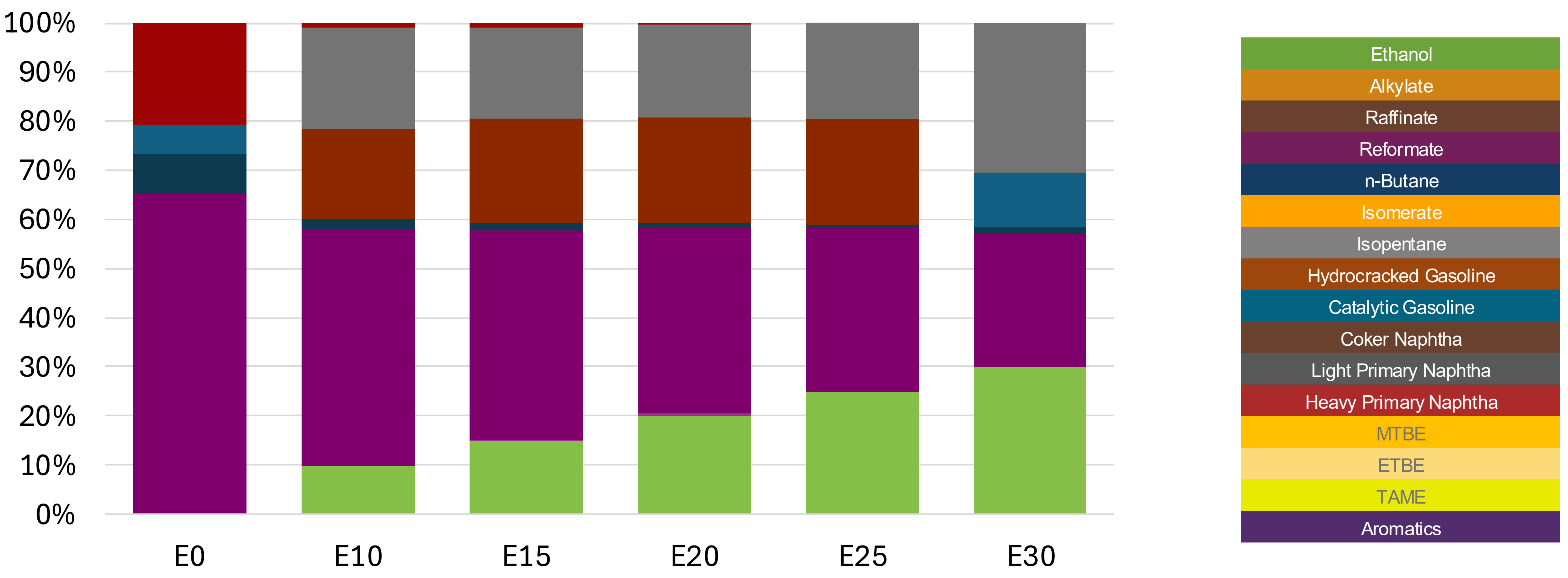


Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US$/gal)	$ 2.542	$ 2.416	$ 2.322	$ 2.228	$ 2.105	$ 1.983

Average CIF 2024 Prices

Source: Faro90

Vietnam – Gasoline 92 – Increased Octane

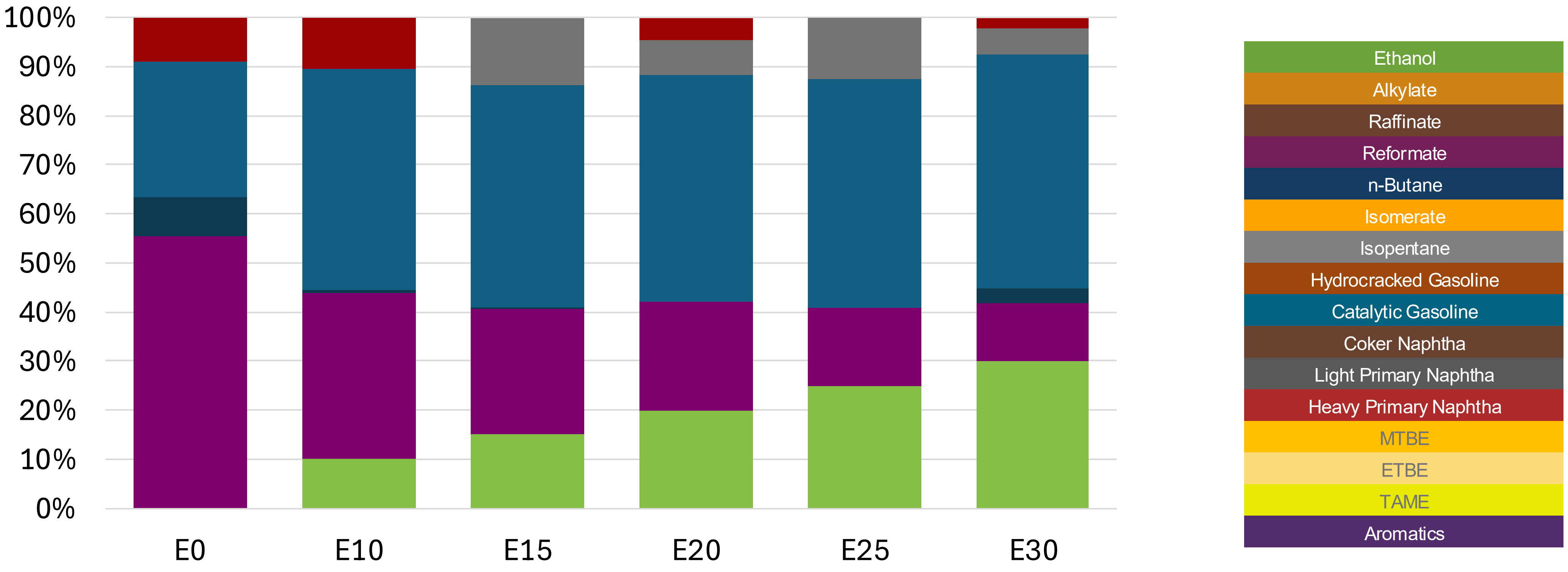


Octane (RON)	92.0	94.1	95.7	97.4	99.0	101.0
Price (US$/gal)	$ 2.280	$ 2.280	$ 2.280	$ 2.280	$ 2.280	$ 2.280

Average CIF 2024 Prices

Source: Faro90

Vietnam – Gasoline 95 – Increased Octane

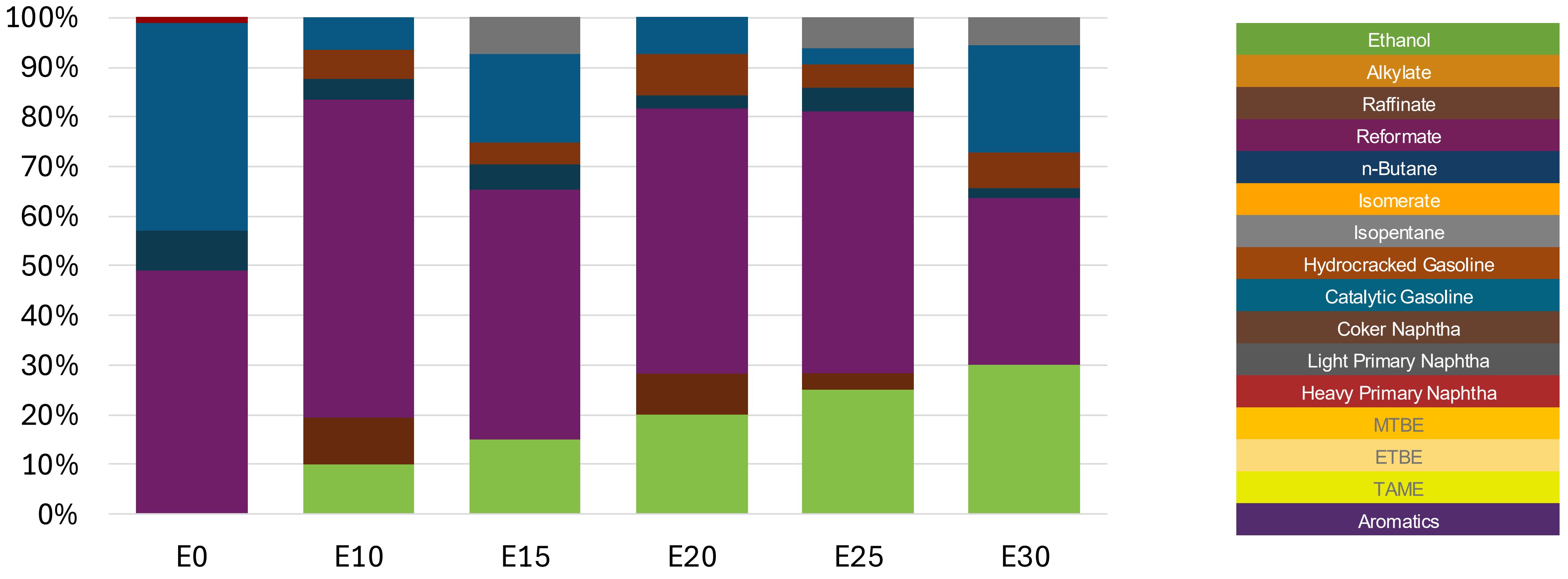


Octane (RON)	95.0	96.8	98.4	100.0	101.6	103.7
Price (US$/gal)	$ 2.436	$ 2.436	$ 2.436	$ 2.436	$ 2.436	$ 2.436

Average CIF 2024 Prices

Source: Faro90

Vietnam – Gasoline 97 – Increased Octane



Octane (RON)	97.0	99.1	101.1	101.9	103.9	104.9
Price (US$/gal)	$ 2.541	$ 2.541	$ 2.541	$ 2.541	$ 2.541	$ 2.541

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Vietnam

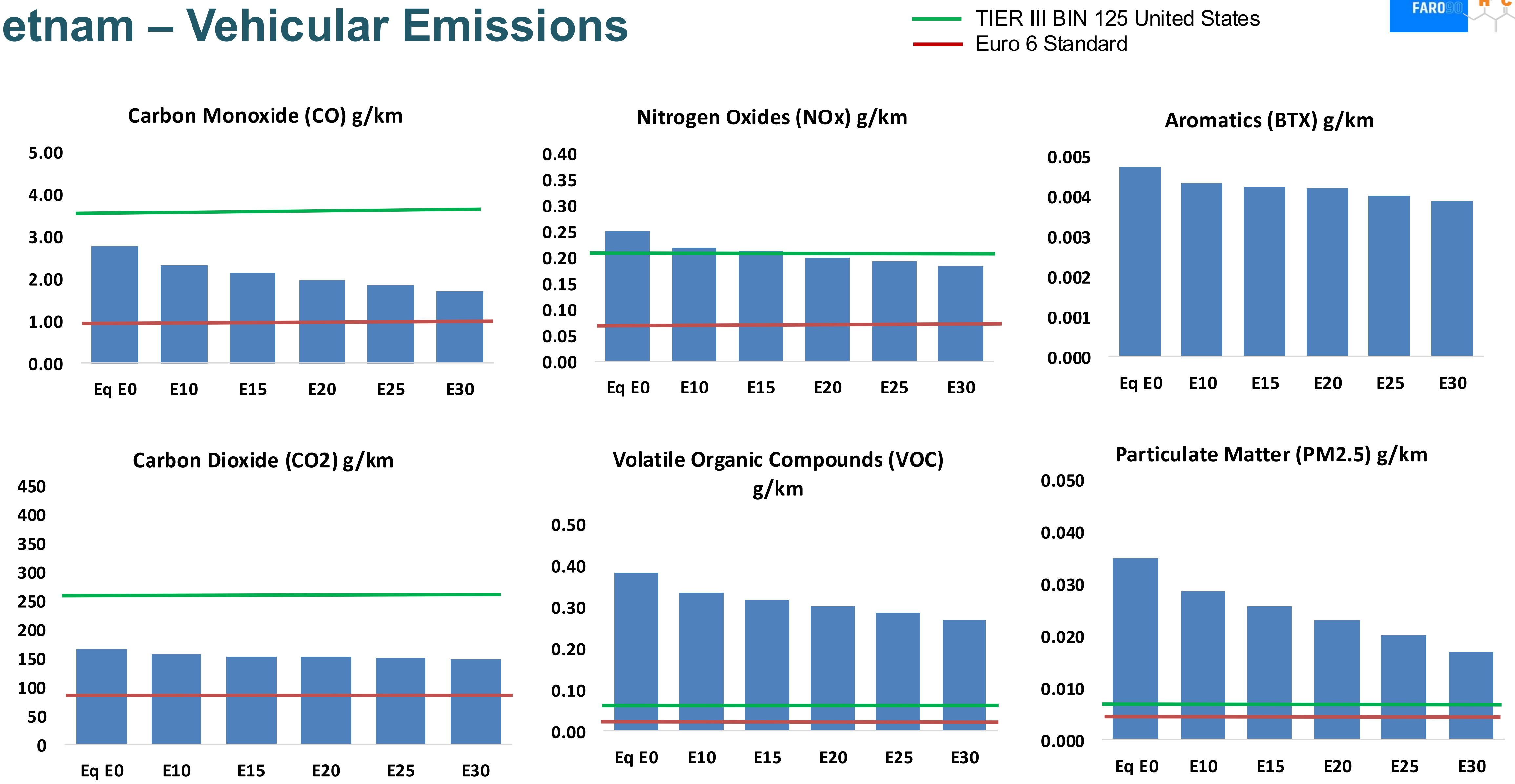
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	492
		Single FI	3-Ways		>8 years	15,070
		Multi FI	3-Ways		>8 years	60,730
			Euro III		>8 years	77,663
			Euro IV		>8 years	240,336
			Euro V		>8 years	159,849
			Euro 6		>8 years	485,445
					4-8 years	986,085
					<4 years	1,380,674
			Hybrid		All	7,022
		Electric			All	171,294
	Medium	Carburator	3-Ways	PCV Tank	>8 years	226
		Single FI	3-Ways		>8 years	6,925
		Multi FI	3-Ways		>8 years	27,904
			Euro III		>8 years	35,685
			Euro IV		>8 years	110,429
			Euro V		>8 years	73,447
			Euro 6		>8 years	223,051
					4-8 years	453,085
					<4 years	634,390
			Hybrid		All	3,226
		Electric			All	78,706
Sml Engines	Light	4-Cycle	None	PCV	>8 years	787,212
			Euro III		>8 years	801,358
			Euro IV		>8 years	2,377,973
			Euro V		>8 years	1,481,363
			Euro 6		>8 years	4,266,669
					4-8 years	10,604,608
					<4 years	4,191,107
			Hybrid		All	0
		Electric			All	2,059,748
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	394
		Fuel-Injection	3-Ways		>8 years	102,199
			Euro III		>8 years	139,582
			Euro IV		>8 years	443,080
			Euro V		>8 years	260,838
			Euro 6		>8 years	713,755
					4-8 years	1,186,849
					<4 years	1,858,854
			Hybrid		All	0
		Electric			All	1,534

0 2,000,000 4,000,000 6,000,000 8,000,000 10,000,000 12,000,000

Vehicle Fleet: 35,508,859
Gasoline Vehicle Fleet: 25,550,917

Source: MoIT

Vietnam – Vehicular Emissions



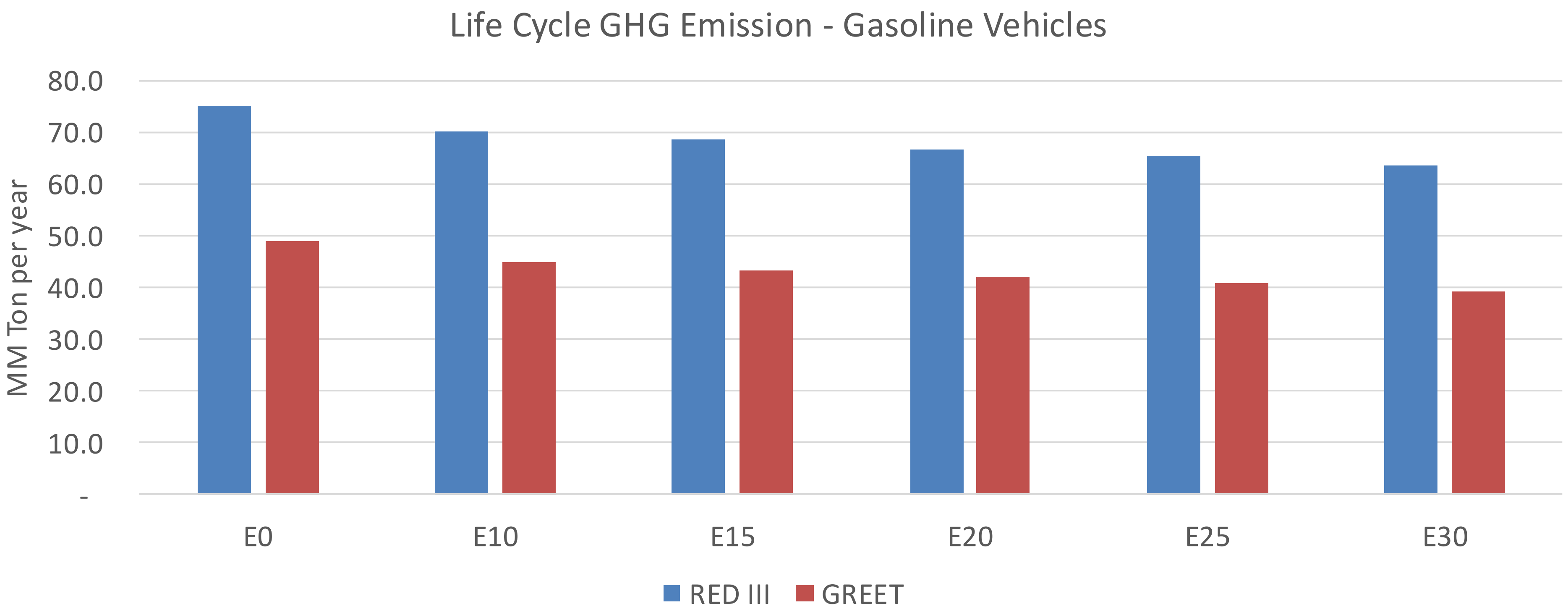
Source: Faro90

Vietnam – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,359,820	1,252,428	1,145,036	1,056,476	967,099	898,296	825,989	-16%	-29%
CO2	80,379,150	78,312,257	76,360,192	74,824,950	74,065,253	73,754,180	72,394,679	-5%	-8%
VOC	186,910	175,504	164,098	155,329	146,771	140,271	131,313	-12%	-21%
NOx	122,884	113,463	107,004	103,717	98,418	94,266	89,376	-13%	-20%
SOx	872	806	740	685	631	573	512	-15%	-28%
NH3	33,331	33,001	32,671	32,826	33,195	33,455	33,671	-2%	0%
N2O	1,350	1,342	1,337	1,343	1,371	1,386	1,402	-1%	2%
CH4	47,919	47,919	47,919	48,637	49,692	50,602	51,465	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,465	1,391	1,318	1,265	1,216	1,179	1,126	-10%	-17%
Acetaldehyde	3,770	4,327	6,348	9,668	13,171	15,051	17,784	68%	249%
Formaldehyde	14,599	14,194	16,546	19,429	20,354	22,517	24,513	13%	39%
Benzene	2,328	2,237	2,120	2,086	2,069	1,979	1,915	-9%	-11%
PM2.5	5,474	4,977	4,479	4,041	3,612	3,153	2,672	-18%	-34%
PM10	11,540	10,491	9,442	8,519	7,615	6,647	5,634	-18%	-34%

Source: Faro90

Vietnam – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	489.9
Target @ 2035 (MMT/year)	330.6
Delta	159.3

vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.6%	11.5%	14.4%	16.7%	20.0%
RED II/III	0.0%	6.5%	8.9%	11.1%	13.0%	15.5%

Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.6%	3.5%	4.4%	5.1%	6.1%
RED II/III	0.0%	3.1%	4.2%	5.2%	6.1%	7.3%

Source: Greet, RED II/III, climateactiontracker.org, F90